US012411392B2

(12) **United States Patent**
Shabtay et al.

(10) **Patent No.: US 12,411,392 B2**
(45) **Date of Patent: *Sep. 9, 2025**

(54) **SLIM POP-OUT CAMERAS AND LENSES FOR SUCH CAMERAS**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **19/010,050**

(22) Filed: **Jan. 5, 2025**

(65) **Prior Publication Data**

US 2025/0138396 A1 May 1, 2025

Related U.S. Application Data

(63) Continuation of application No. 18/768,244, filed on Jul. 10, 2024, now Pat. No. 12,292,671, which is a

(Continued)

(51) **Int. Cl.**

G03B 17/04 (2021.01)
G02B 7/02 (2021.01)

(Continued)

(52) **U.S. Cl.**

CPC ***G03B 17/04*** (2013.01); ***G02B 7/021*** (2013.01); ***G02B 7/026*** (2013.01); ***G02B 13/02*** (2013.01);

(Continued)

(58) **Field of Classification Search**

CPC H04N 23/54; H04N 23/55; H04N 23/57; G02B 7/021; G02B 7/026; G02B 13/02;

(Continued)

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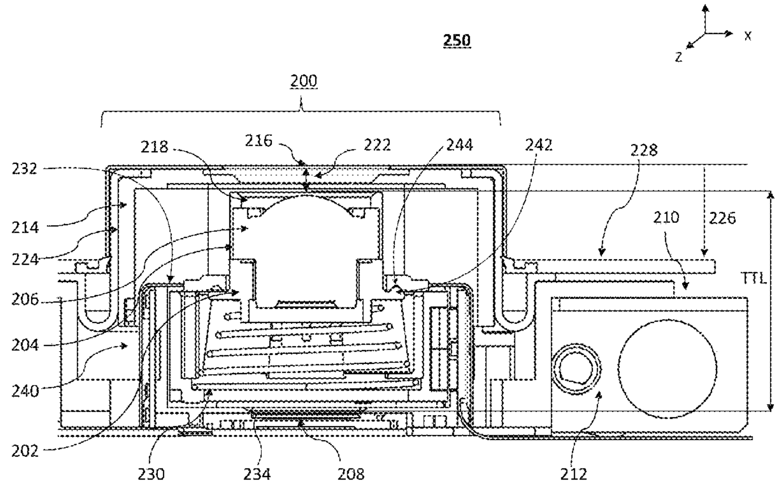
(57) **ABSTRACT**

Digital cameras comprising a lens assembly comprising N lens elements L_1-L_N starting with L_1 on an object side, wherein N is ≥4, an image sensor having a sensor diagonal S_D, and a pop-out mechanism that controls a largest air-gap d between two consecutive lens elements within lens elements L_1 and L_N to bring the camera to an operative pop-out state and a collapsed state, wherein the lens assembly has a total track length TTL in the operative pop-out state and a collapsed total track length cTTL in the collapsed state, wherein S_D is in the range of 7-20 mm and wherein $cTTL/S_D<0.6$.

20 Claims, 48 Drawing Sheets



Related U.S. Application Data

continuation of application No. 17/567,417, filed on Jan. 3, 2022, now Pat. No. 12,066,747, which is a continuation of application No. 17/460,231, filed on Aug. 29, 2021, now Pat. No. 12,072,609, which is a continuation-in-part of application No. 17/291,475, filed as application No. PCT/IB2020/058697 on Sep. 18, 2020, now abandoned.

(60) Provisional application No. 63/037,836, filed on Jun. 11, 2020, provisional application No. 63/026,317, filed on May 18, 2020, provisional application No. 62/904,913, filed on Sep. 24, 2019.

(51) **Int. Cl.**

G02B 13/02	(2006.01)
G02B 13/06	(2006.01)
G03B 17/12	(2021.01)
G03B 30/00	(2021.01)
H04N 23/54	(2023.01)
H04N 23/55	(2023.01)
H04N 23/57	(2023.01)

(52) **U.S. Cl.**
CPC ***G02B 13/06*** (2013.01); ***G03B 30/00*** (2021.01); ***H04N 23/54*** (2023.01); ***H04N 23/55*** (2023.01); ***H04N 23/57*** (2023.01); *G03B 17/12* (2013.01)

(58) **Field of Classification Search**
CPC G02B 13/06; G03B 30/00; G03B 17/04; G03B 17/12
See application file for complete search history.

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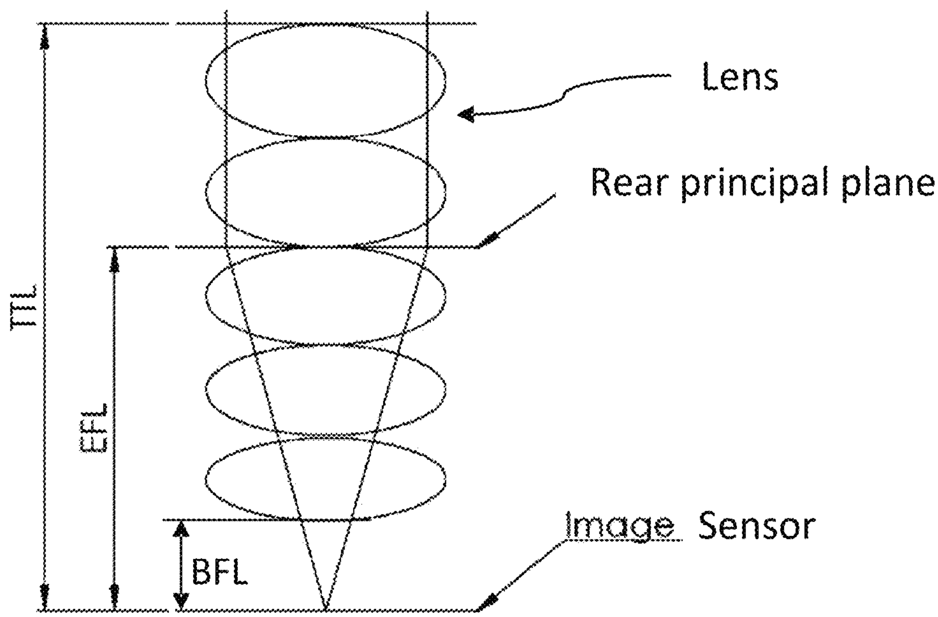

Lens
Rear principal plane
TTL
EFL
BFL
Image Sensor

FIG.1A

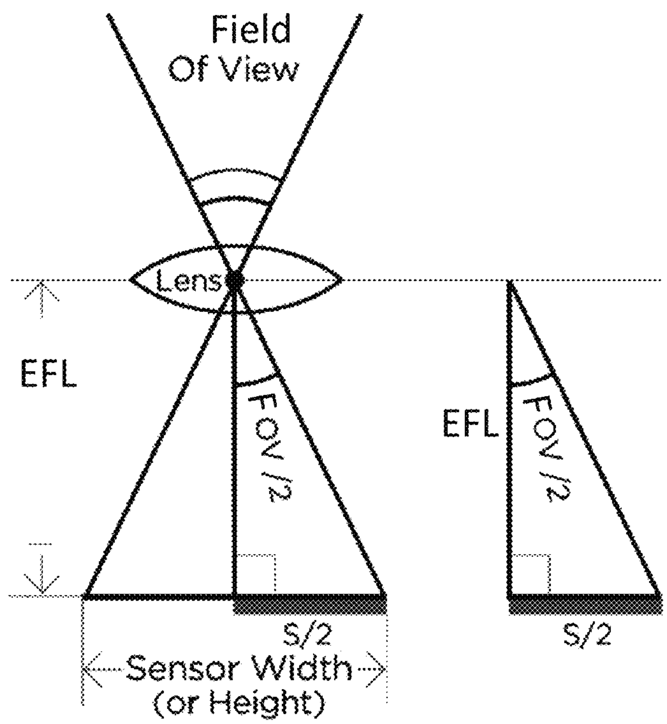
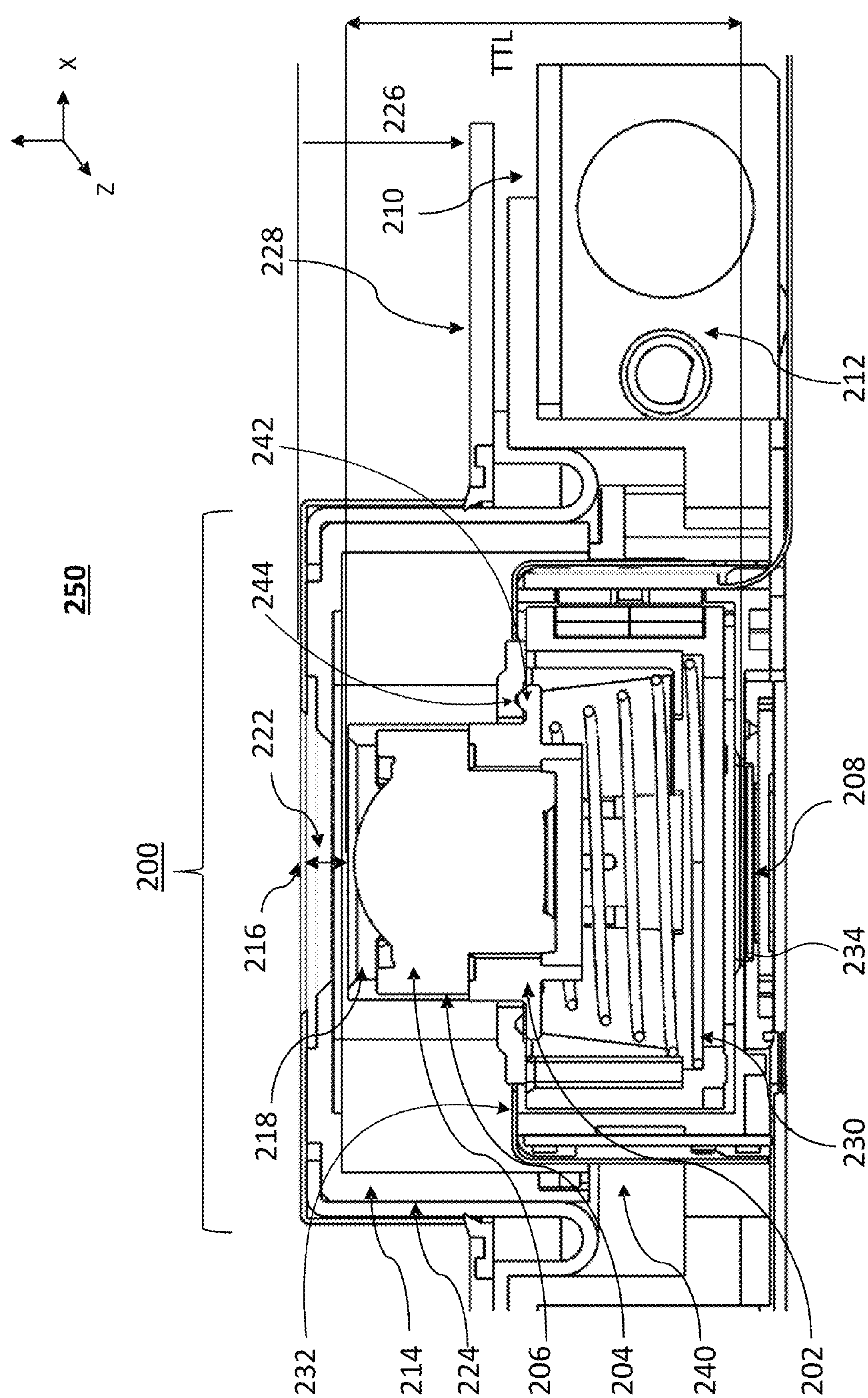

Field
Of View
Lens
EFL
FOV /2
EFL
FOV /2
S/2
S/2
Sensor Width
(or Height)

FIG.1B



X
Z
TTL
226
210
228
212
242
250
244
222
208
200
216
234
218
230
232
214
224
206
204
240
202

FIG.2A

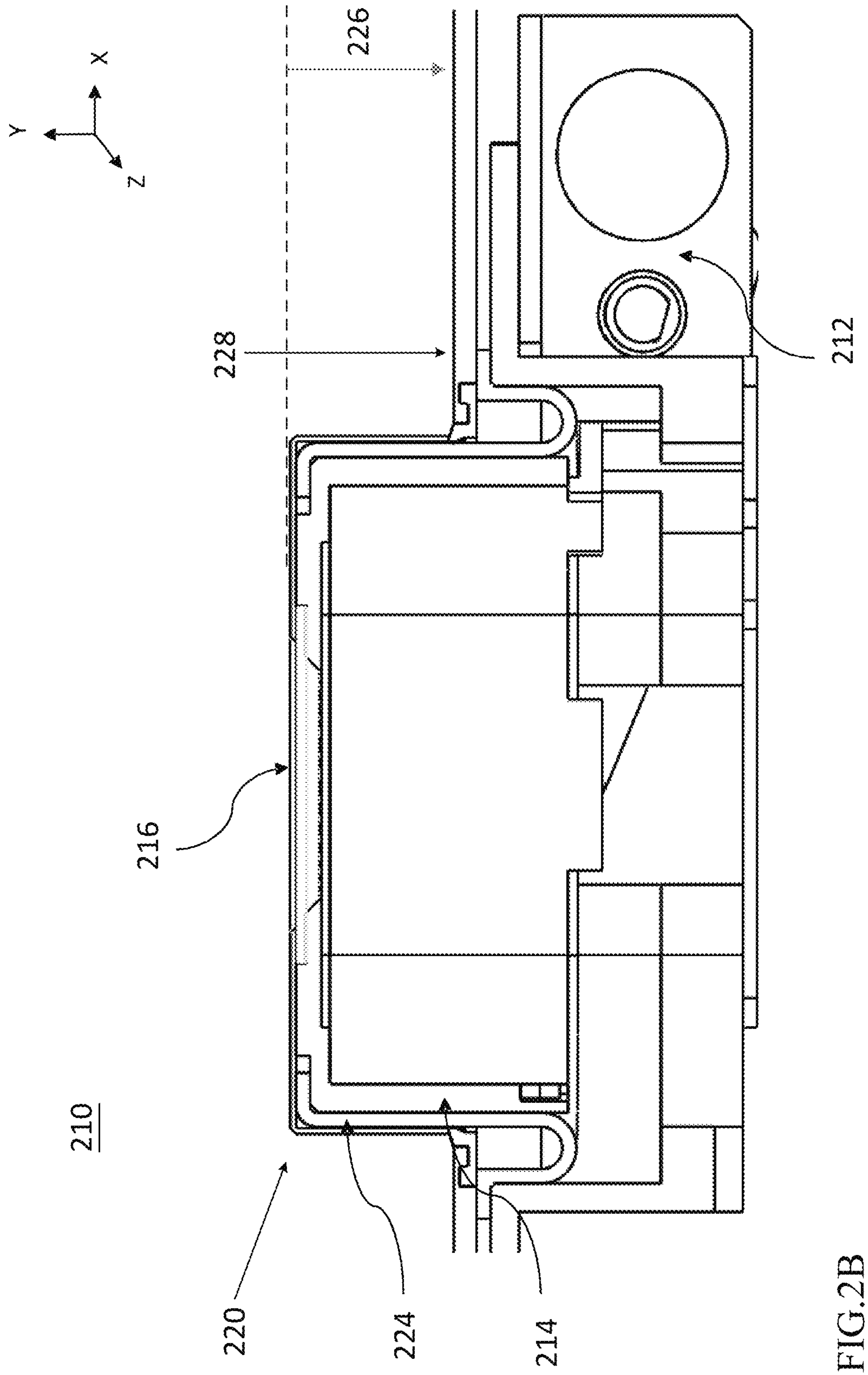


FIG.2B

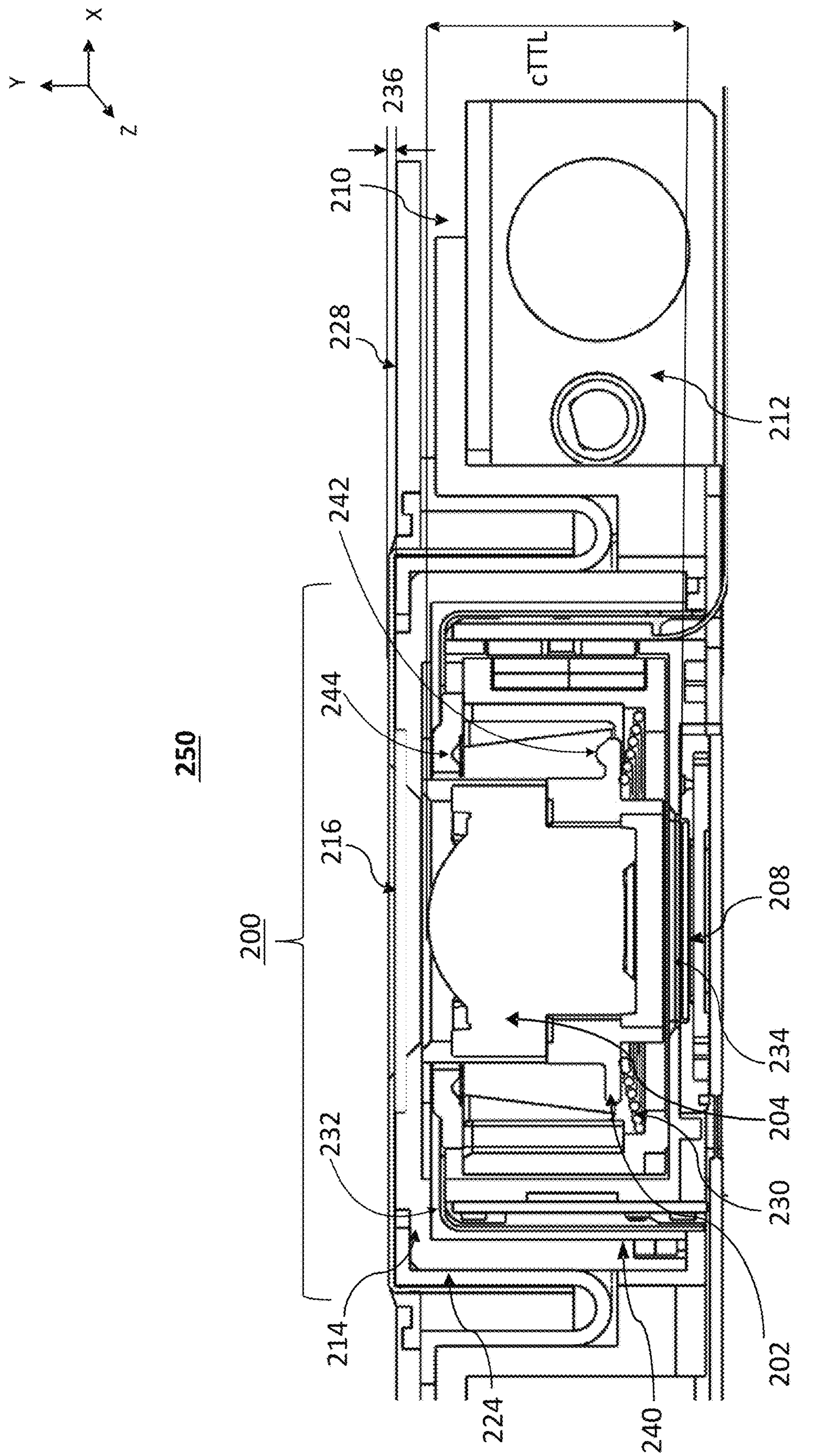

X
Y
Z
cTTL
236
210
228
212
242
244
250
216
200
208
234
204
230
232
214
224
240
202

FIG.2C

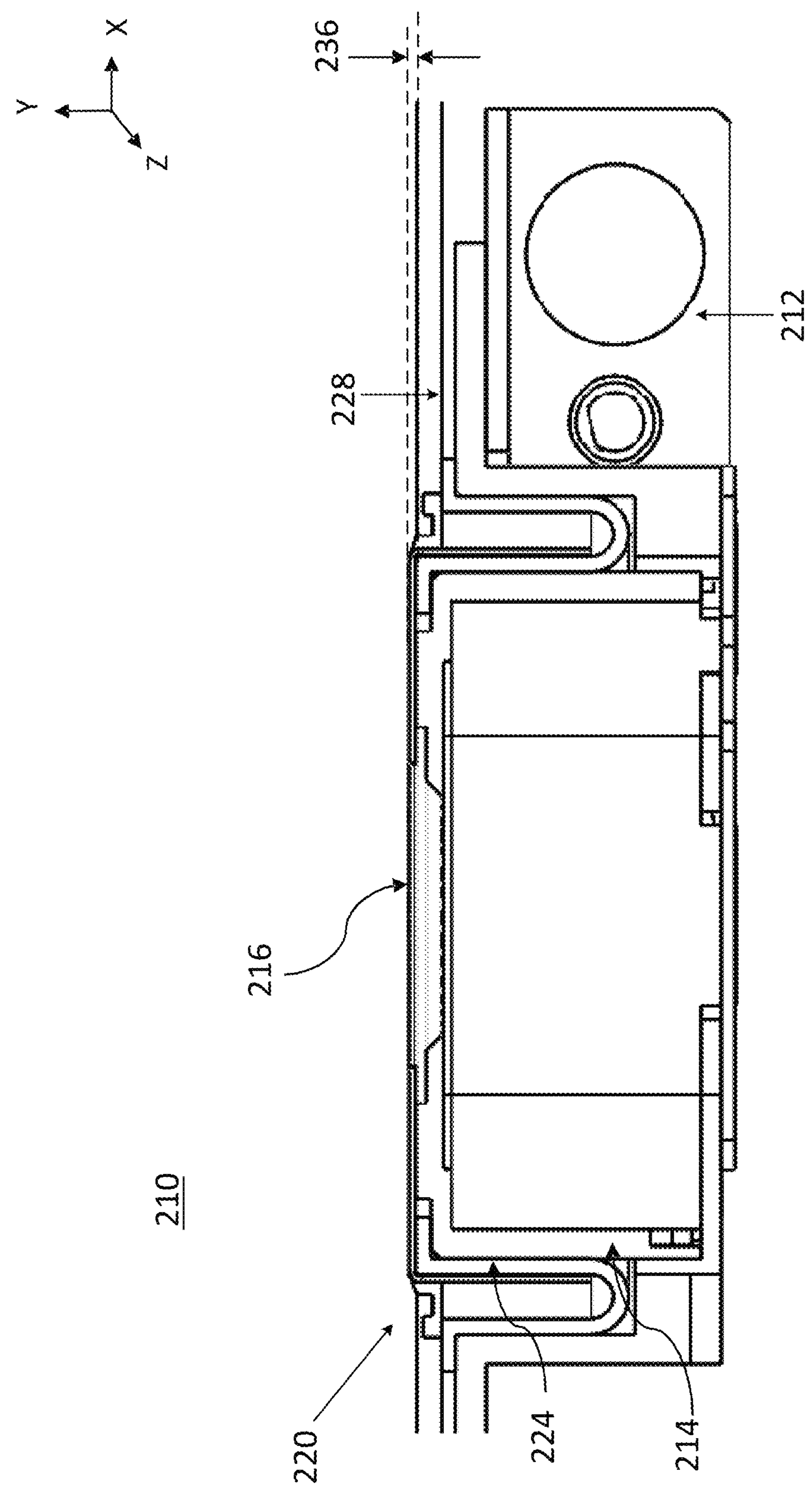


FIG.2D

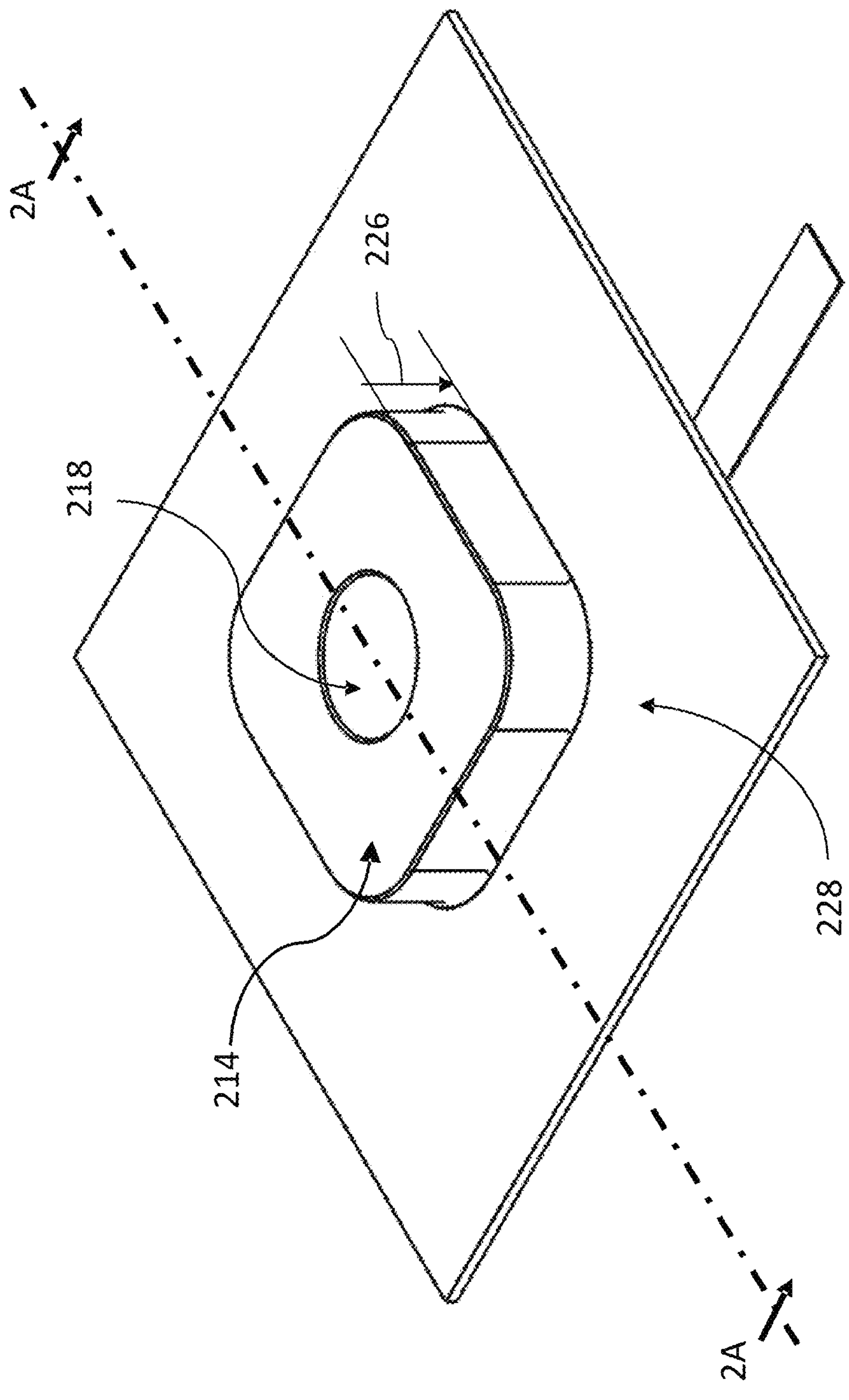


FIG.3A

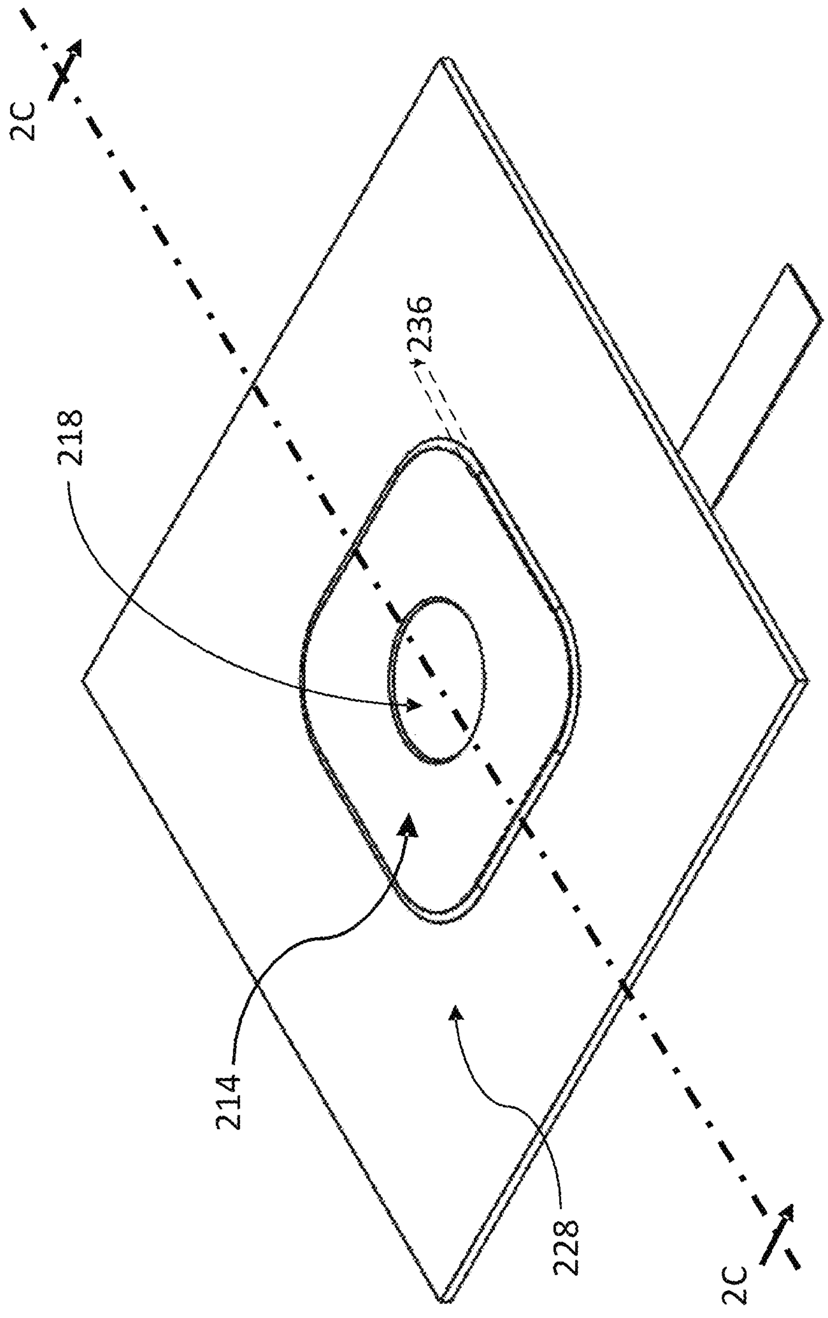


FIG.3B

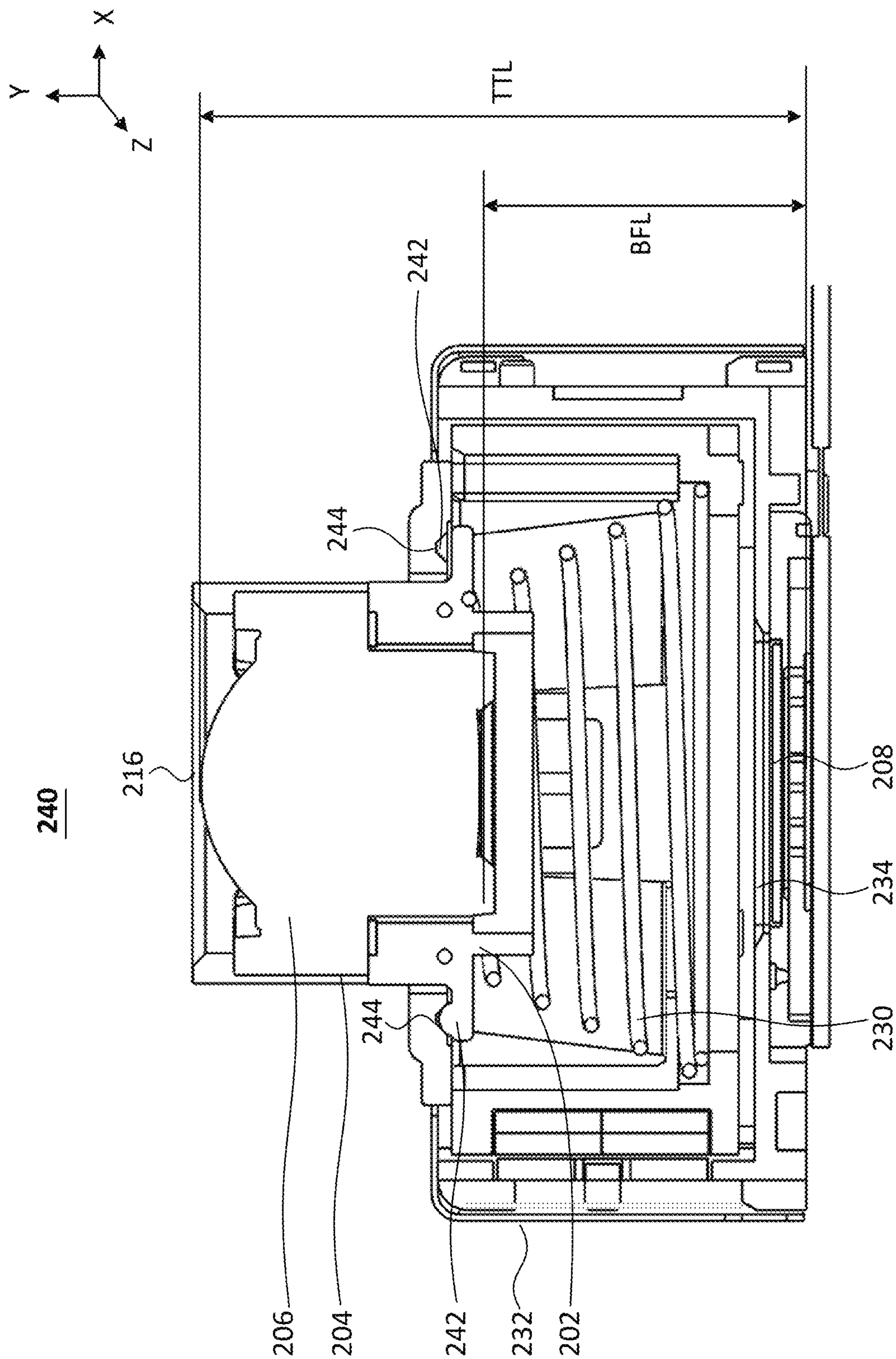


FIG.4A

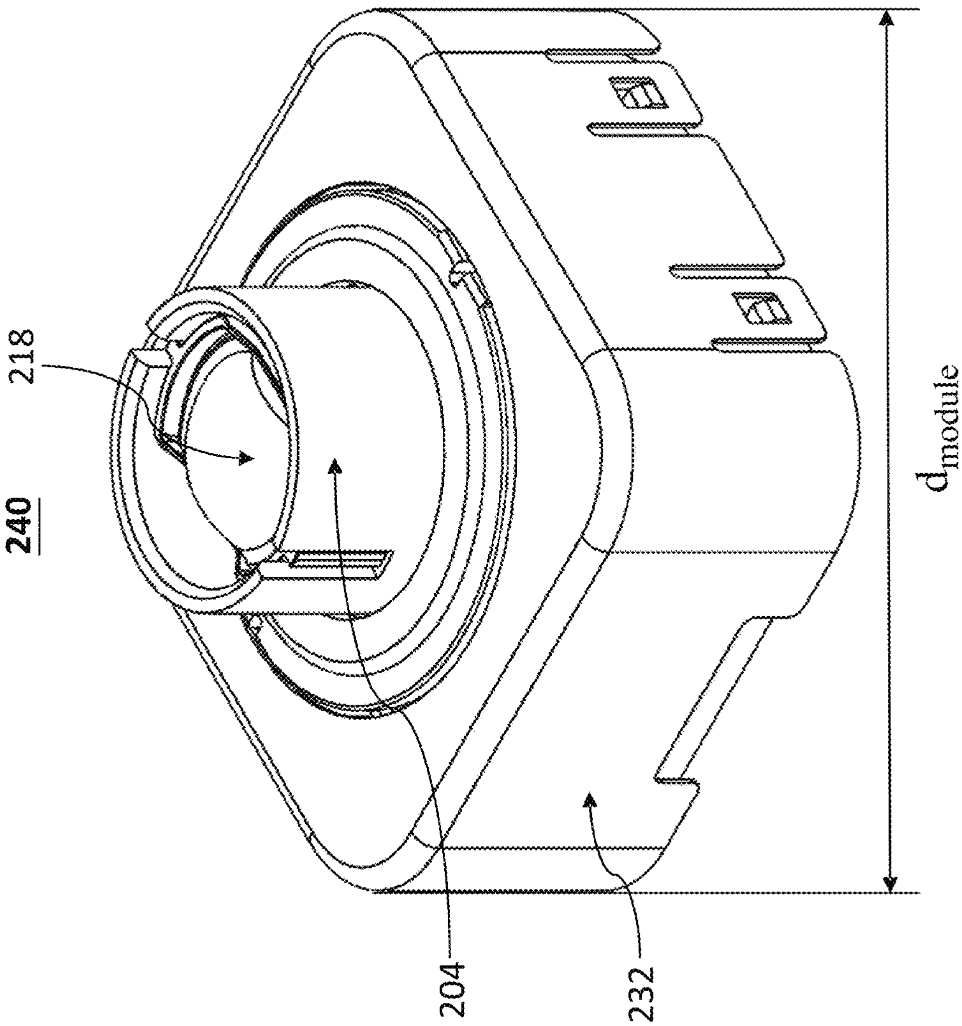


FIG. 4B

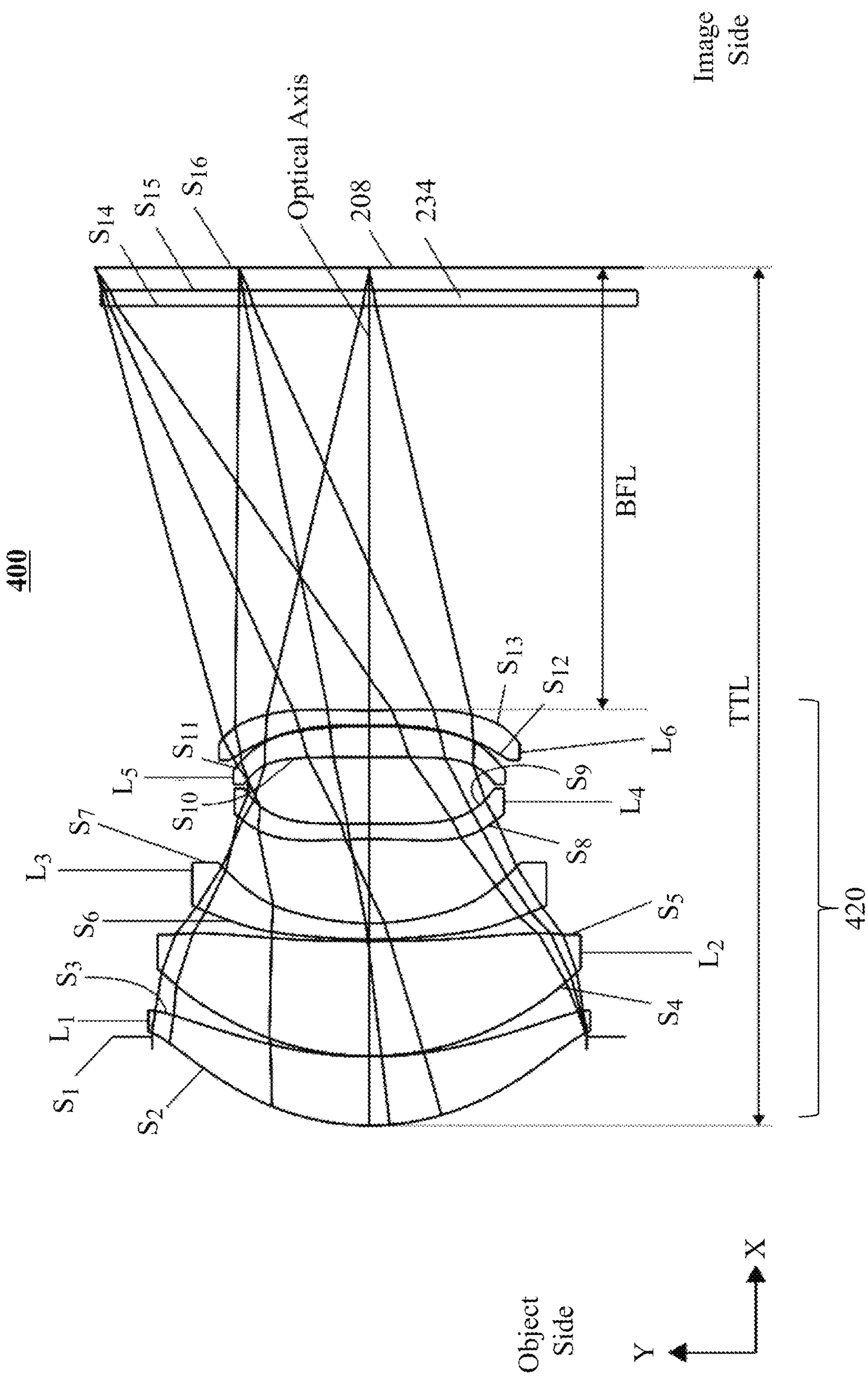

400
Image Side
Object Side
Optical Axis
208
234
S_{14}
S_{15}
S_{16}
S_{13}
S_{12}
S_{11}
S_{10}
S_9
S_8
S_7
S_6
S_5
S_4
S_3
S_2
S_1
L_1
L_2
L_3
L_4
L_5
L_6
BFL
TTL
420
X
Y

FIG.4C

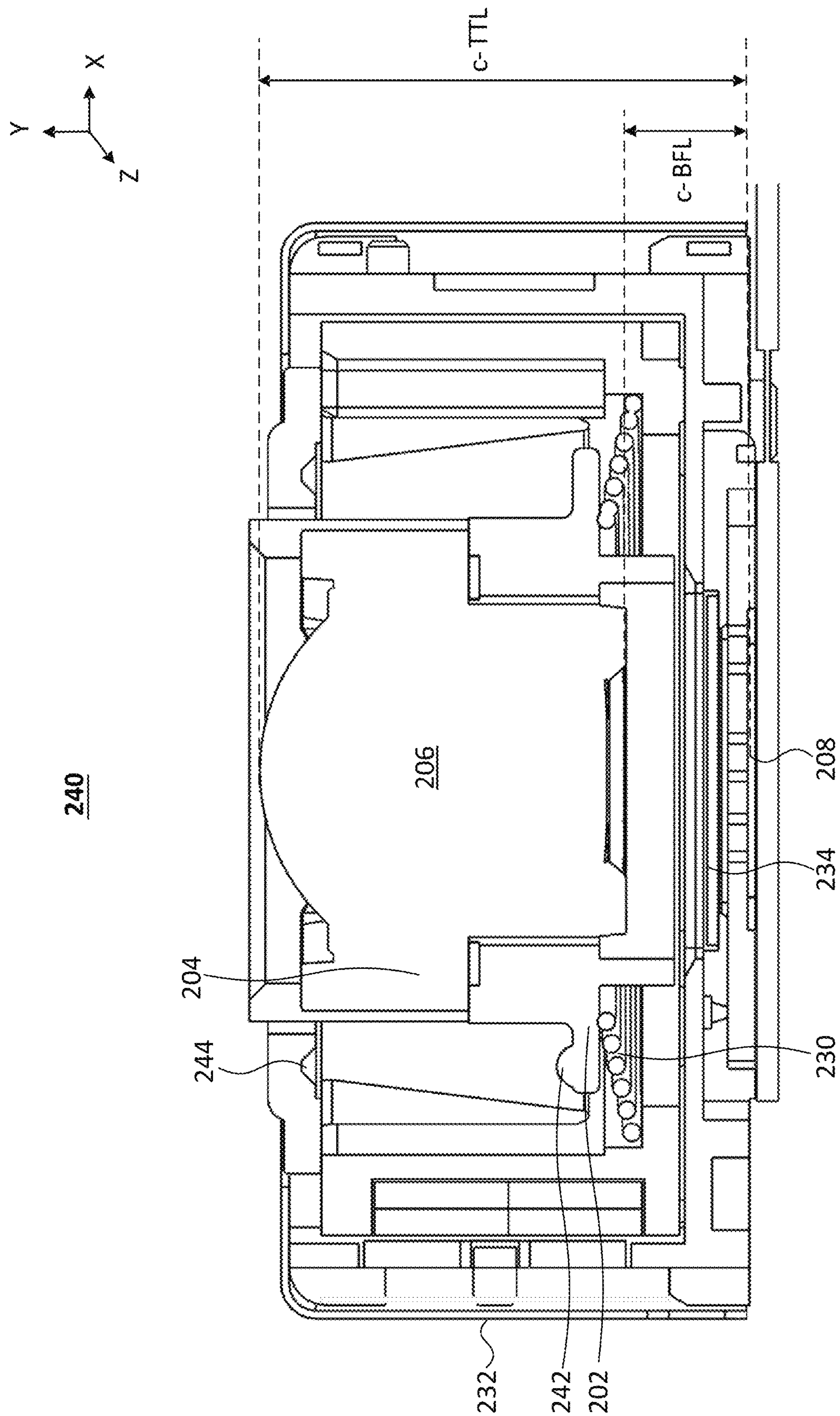


FIG.5A

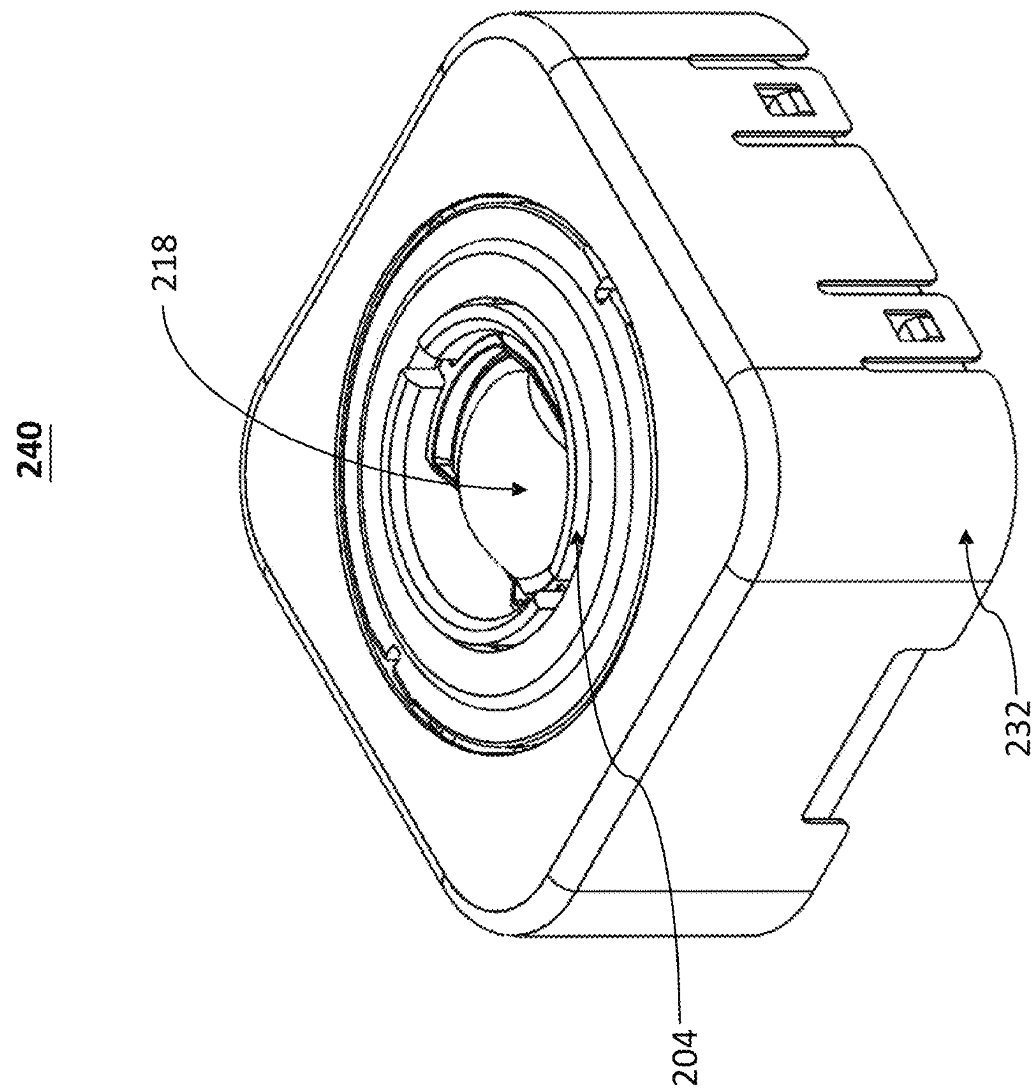


FIG. 5B

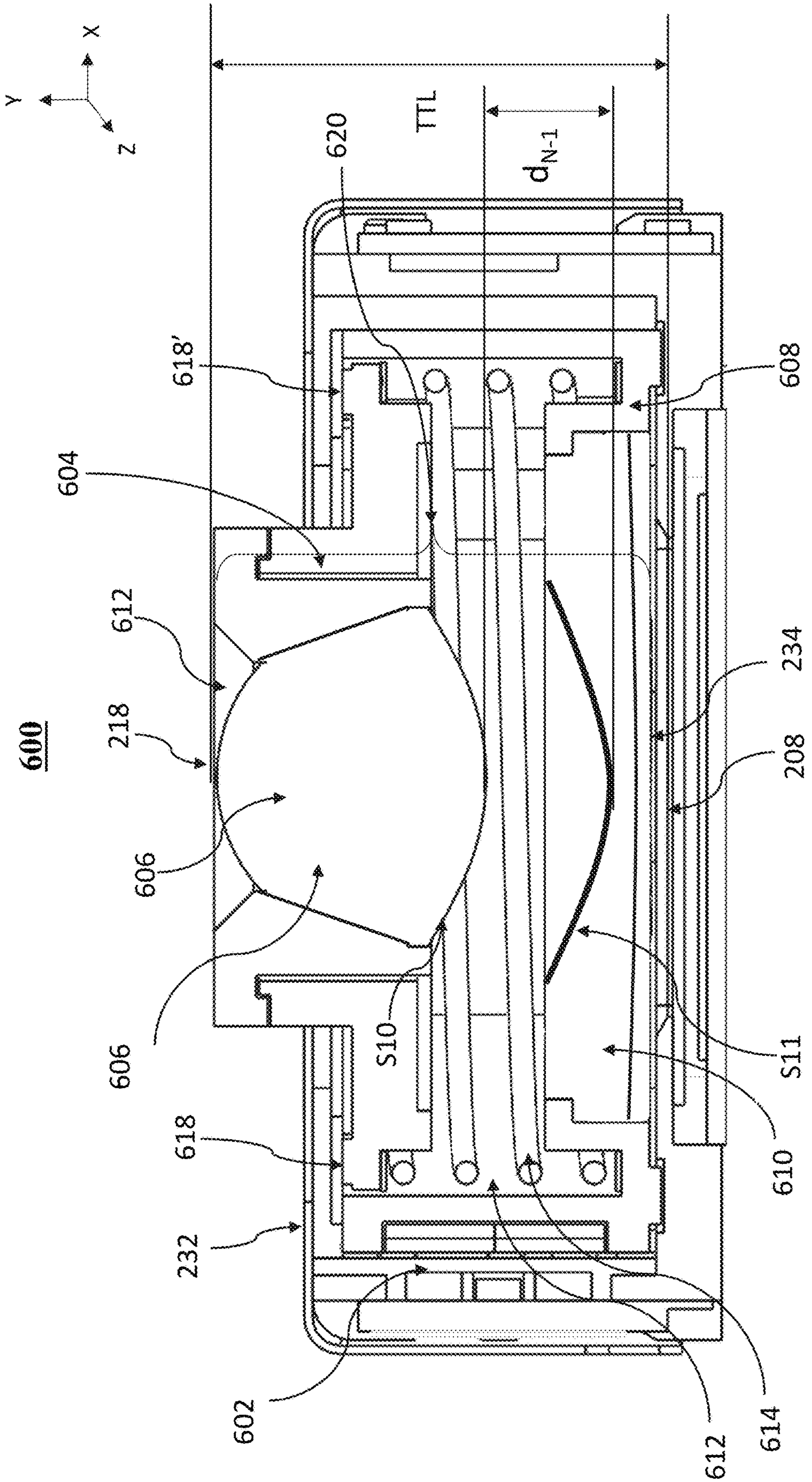

600
X
Y
Z
TTL
d_{N-1}
620
618'
604
612
218
606
606
618
232
602
612
614
608
234
208
S11
610
S10

FIG. 6A

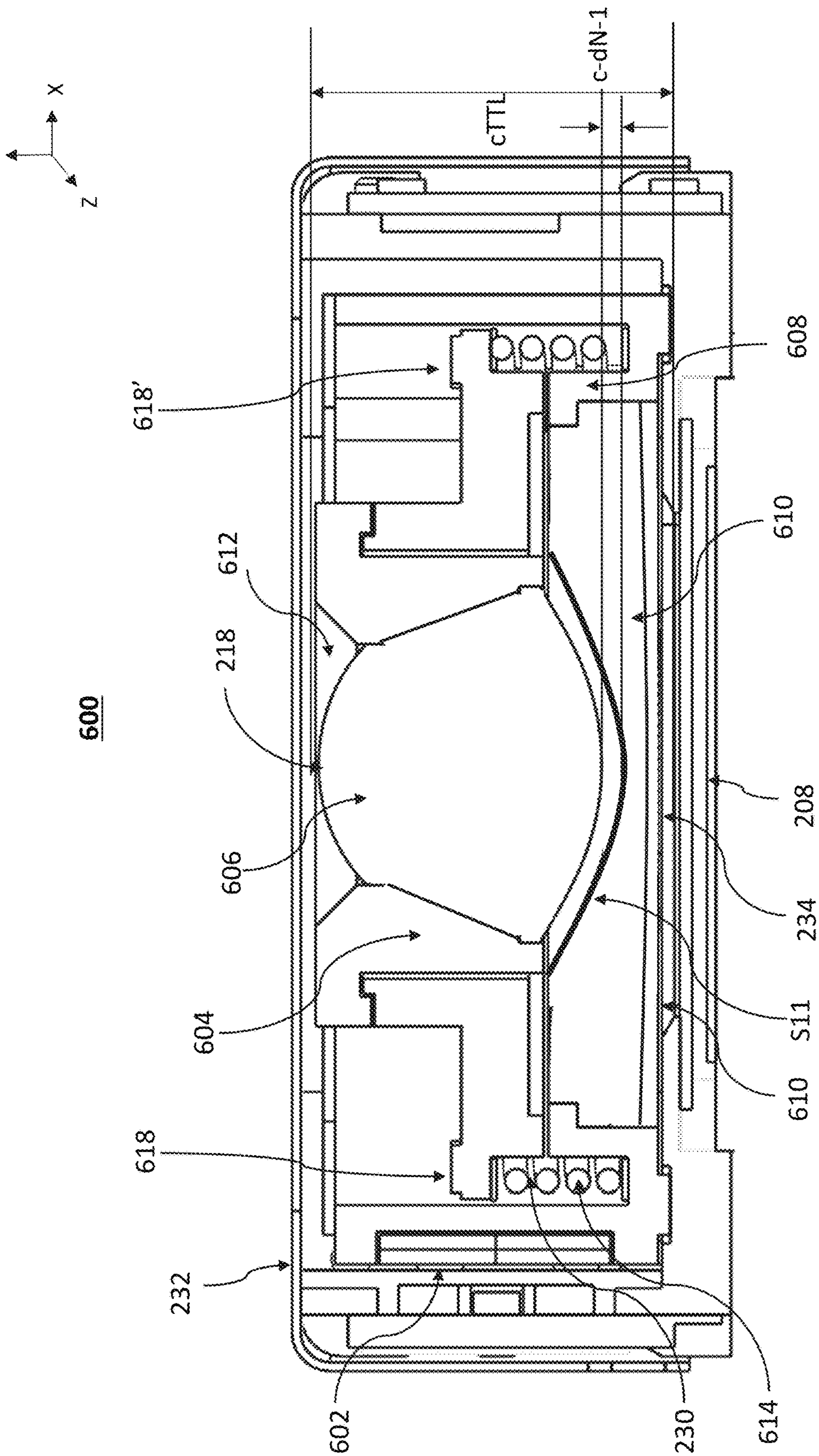

600
x
z
cTTL
c-dN-1
618'
608
612
610
218
606
208
234
604
S11
610
618
232
602
230
614

FIG. 6B

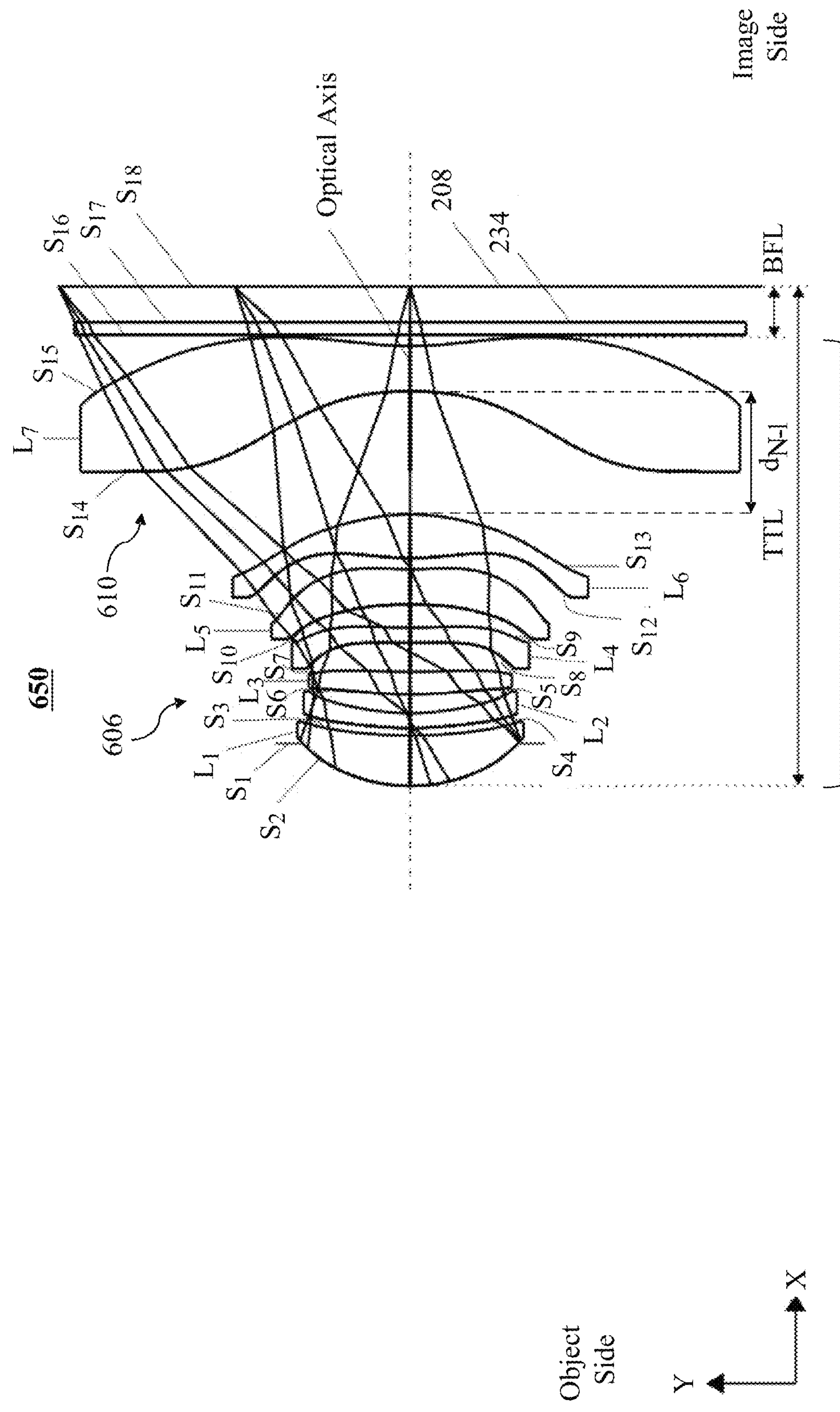

650
606
610
620'
Optical Axis
Image Side
Object Side
208
234
BFL
TTL
d_{N-1}
L_1
L_2
L_3
L_4
L_5
L_6
L_7
S_1
S_2
S_3
S_4
S_5
$S6$
S_7
S_8
S_9
S_{10}
S_{11}
S_{12}
S_{13}
S_{14}
S_{15}
S_{16}
S_{17}
S_{18}
X
Y

FIG. 6C

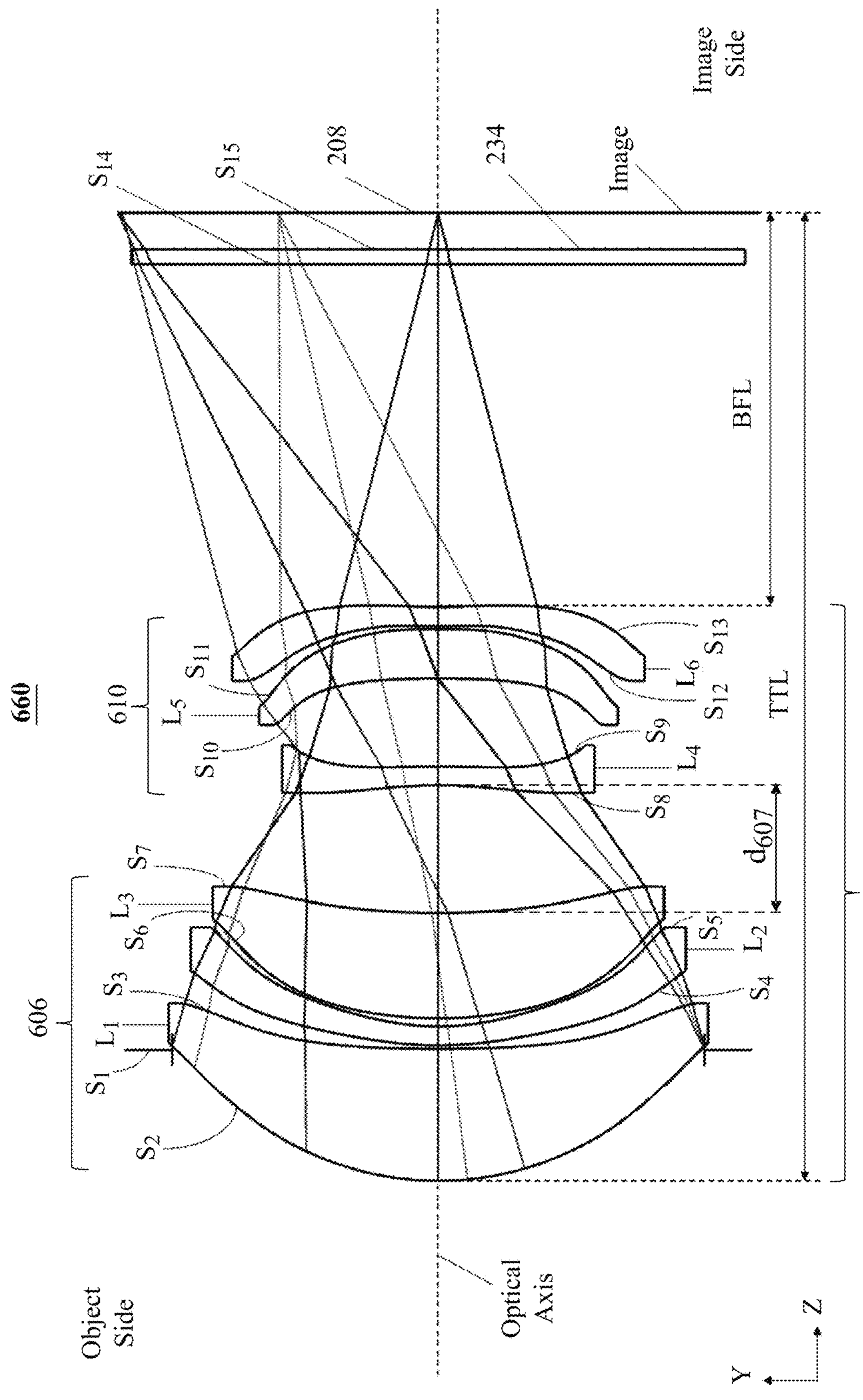

660
Object Side
Image Side
Optical Axis
606
610
620''
208
234
Image
BFL
TTL
d_{607}
S_1
S_2
S_3
S_4
S_5
S_6
S_7
S_8
S_9
S_{10}
S_{11}
S_{12}
S_{13}
S_{14}
S_{15}
L_1
L_2
L_3
L_4
L_5
L_6
Y
Z

FIG.6D

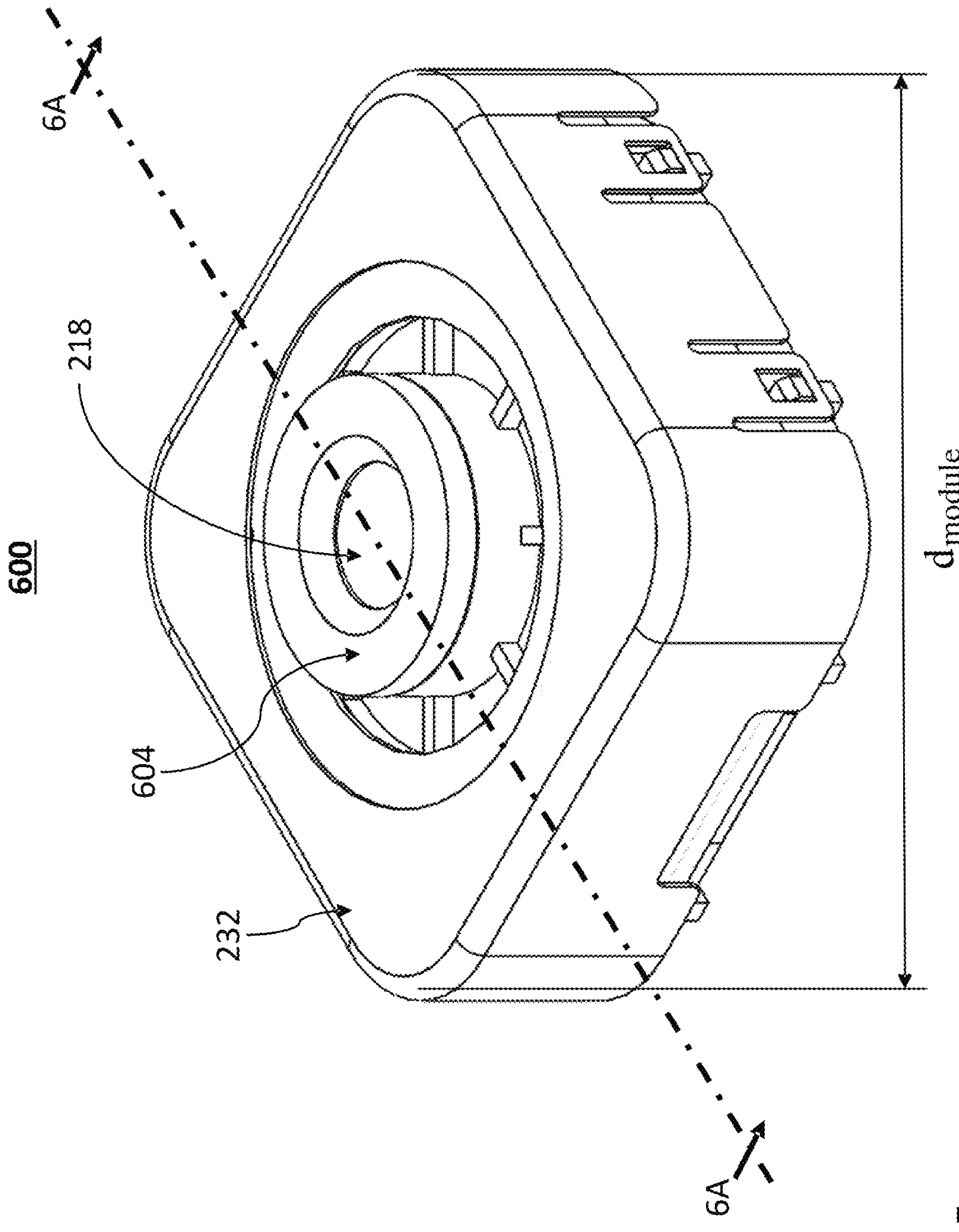


FIG.7

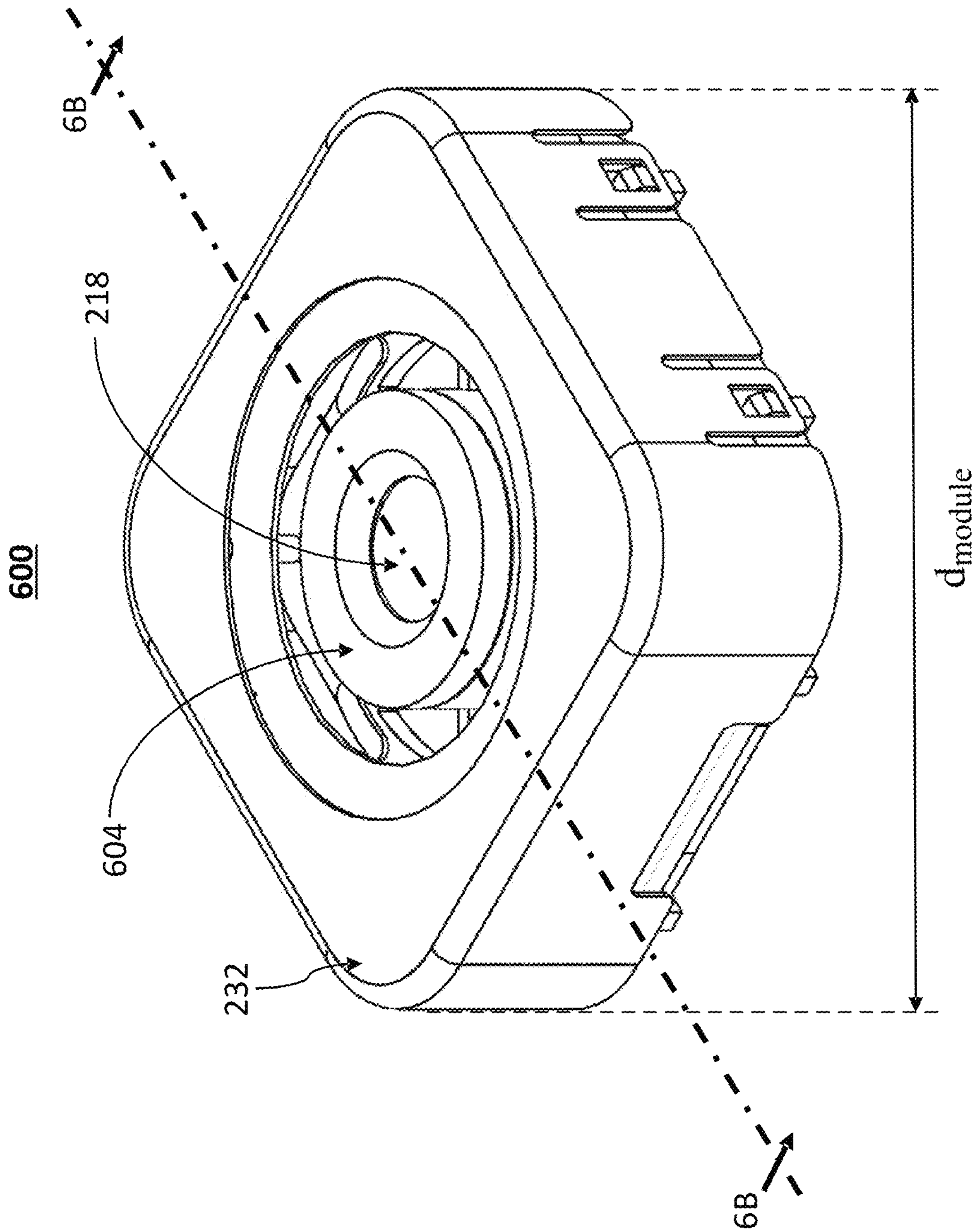


FIG.8

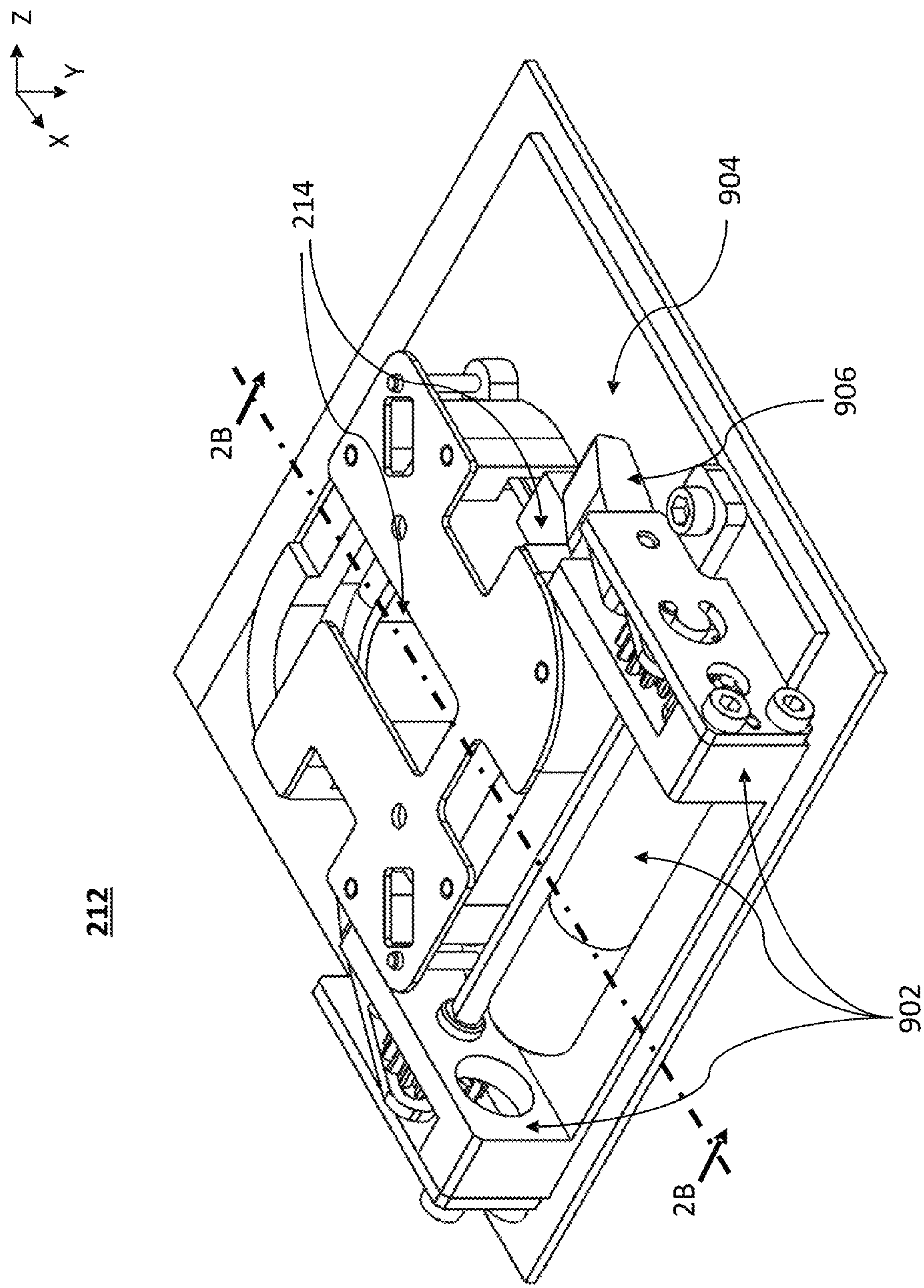
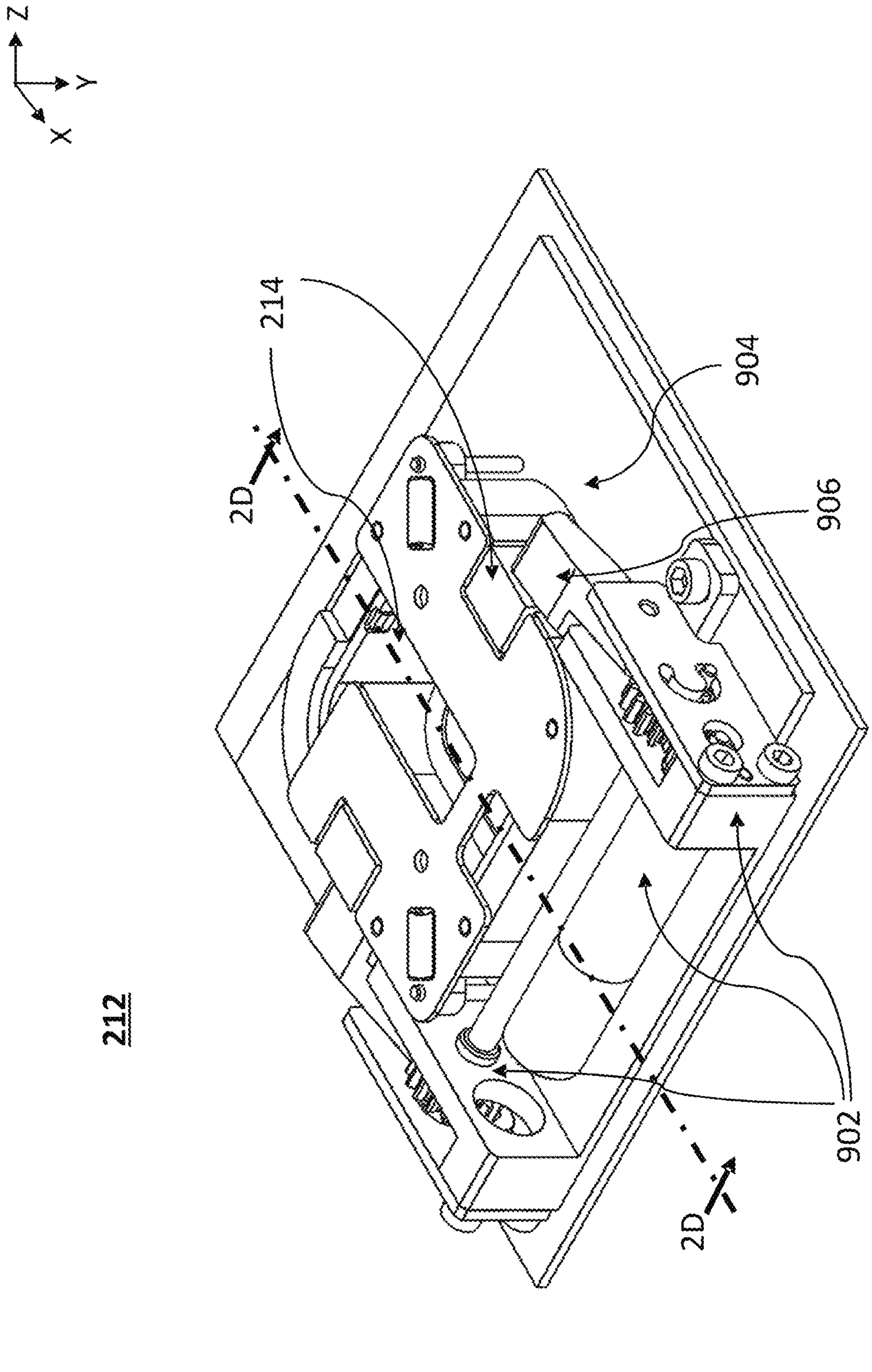


FIG.9A



212

FIG.9B

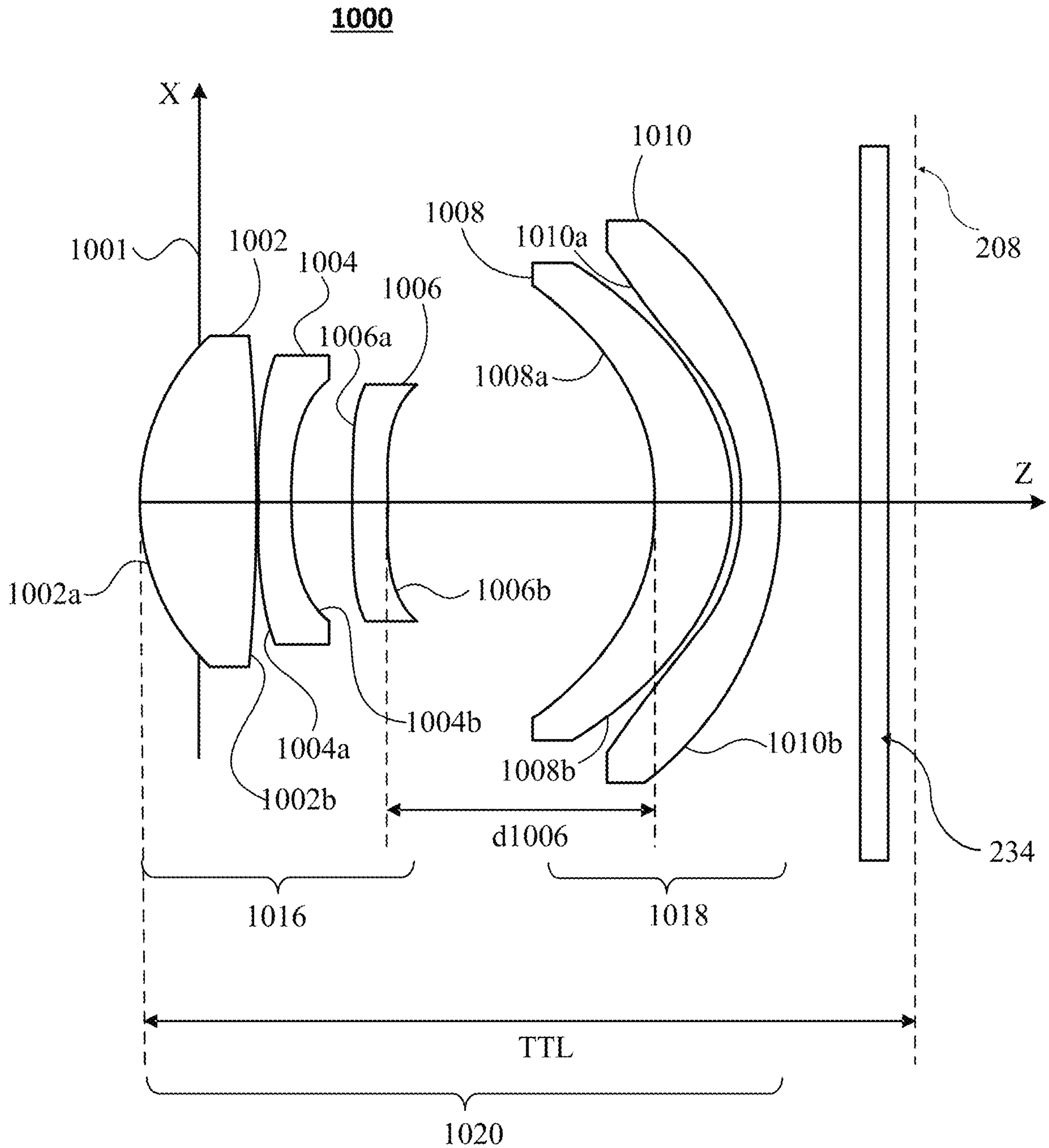

1000
X
Z
1001
1002
1004
1006
1006a
1008
1008a
1010
1010a
208
1002a
1006b
1004b
1004a
1002b
1008b
1010b
234
d1006
1016
1018
TTL
1020

FIG. 10

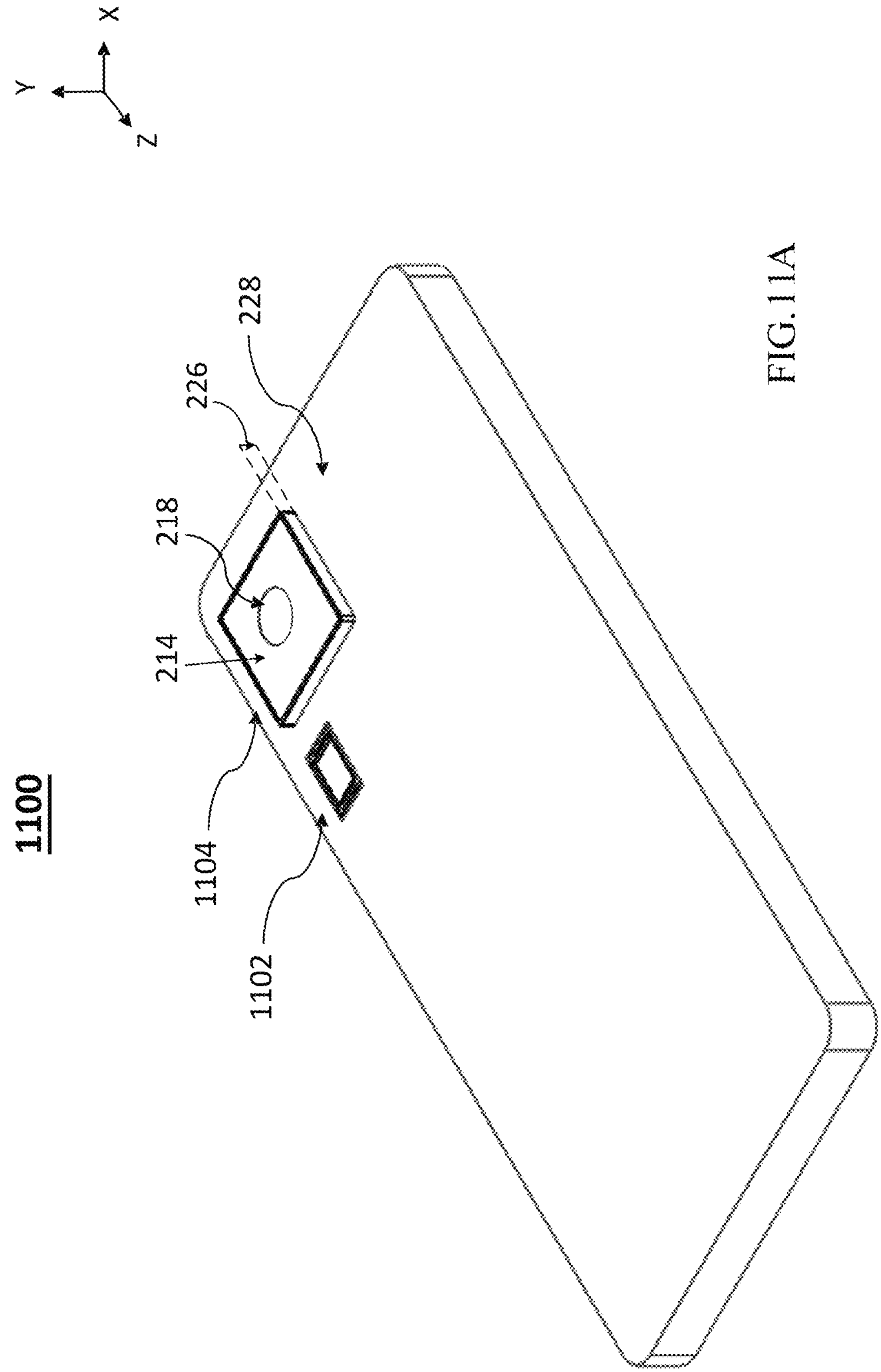


FIG. 11A

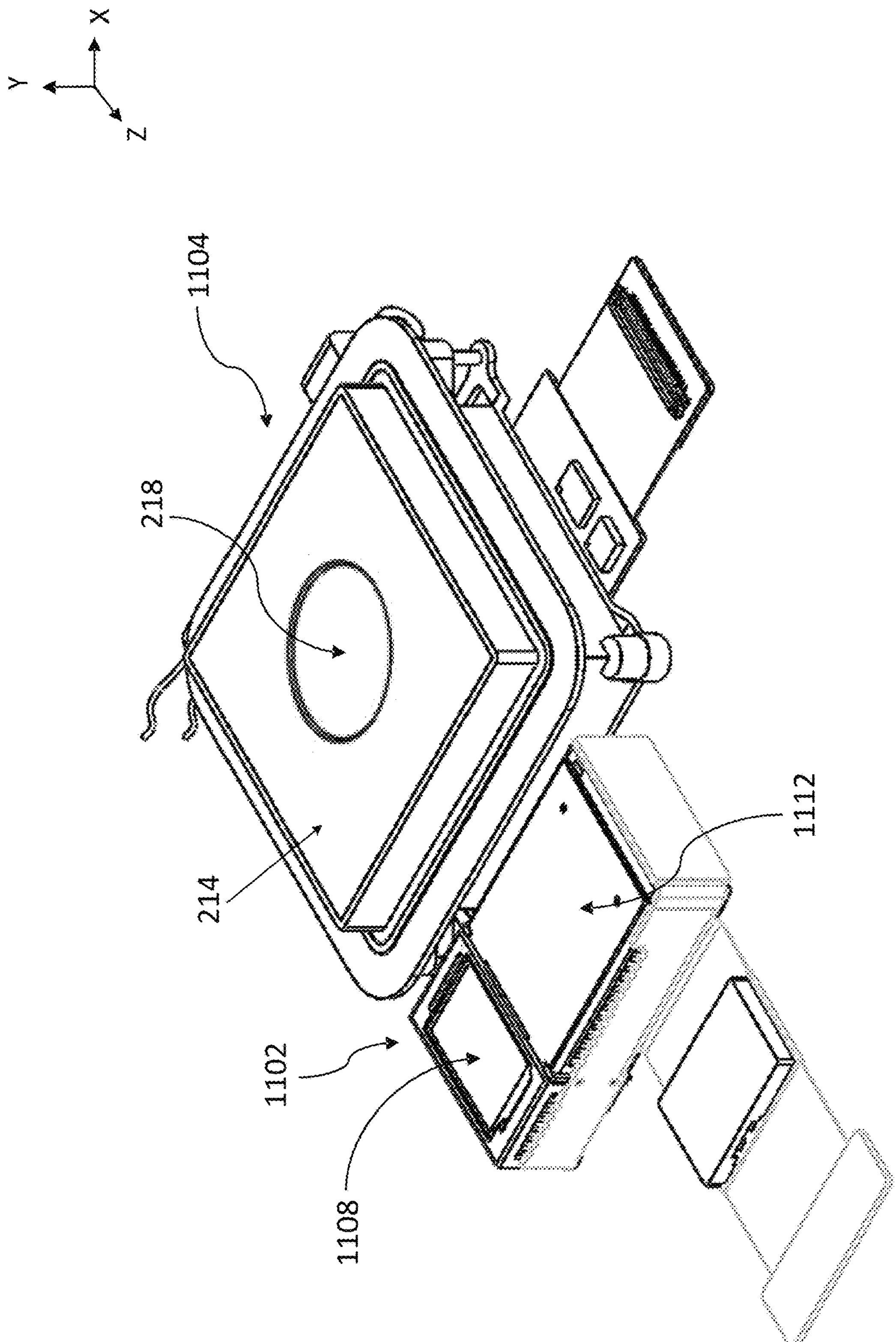


FIG.11B

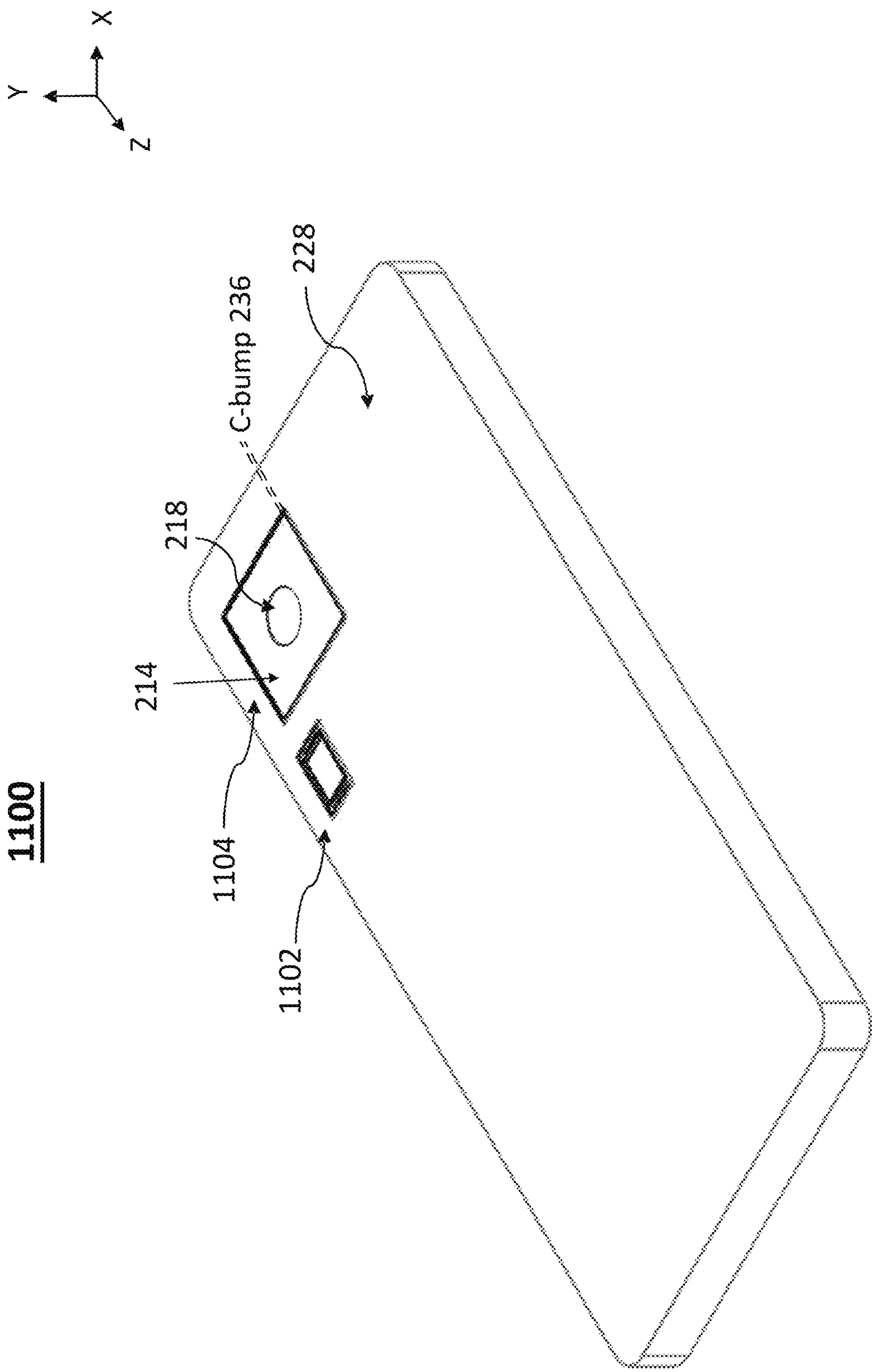


FIG.11C

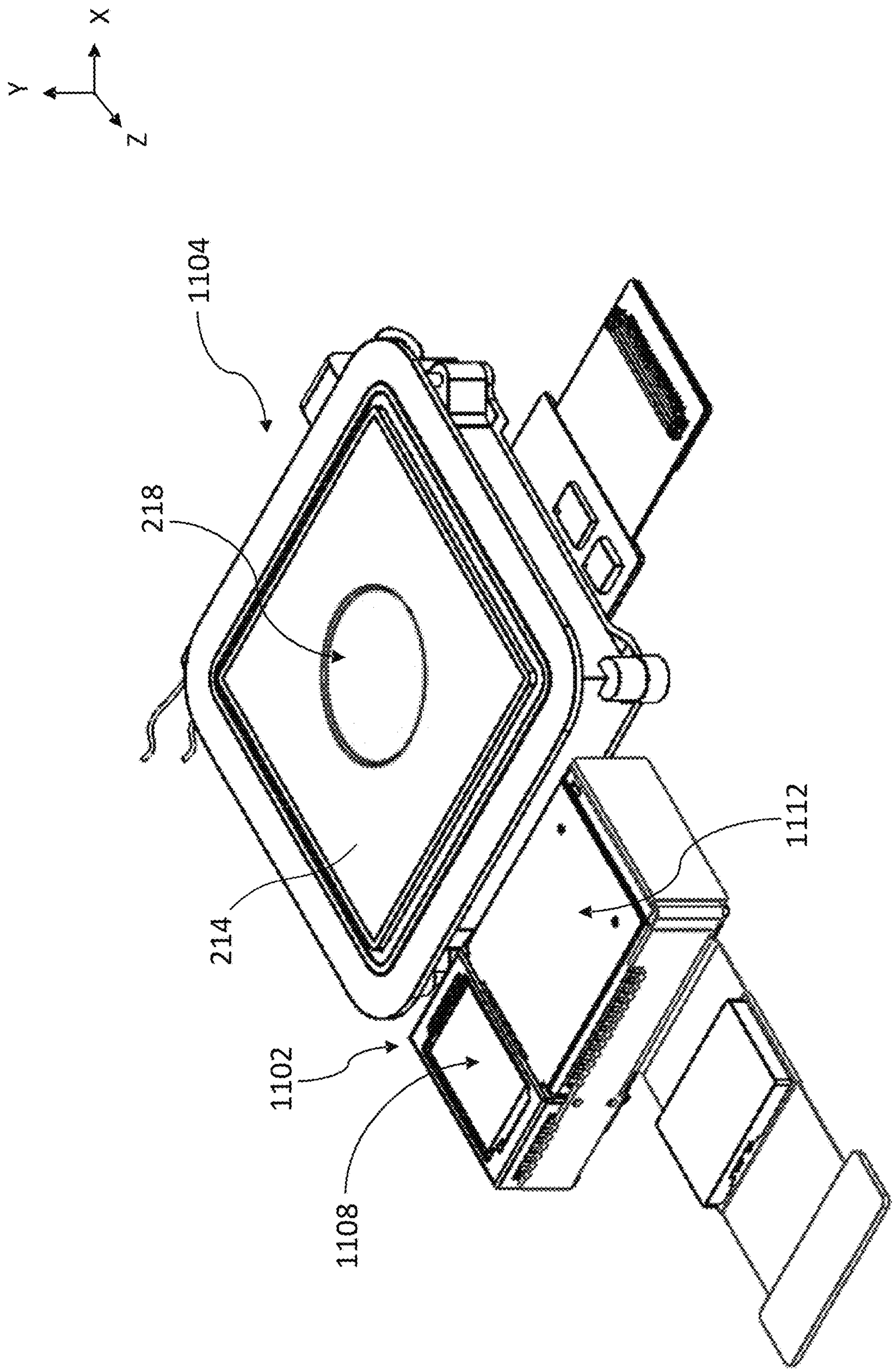


FIG. 11D

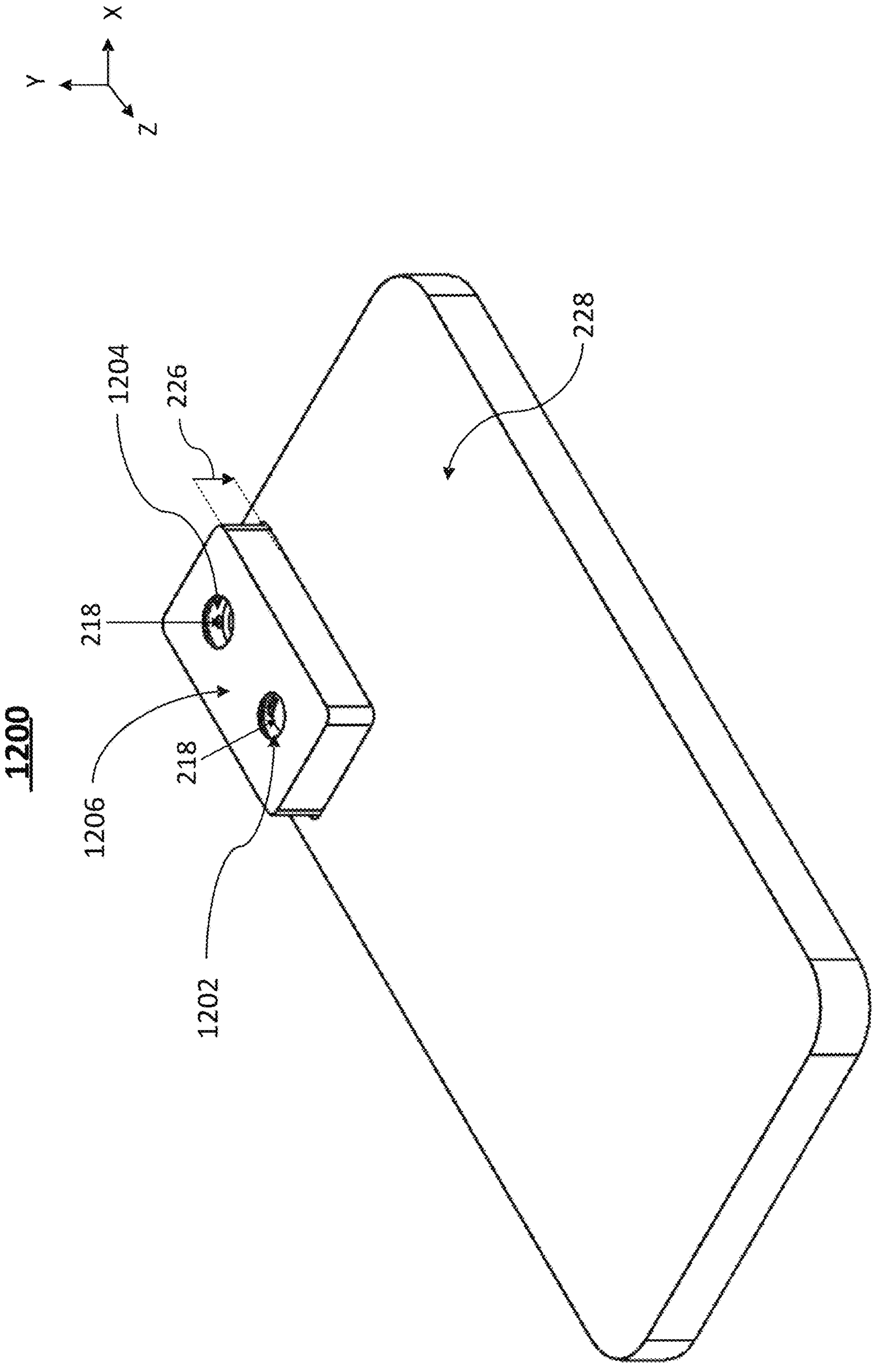


FIG.12A

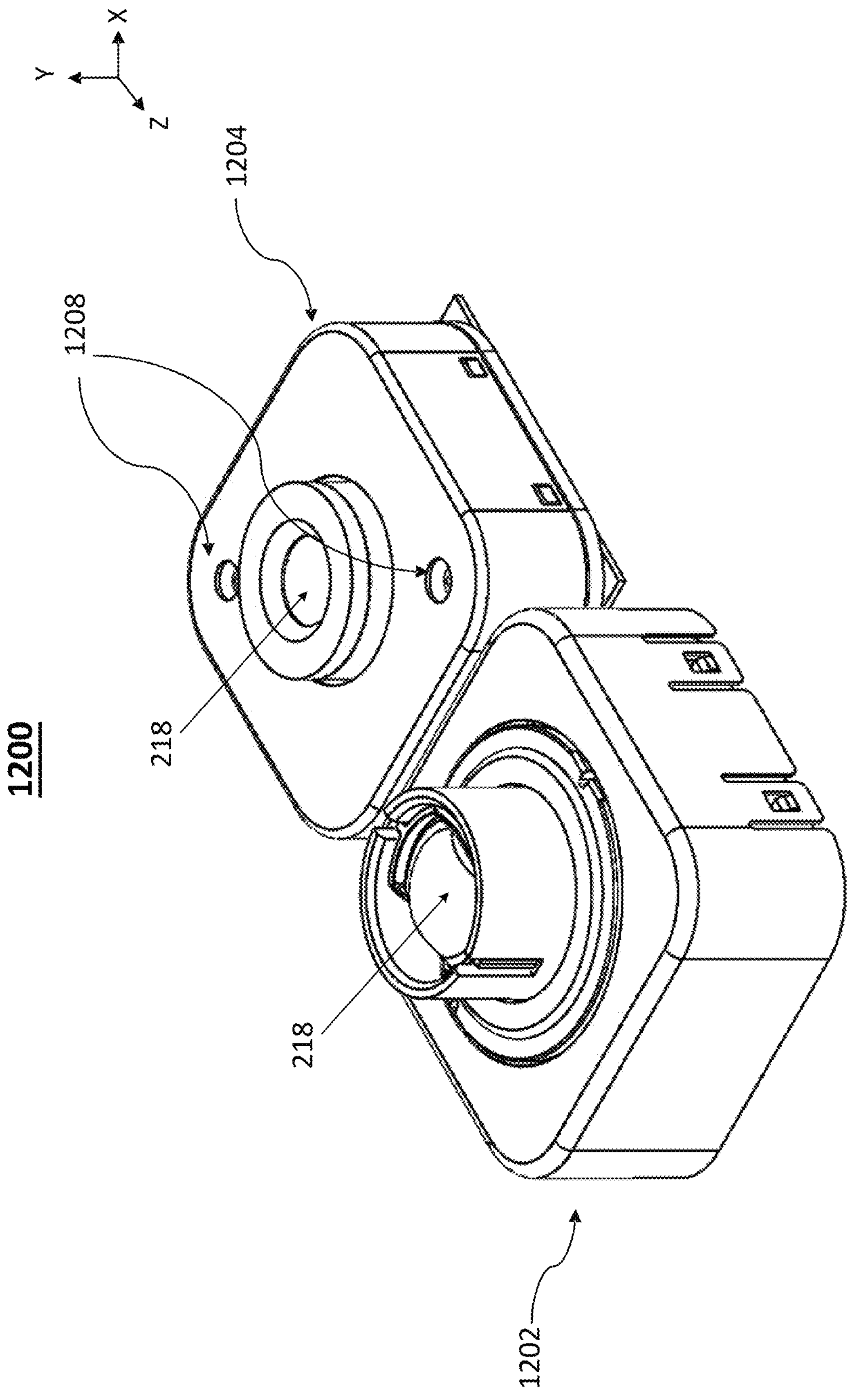


FIG.12B

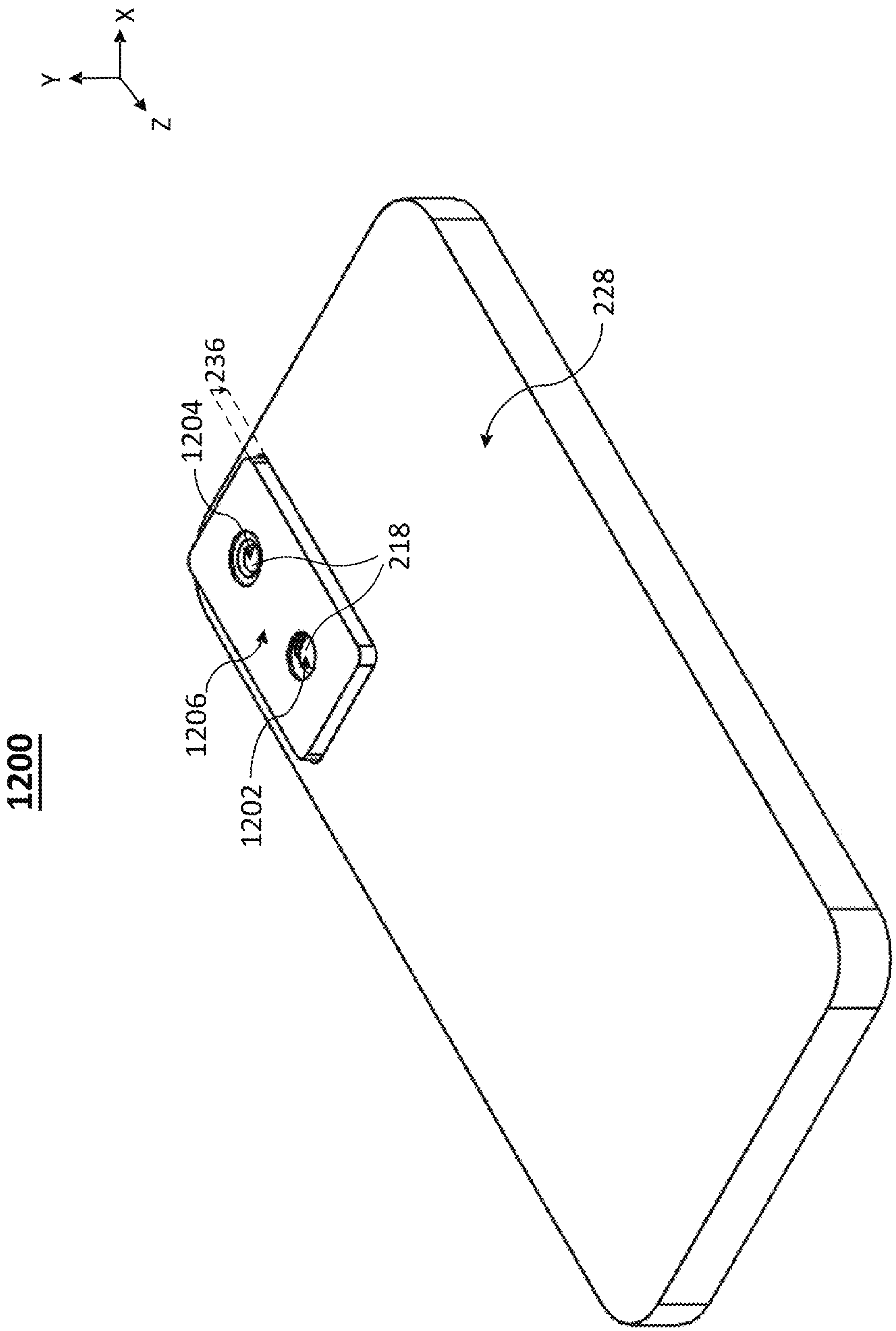


FIG.12C

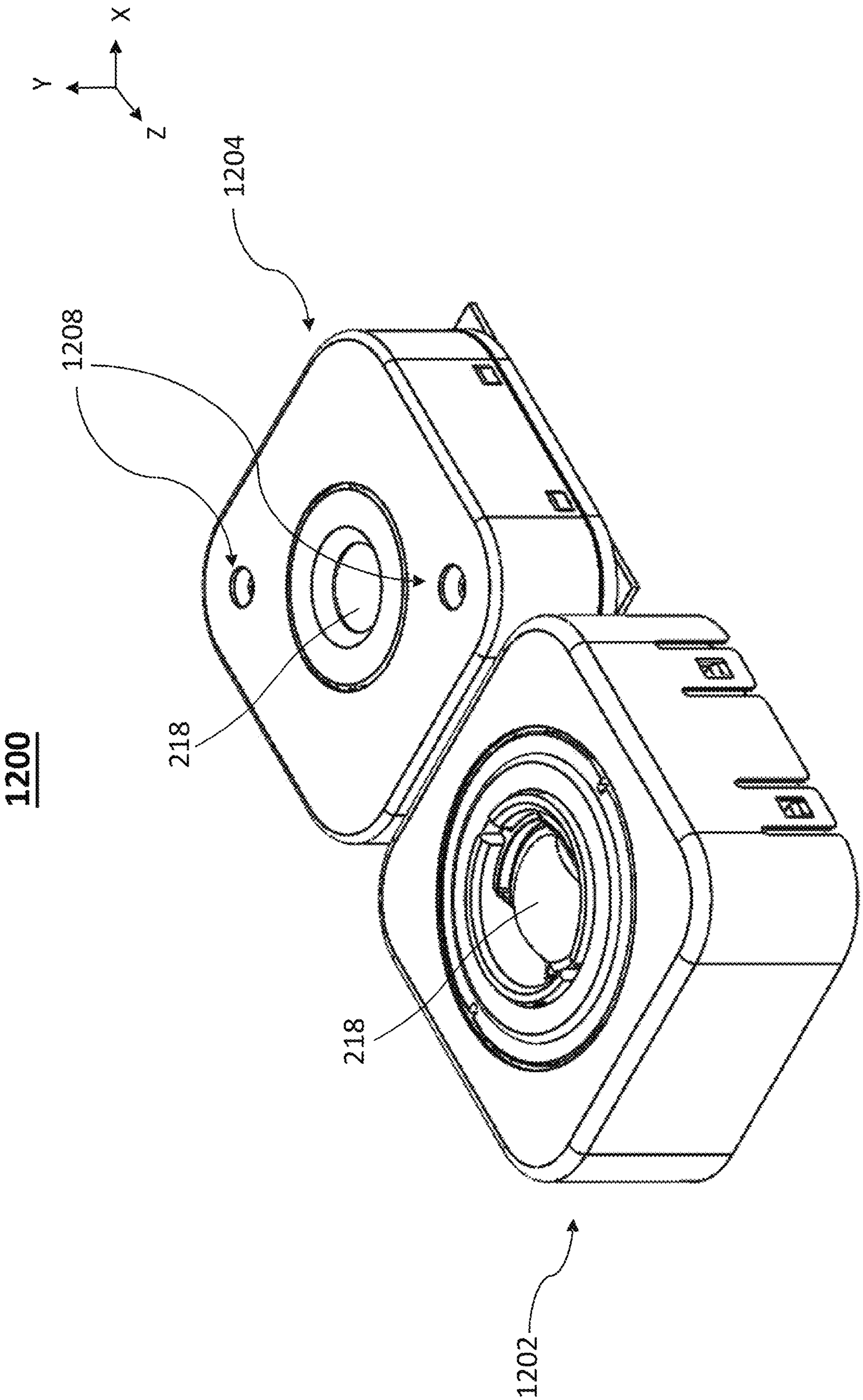


FIG.12D

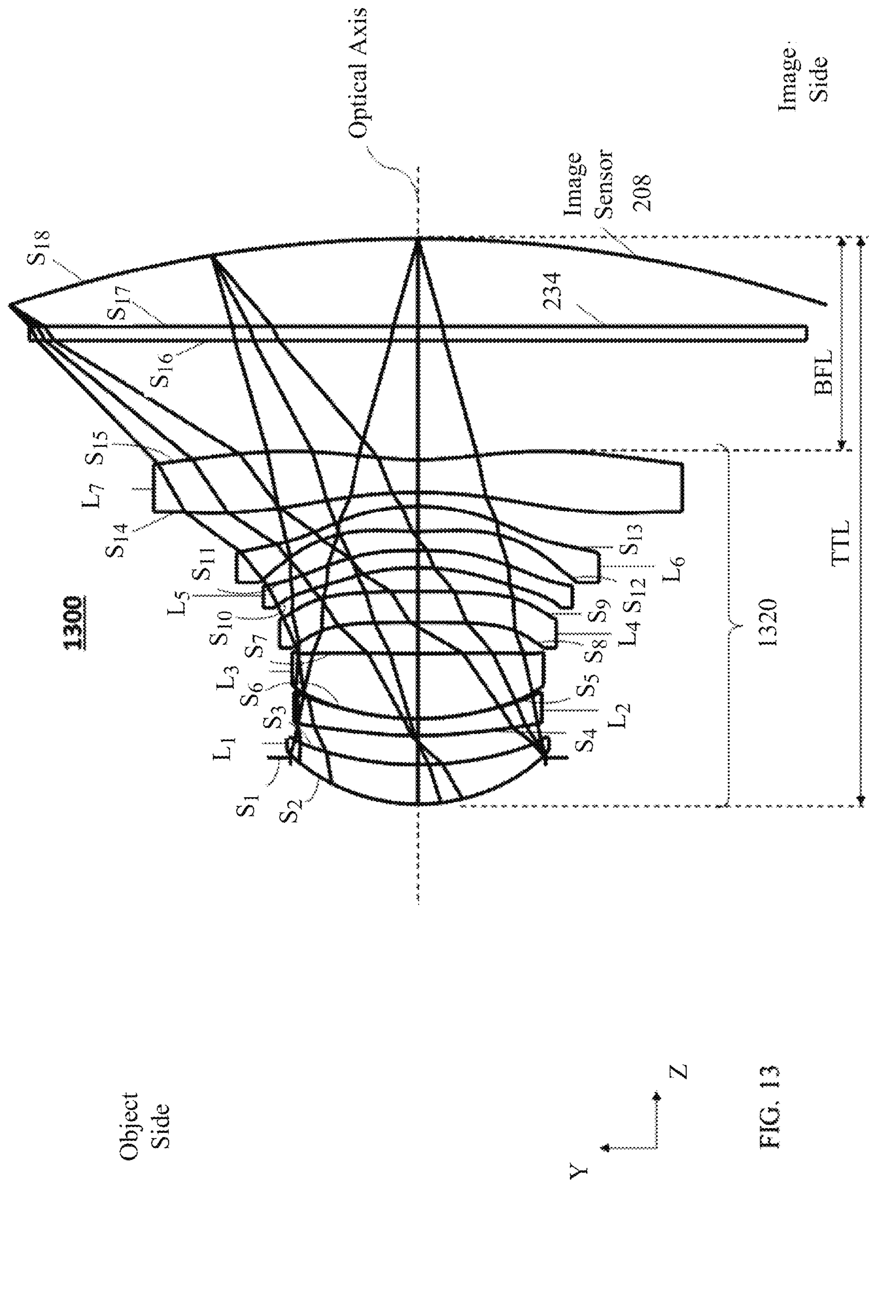

1300
Optical Axis
Image Side
Image Sensor 208
234
S_{18}
S_{17}
S_{16}
S_{15}
L_7
S_{14}
S_{11}
L_5
S_{10}
S_7
L_3
S_6
S_3
L_1
S_1
S_2
S_{13}
L_6
S_{12}
S_9
L_4
S_8
S_5
L_2
S_4
BFL
TTL
1320
Object Side
Z
Y

FIG. 13

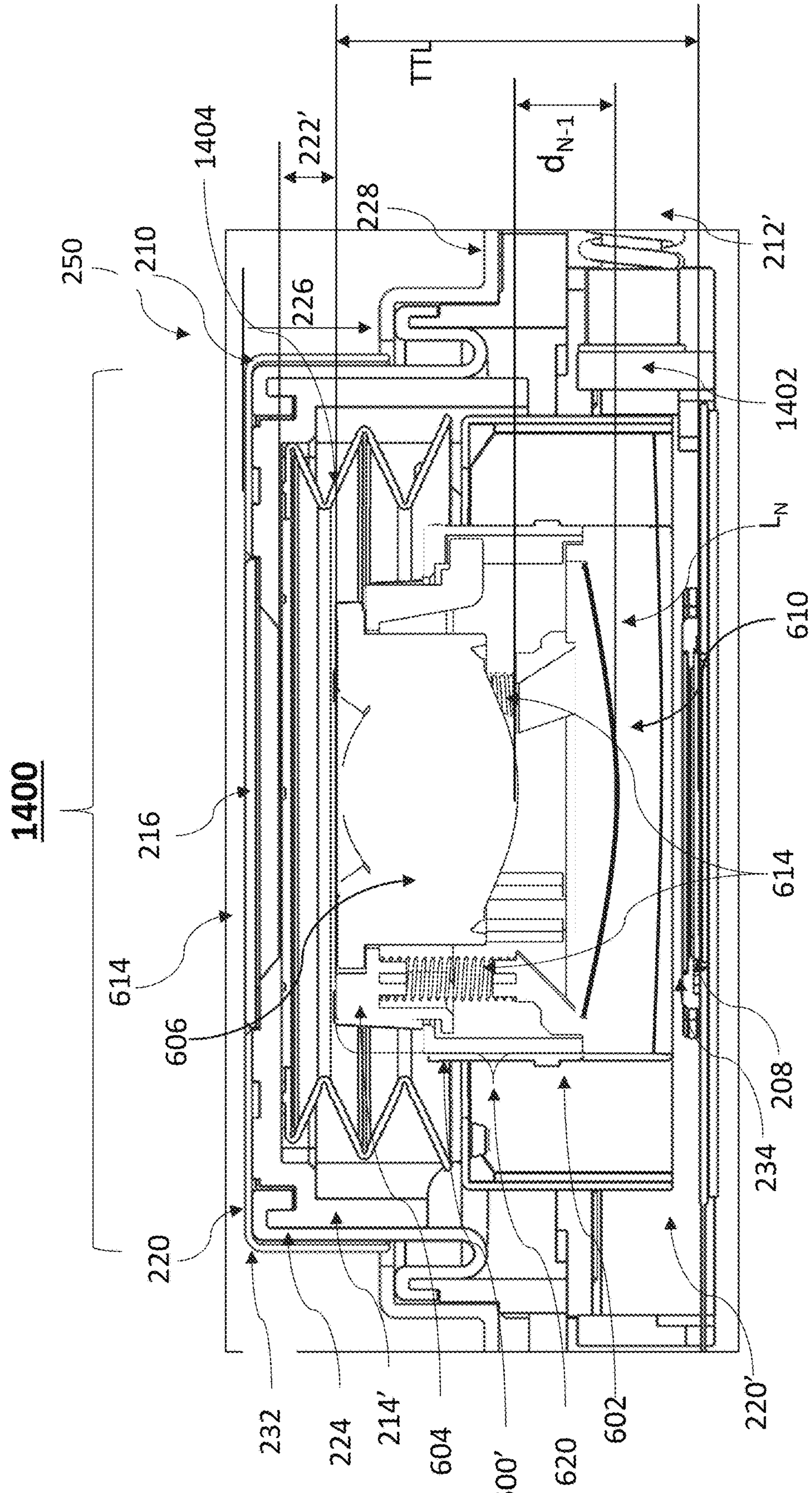

1400
250
210
1404
222'
228
TTL
d_{N-1}
212'
226
1402
L_N
610
614
216
614
606
614
208
234
220
232
224
214'
604
600'
620
602
220'

FIG. 14A

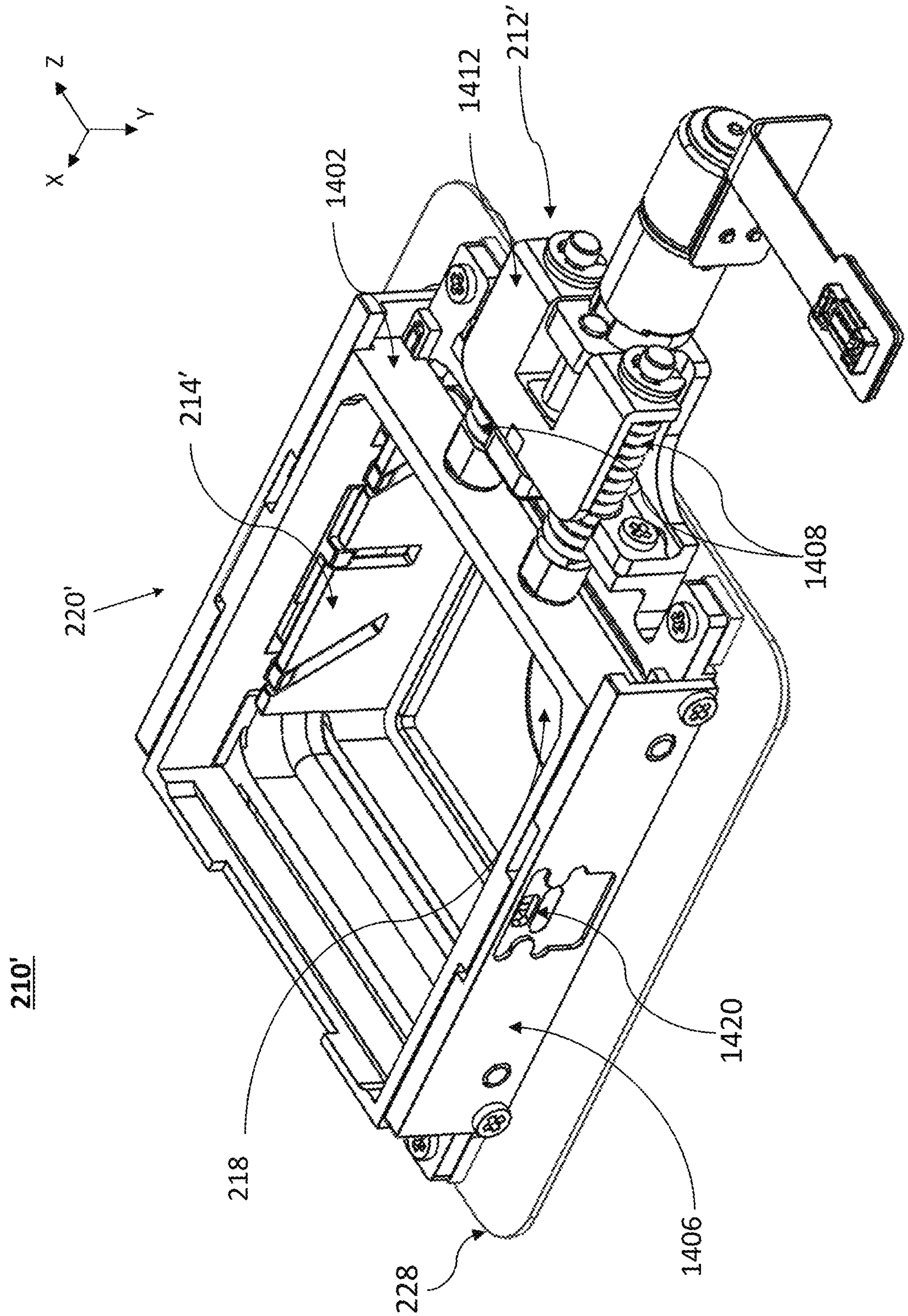
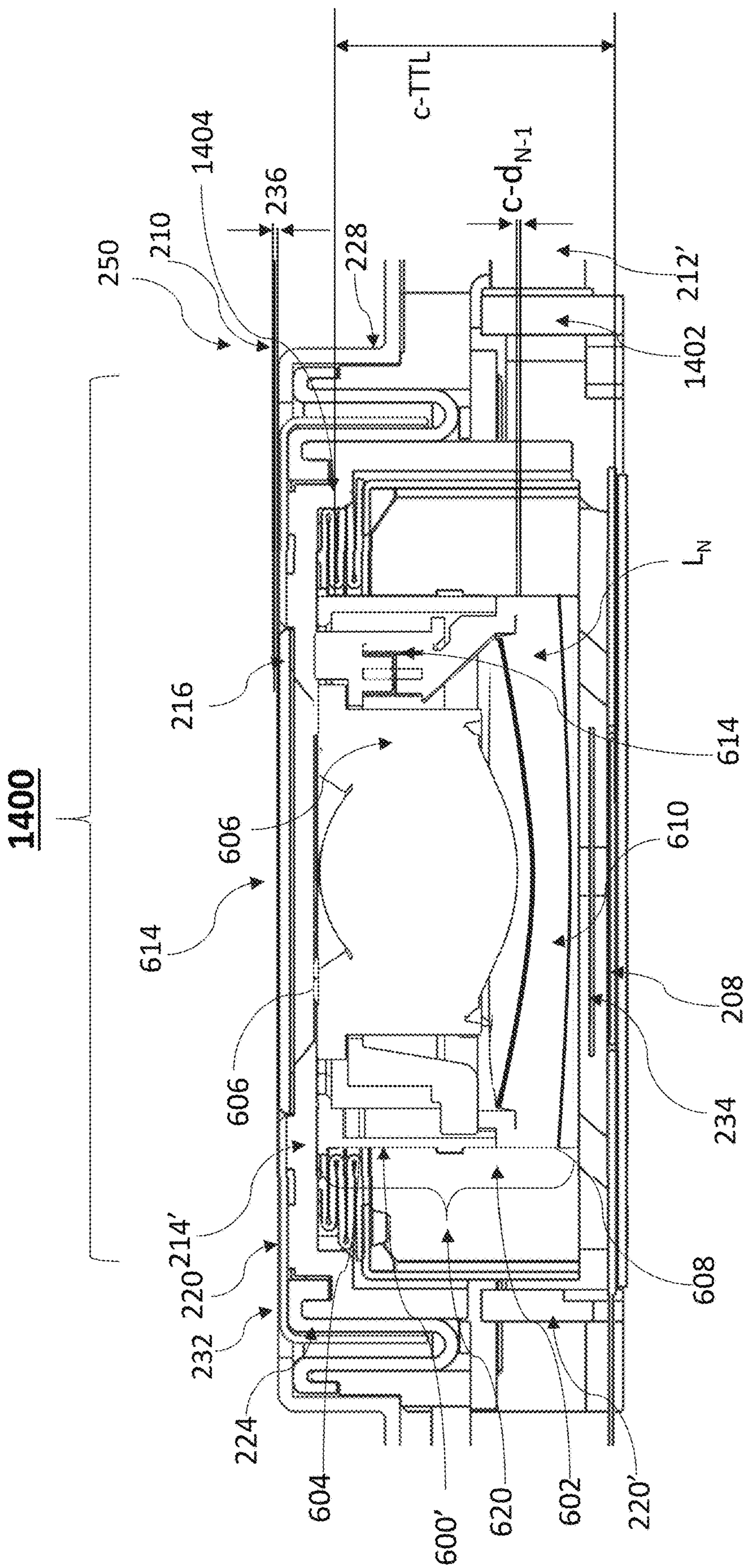


FIG. 14B



1400
c-TTL
c-d_{N-1}
1404
236
210
250
228
212'
1402
L_N
216
614
606
610
208
234
608
214'
220
232
224
604
600'
620
602
220'

FIG. 14C

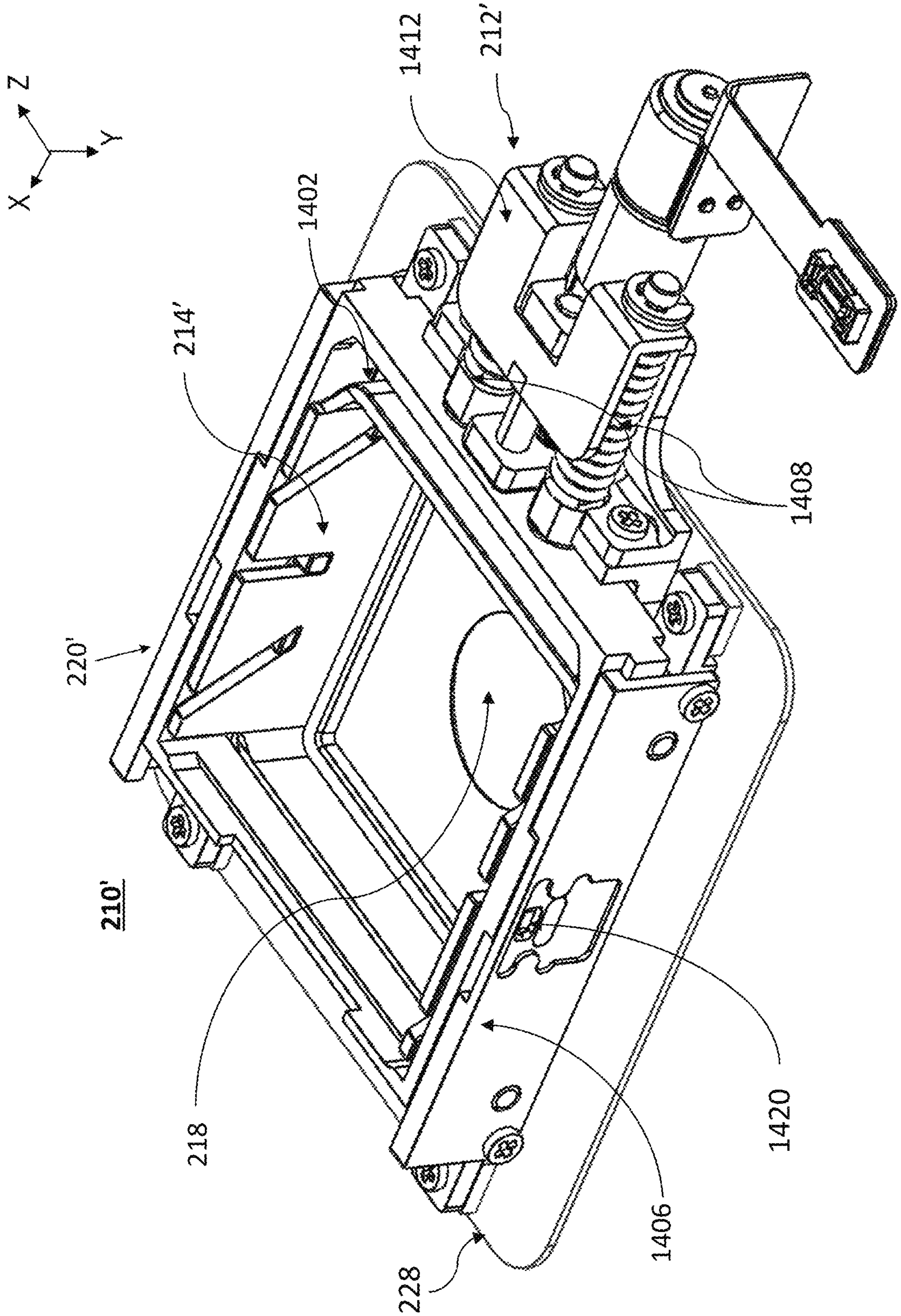


FIG. 14D

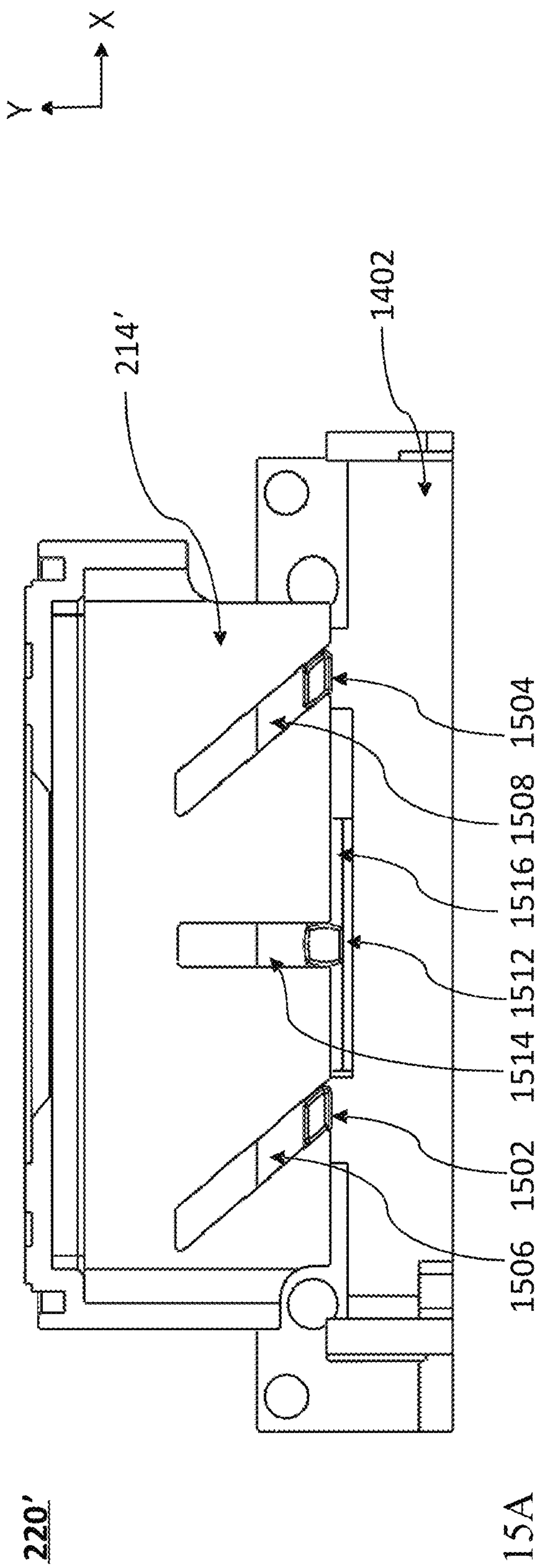


FIG.15A

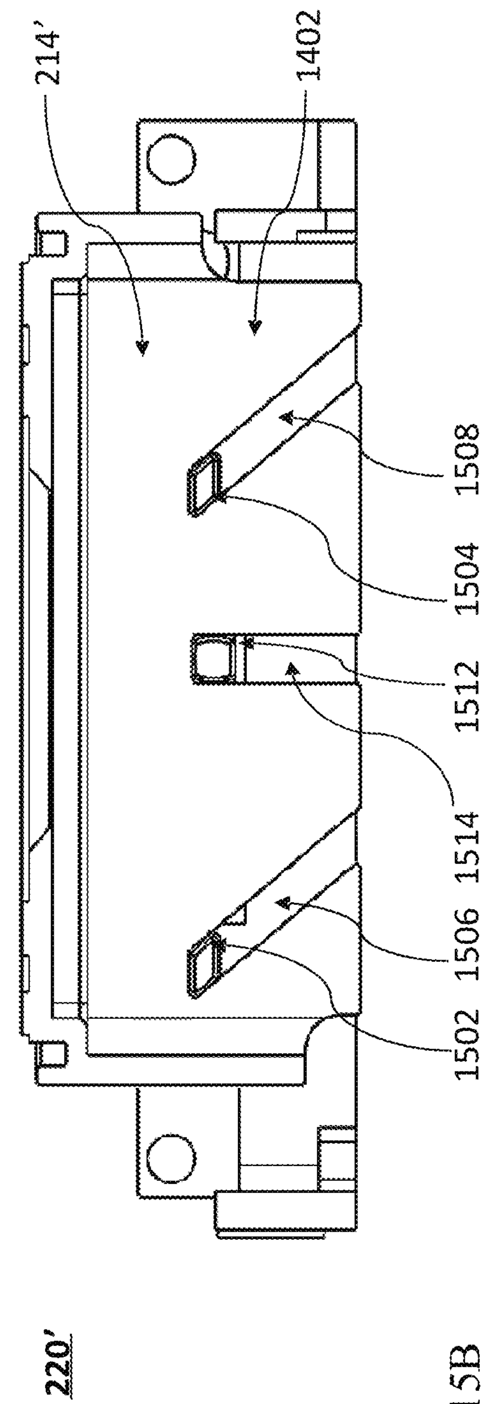


FIG.15B

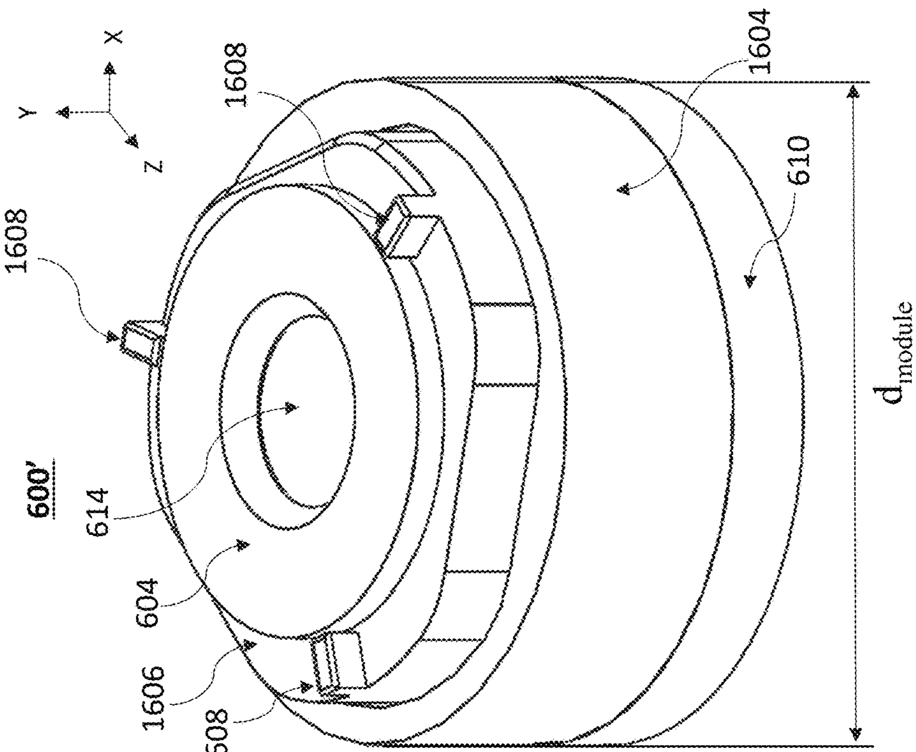


FIG. 16B

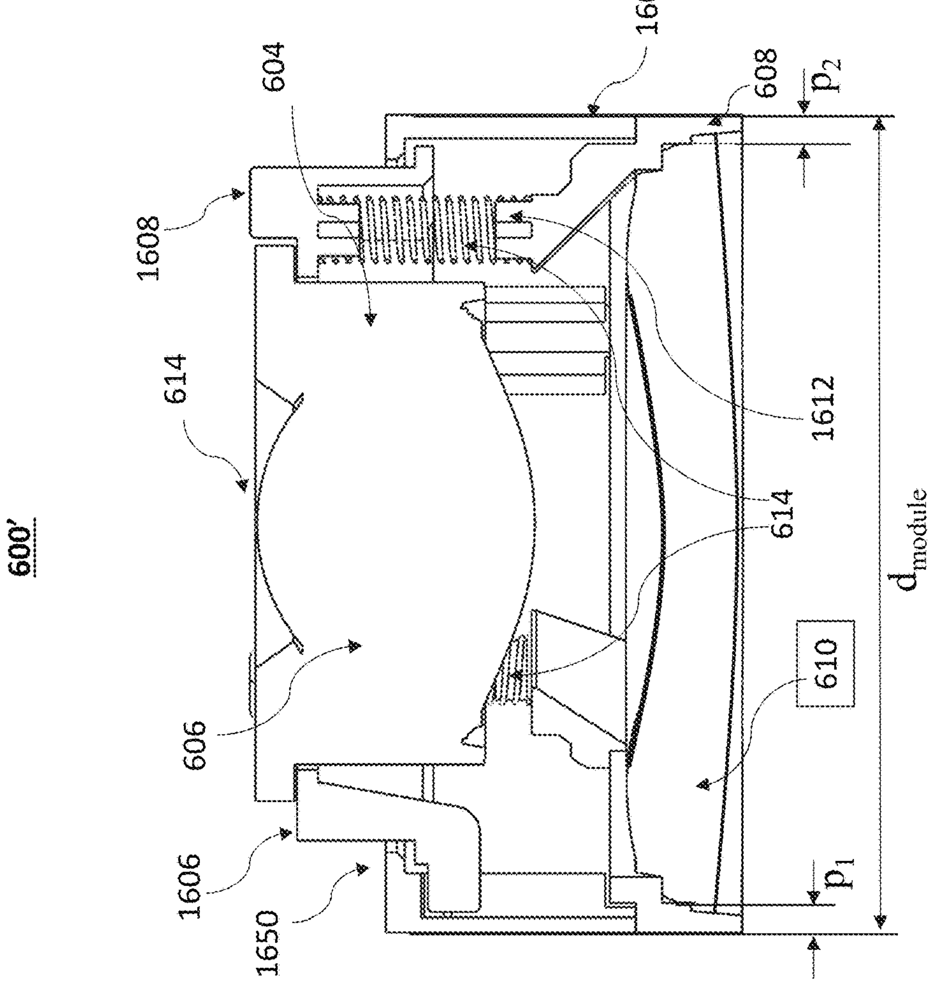


FIG. 16A

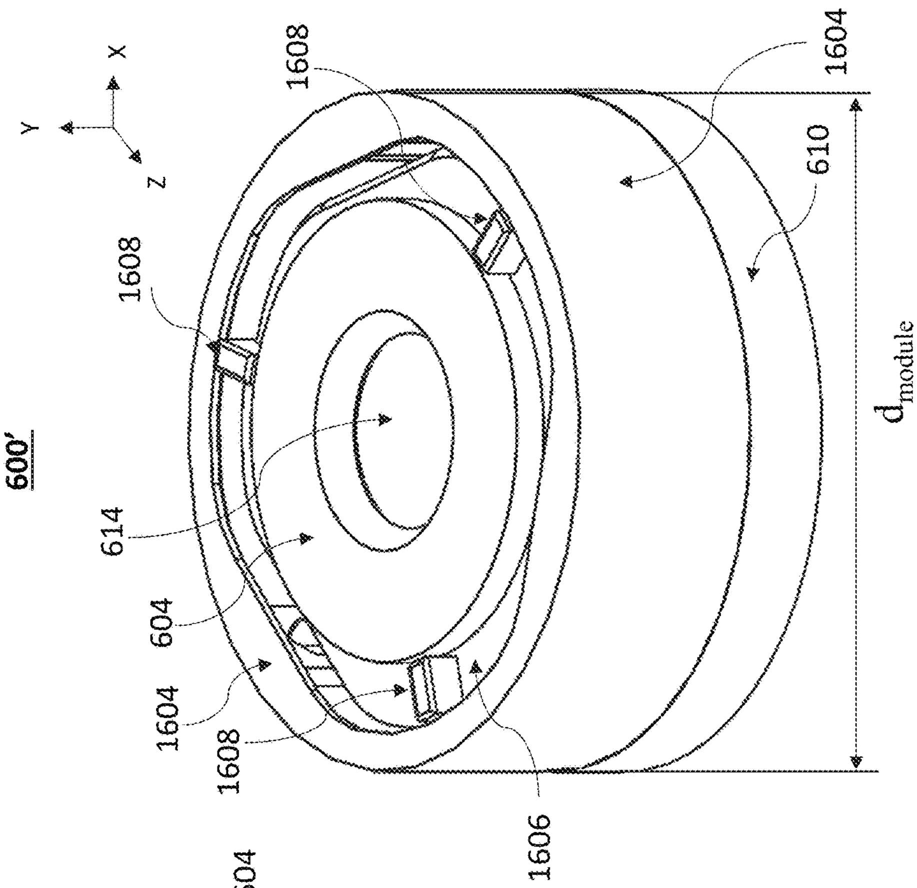


FIG. 17B

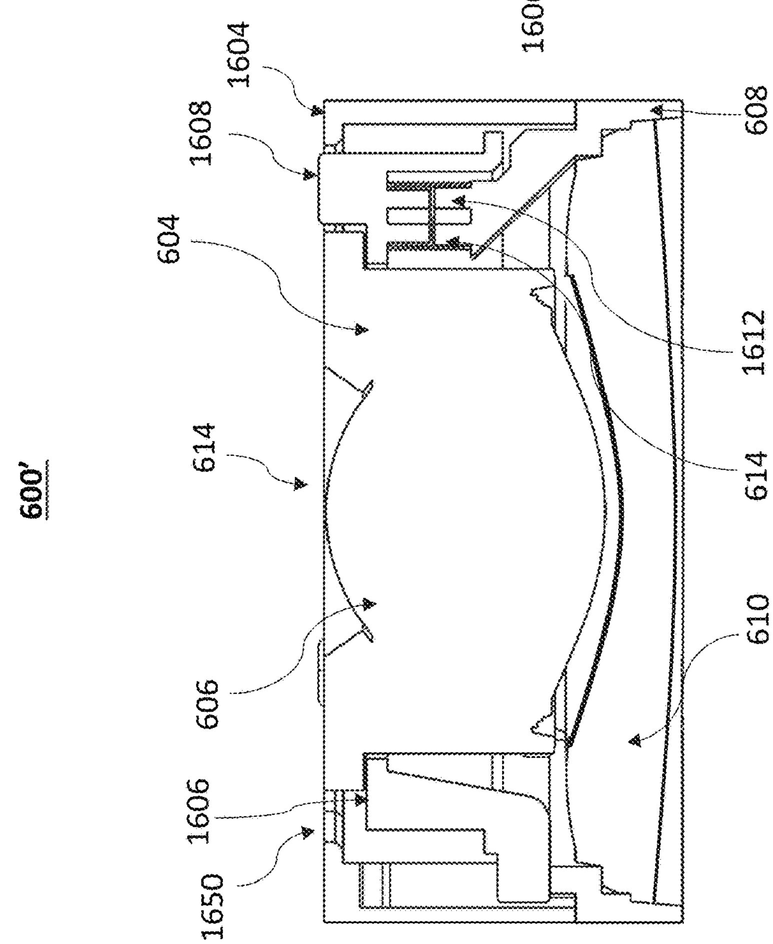


FIG. 17A

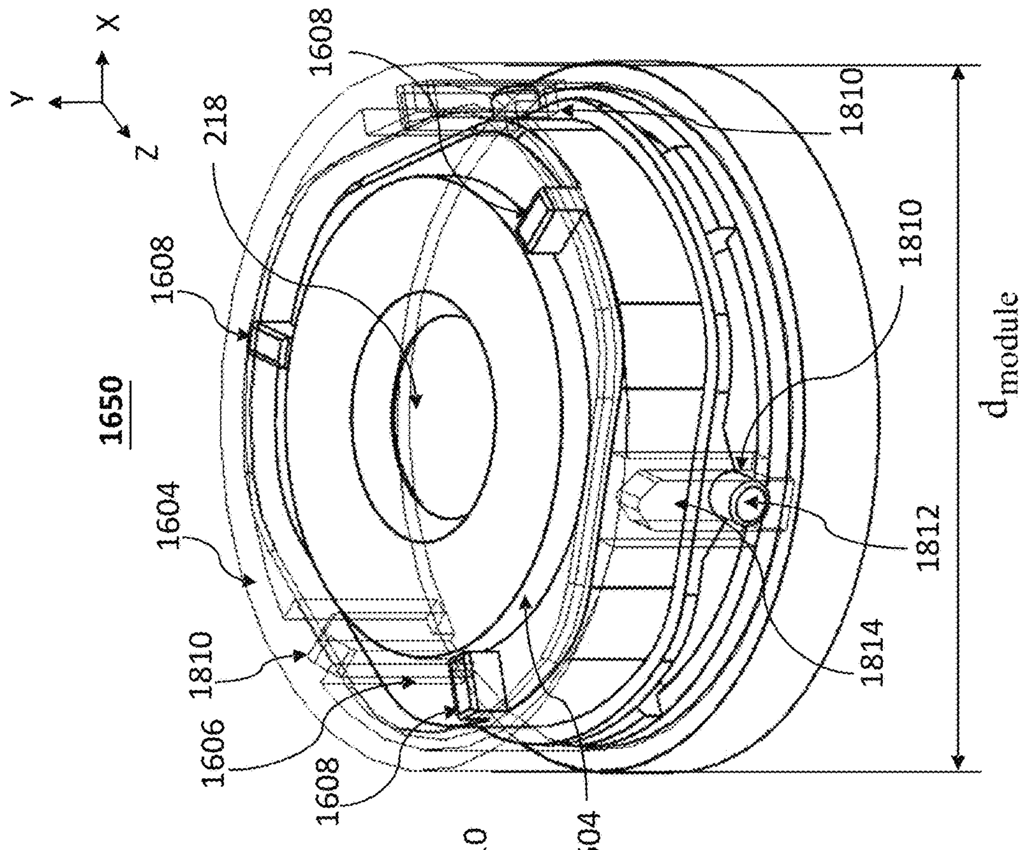


FIG.18B

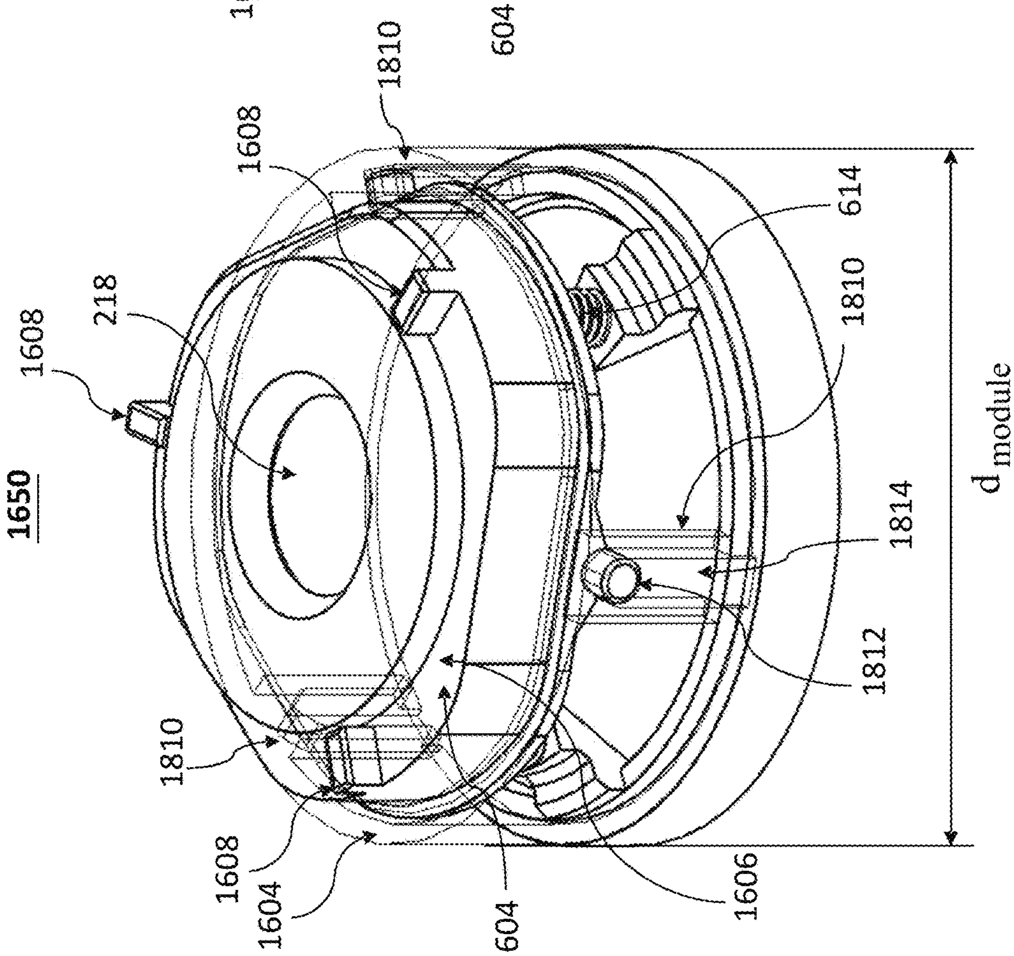


FIG.18A

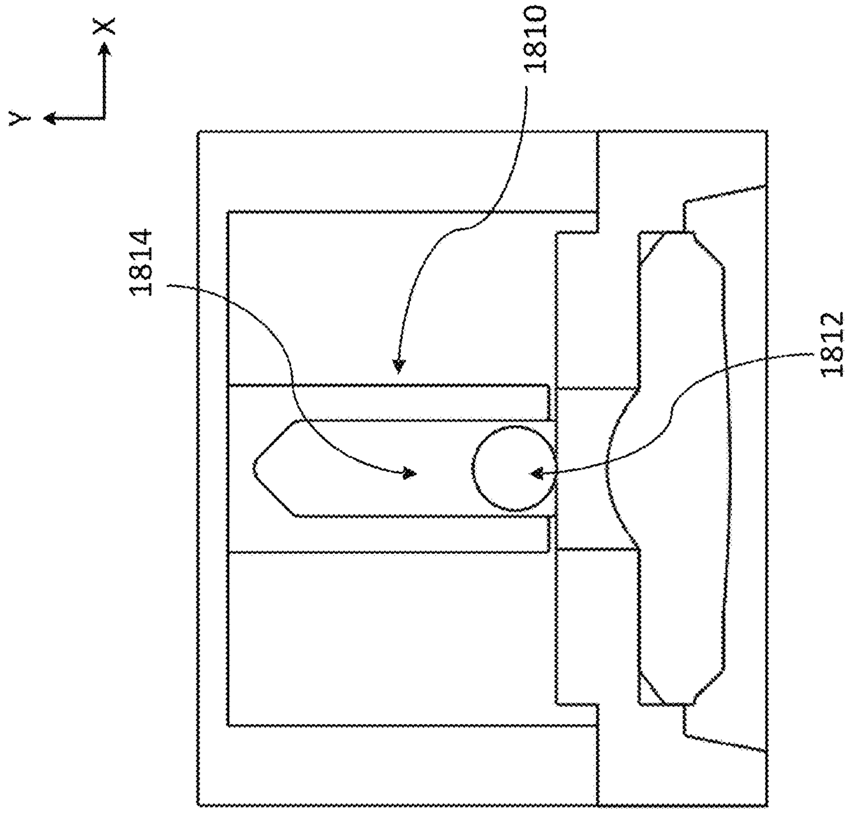


FIG.18D

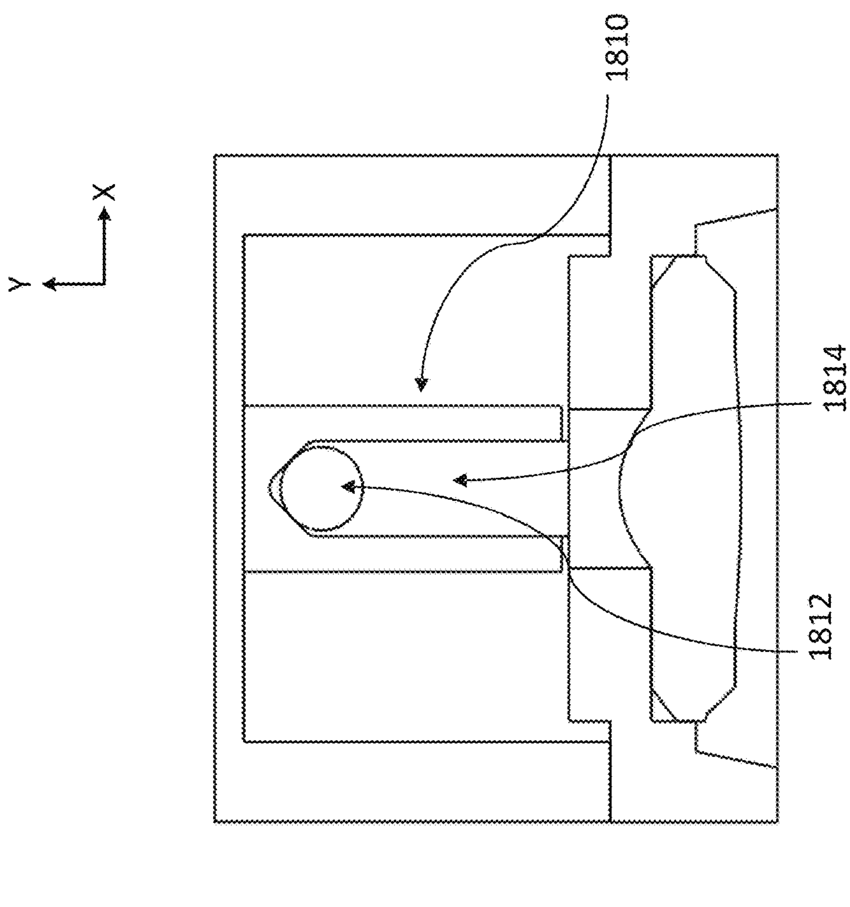


FIG.18C

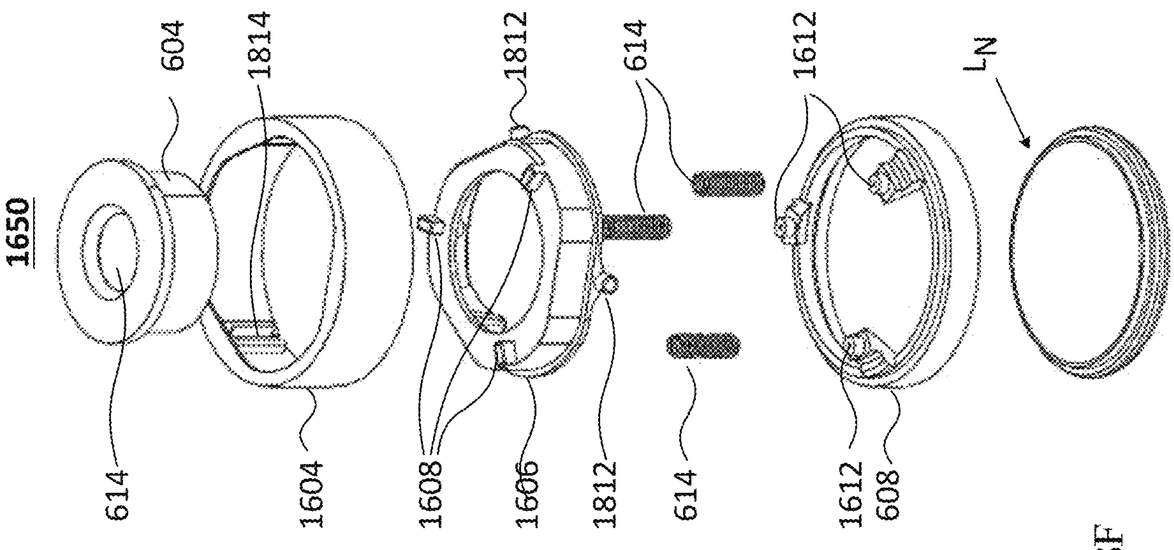


FIG. 18F

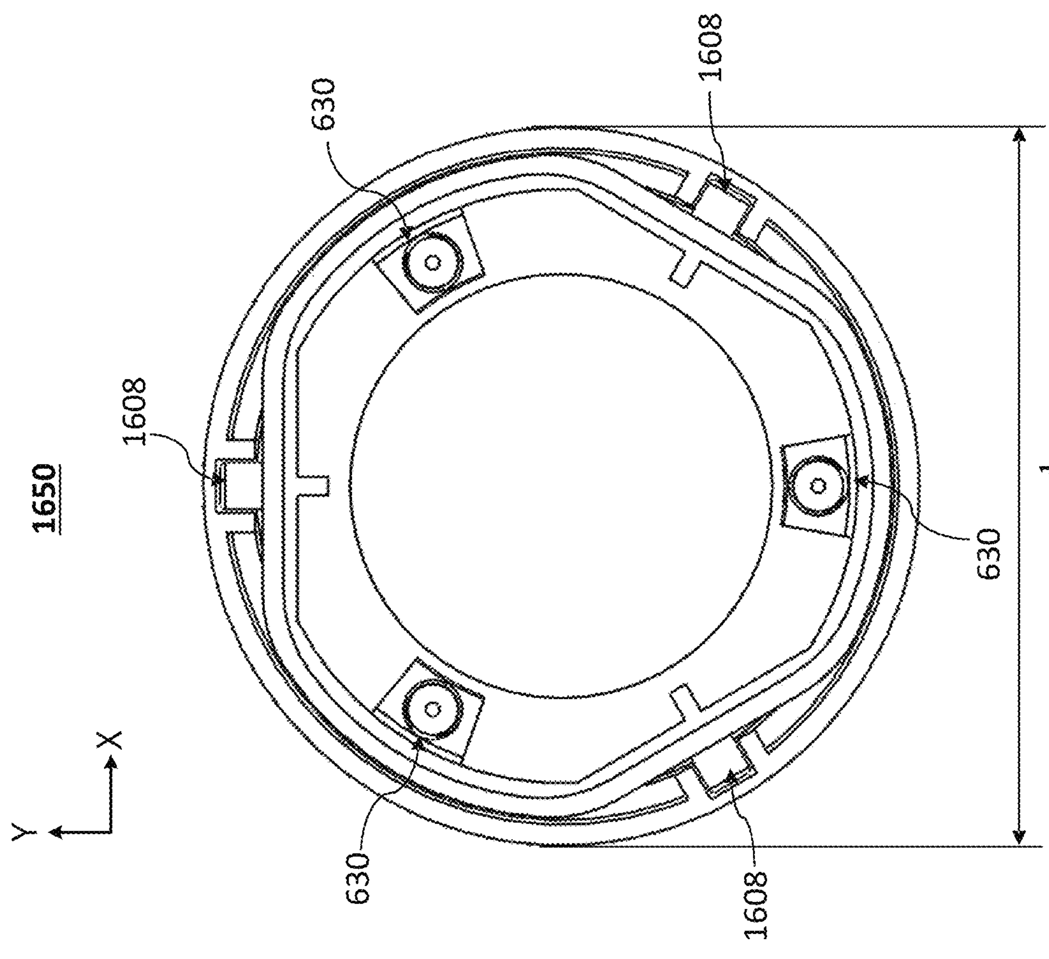


FIG. 18E

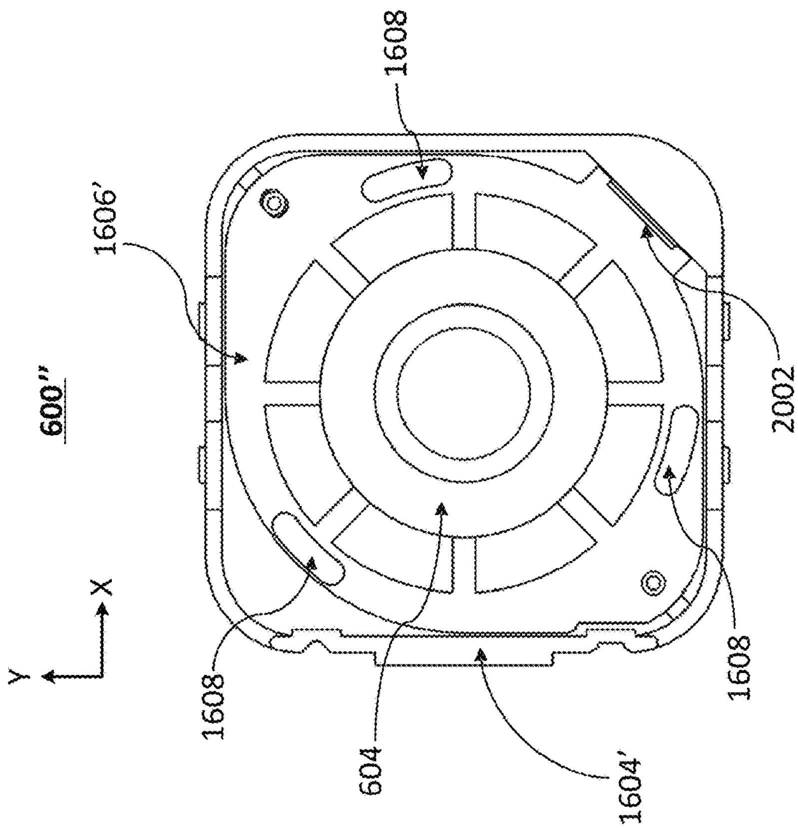


FIG.19B

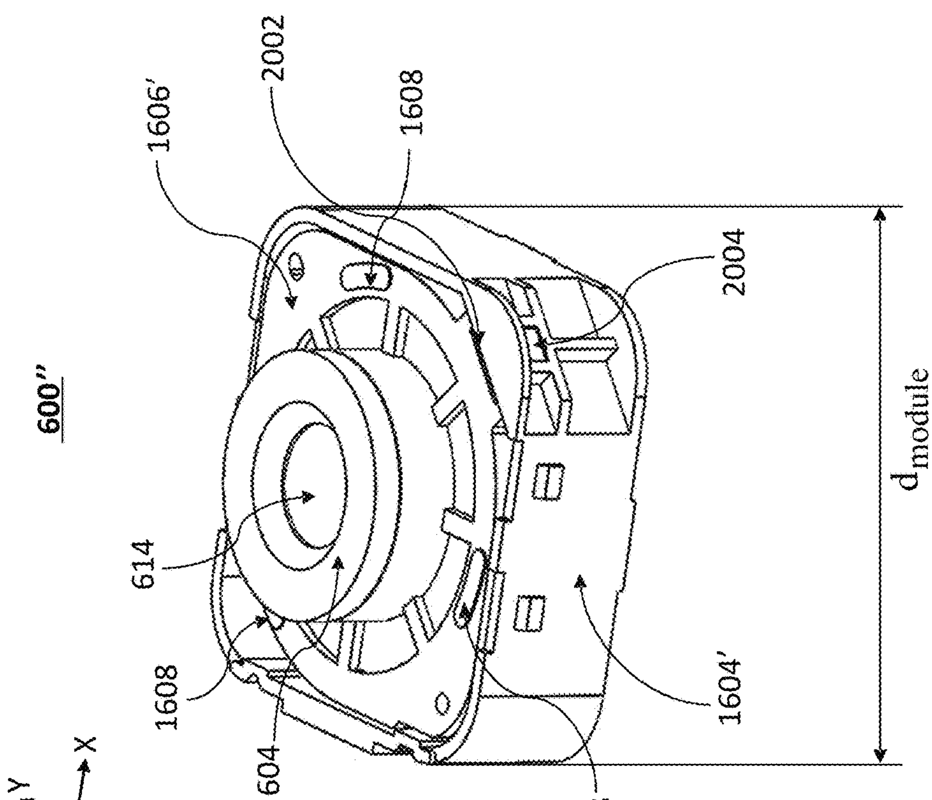


FIG.19A

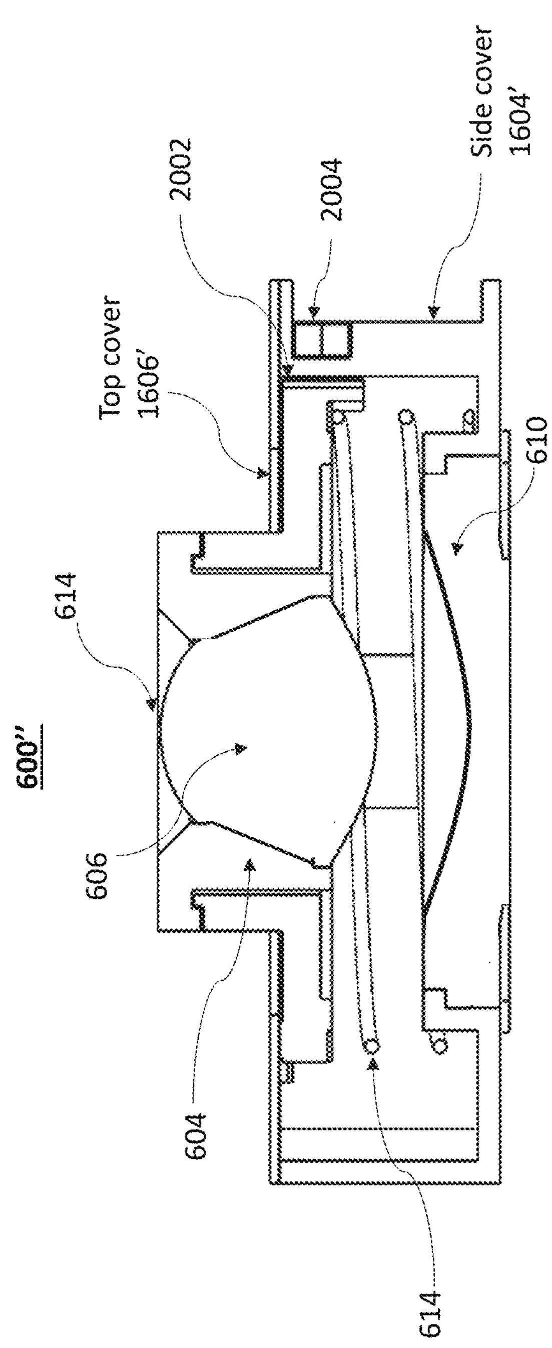


FIG. 19C

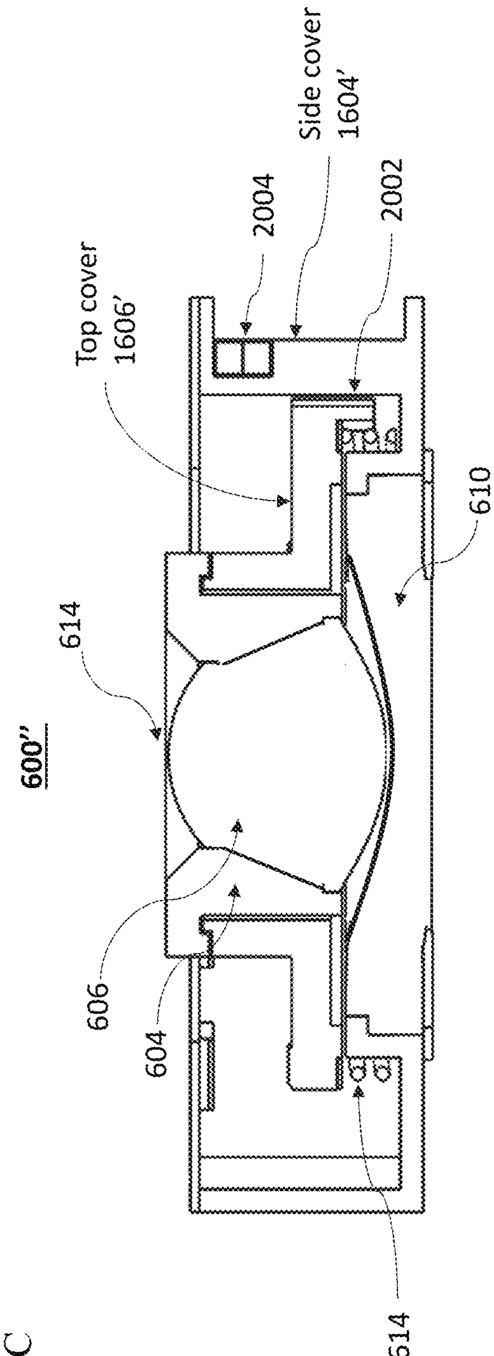


FIG. 19D

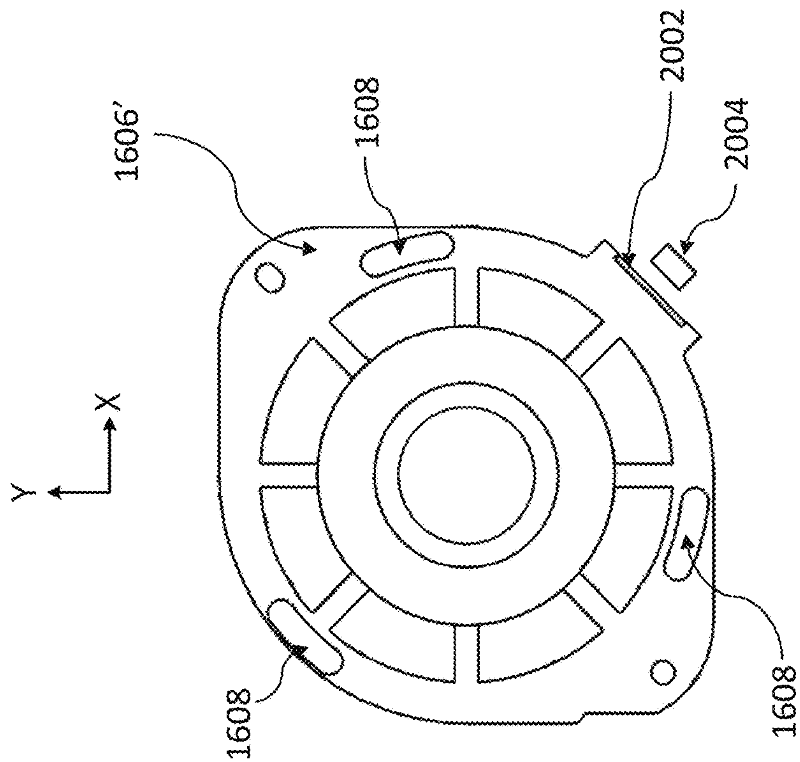


FIG.19F

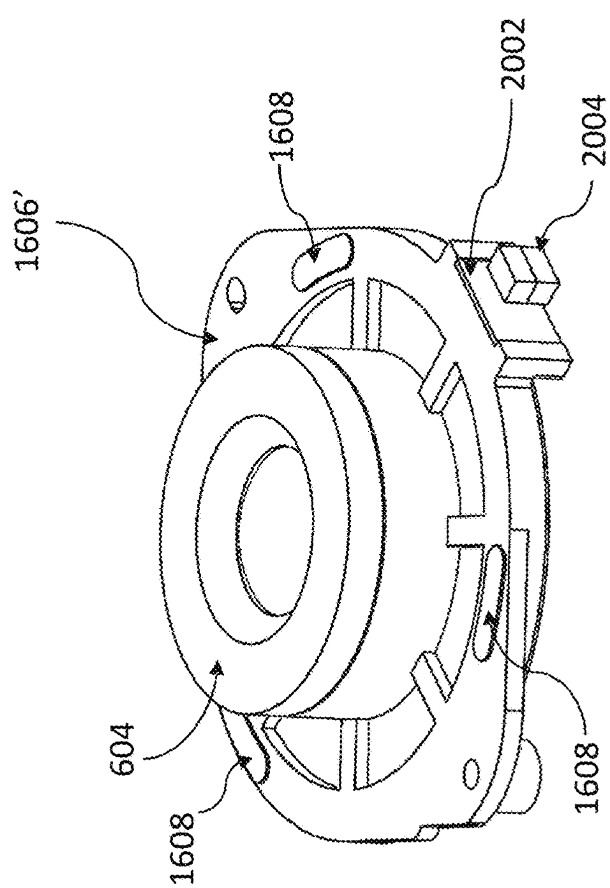


FIG.19E

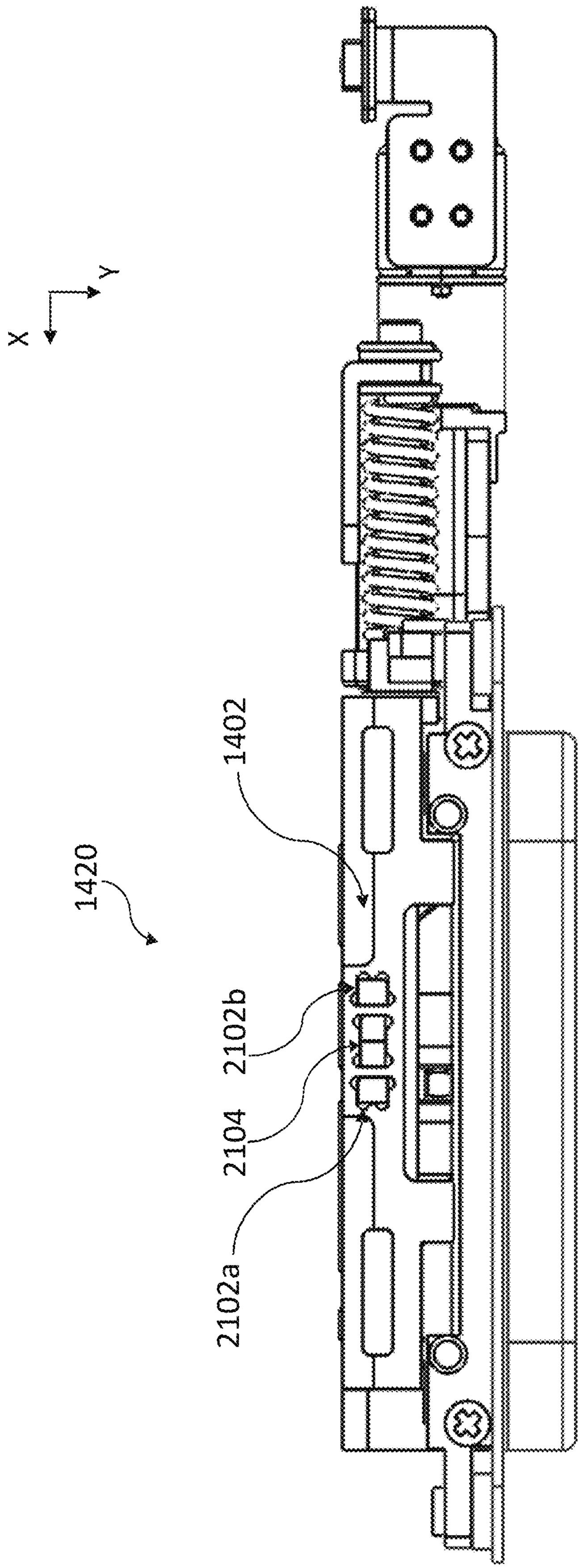


FIG.20A

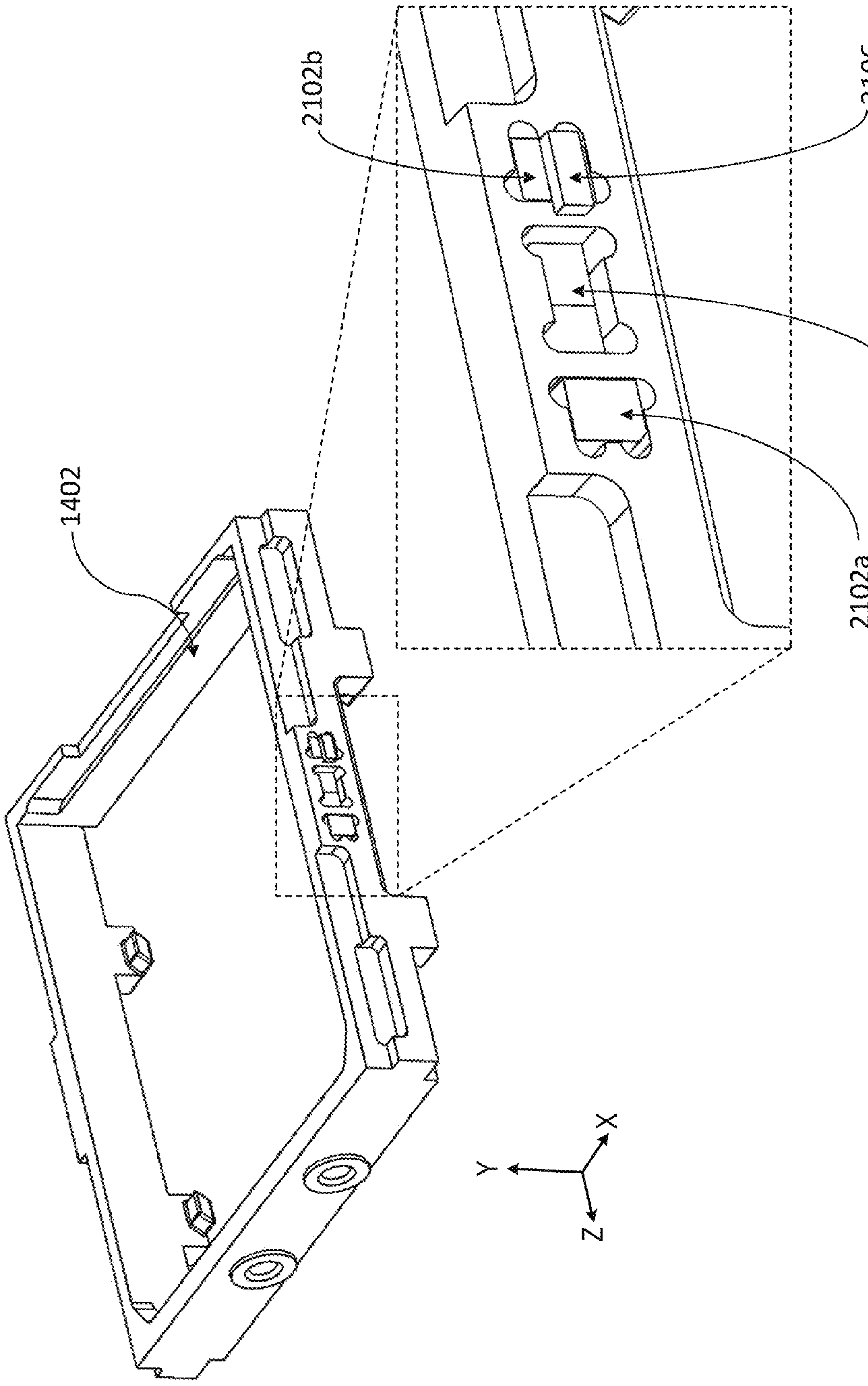


FIG. 20B

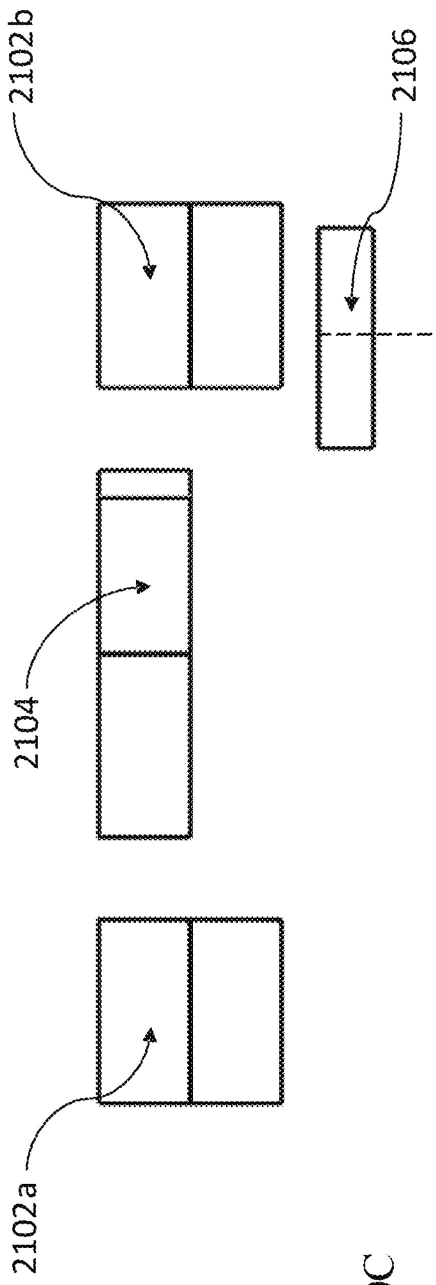


FIG.20C

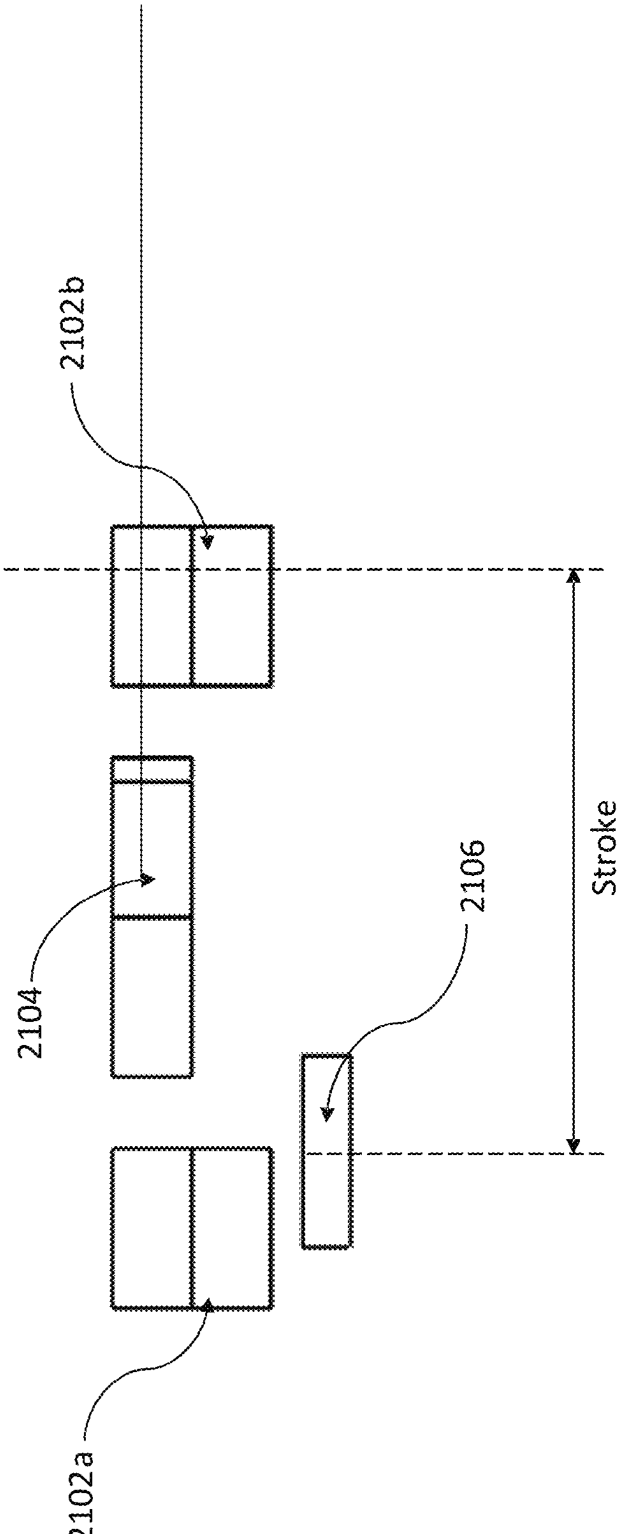
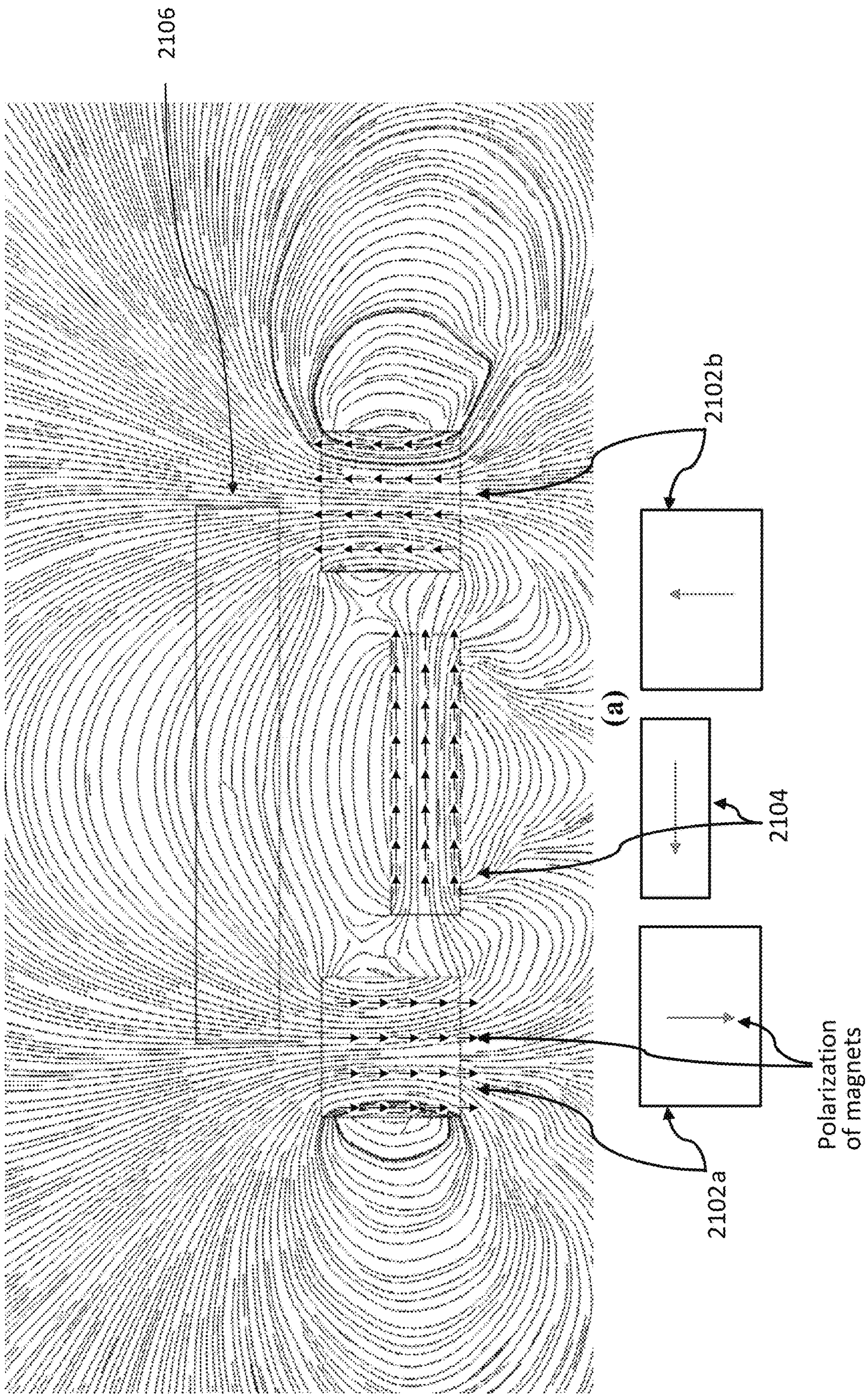


FIG.20D



(b)

FIG.20E

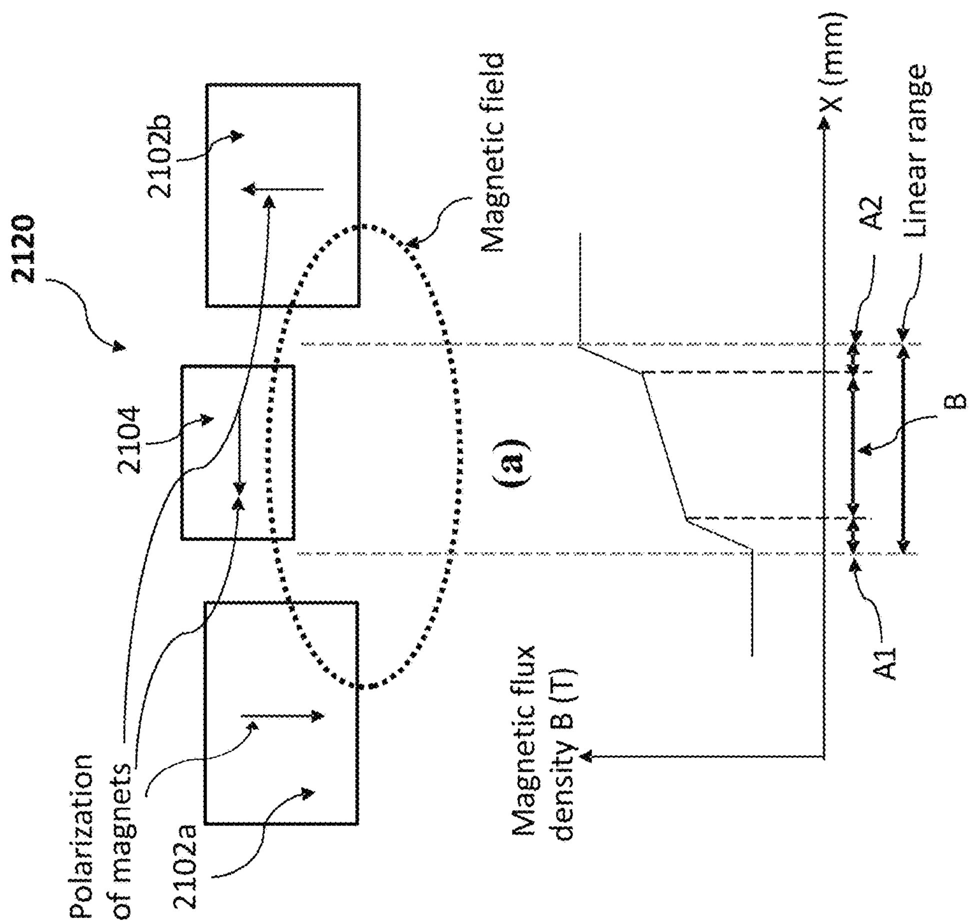


FIG.20G

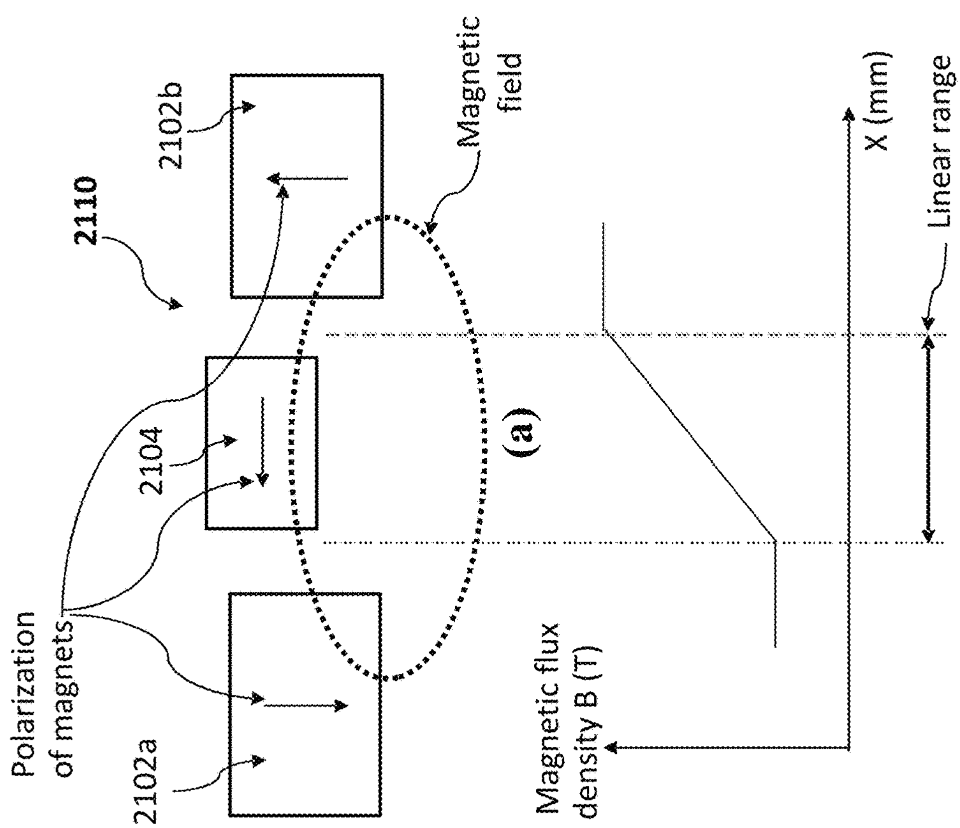


FIG.20F

SLIM POP-OUT CAMERAS AND LENSES FOR SUCH CAMERAS

CROSS REFERENCE TO RELATED APPLICATIONS

This is a continuation of U.S. patent application Ser. No. 18/768,244 filed Jul. 10, 2024 (now allowed), which was a continuation of U.S. patent application Ser. No. 17/567,417 filed Jan. 3, 2022 (now U.S. Pat. No. 12,066,747), which was a continuation of U.S. patent application Ser. No. 17/460,231 filed Aug. 29, 2021 (now U.S. Pat. No. 12,072, 609), which was a continuation in part (CIP) of U.S. patent application Ser. No. 17/291,475 filed May 5, 2021, which was a 371 from international application PCT/IB2020/058697 filed Sep. 18, 2020, and is related to and claims priority from U.S. Provisional Patent Applications No. 62/904,913 filed Sep. 24, 2019, 63/026,317 filed May 18, 2020 and 63/037,836 filed Jun. 11, 2020, all of which are incorporated herein by reference in their entirety.

FIELD

The present disclosure relates in general to digital cameras, and more particularly, to digital cameras with pop-out mechanisms and lenses.

BACKGROUND

Compact multi-aperture digital cameras (also referred to as "multi-lens cameras" or "multi-cameras") and in particular dual-aperture (or "dual-camera") and triple-aperture (or "triple-camera") digital cameras are known. Miniaturization technologies allow incorporation of such cameras in compact portable electronic devices such as tablets and mobile phones (the latter referred to hereinafter generically as "smartphones"), where they provide advanced imaging capabilities such as zoom, see e.g. co-owned PCT patent application No. PCT/IB2063/060356, which is incorporated herein by reference in its entirety. A typical triple-camera system (exemplarily including an ultra-wide-angle (or "Ultra-Wide" or "UW") camera, wide-angle (or "Wide") camera and a telephoto (or "Tele") camera.

A challenge with dual-aperture zoom cameras relates to camera height and size of image sensor ("Sensor Diagonal" or S_D). There is a large difference in the height (and also of the total track length or "TTL") of the Tele and Wide cameras. FIG. **1**A illustrates schematically the definition of various entities such as TTL, effective focal length (EFL) and back focal length (BFL). The TTL is defined as the maximal distance between the object-side surface of a first lens element and a camera image sensor plane. The BFL is defined as the minimal distance between the image-side surface of a last lens element and a camera image sensor plane. In the following, "W" and "T" subscripts refer respectively to Wide and Tele cameras. The EFL has a meaning well known in the art. In most miniature lenses, the TTL is larger than the EFL, as in FIG. **1**A.

FIG. **1**B shows an exemplary camera system having a lens with a field of view (FOV), an EFL and an image sensor with a sensor width S. For fixed width/height ratios of a (normally rectangular) image sensor, the sensor diagonal is proportional to the sensor width and height. The horizontal FOV relates to EFL and sensor width as follows:

$$\tan\left(\frac{FOV}{2}\right) = \frac{s}{2}/EFL$$

This shows that for realizing a camera with a larger image sensor width (i.e. larger sensor diagonal) but same FOV, a larger EFL is required.

In mobile devices, typical Wide cameras have 35 mm equivalent focal lengths ("35 eqFL"). ranging from 22 mm to 28 mm. Image sensors embedded in mobile cameras are smaller than full frame sensors and actual focal lengths in Wide cameras range from 3.2 mm to 7 mm, depending on the sensor size and FOV. In most lenses designed for such cameras, the TTL/EFL ratio is larger than 1.0 and is typically between 1.0 and 1.3. Another characteristic of these lenses is that their TTL-to-sensor diagonal ratio TTL/SD is typically in the range of 0.6 to 0.7. Embedding larger sensors in Wide cameras is desirable, but require larger EFL for maintaining the same FOV, resulting in larger TTL, which is undesirable.

Many mobile devices include now both Tele and Wide cameras. The Tele camera enables optical zoom and other computational photography features such as digital Bokeh. Depending on the Wide camera characteristics and permissible module height, the 35 eqFL of mobile device Tele cameras ranges from 45 mm to 100 mm. The TTL of lenses designed for Tele cameras is smaller than the EFL of such lenses, typically satisfying 0.7<TTL/EFL<1.0. Typical Tele EFL values range from 6 mm to 10 mm (without applying 35 mm equivalence conversion) in vertical (non-folded) Tele cameras and from 10 mm to 30 mm in folded Tele cameras. Larger EFL is desirable for enhancing the optical zoom effect but it results in larger TTL, which is undesirable.

In a continuous attempt to improve the obtained image quality, there is a need to incorporate larger image sensors into the Wide and Tele cameras. Larger sensors allow for improved low-light performance and larger number of pixels, hence improving the spatial resolution as well. Other image quality characteristics, such as noise characteristics, dynamic range and color fidelity may also improve as the sensor size increases.

As the Wide camera sensor becomes larger, the required EFL is larger (for the same 35 mm equivalent focal length), the lens TTL increases and the camera module height becomes larger, resulting in a limit on the permissible sensor size when considering the allowed mobile device thickness or other industrial design constraints. In Wide cameras of most mobile devices, the sensor pixel array size full diagonal ranges from about 4.5 mm (typically referred to as ¼" sensor) to 16 mm (typically referred to 1" sensor).

It would be beneficial to have Wide and/or Tele lens designs that support large EFLs for large sensor diagonals (optical zoom) while still having small TTL for slim design. The latter is presented for example in co-owned U.S. provisional patent application No. 62/904,913.

SUMMARY

In various examples, there are provided digital cameras comprising: an optics module comprising a lens assembly that includes N lens elements L_1-L_N starting with L_1 on an object side, wherein N≥4; an image sensor having a sensor diagonal S_D in the range of 5-20 mm; and a pop-out mechanism configured to control at least one air-gap between lens elements or between a lens element and the image sensor to bring the camera to an operative pop-out state and to a collapsed state, wherein the lens assembly has a total track length TTL in the operative pop-out state and a collapsed total track length cTTL in the collapsed state, and wherein $cTTL/S_D<0.6$.

For simplicity, in the description below, "lens" may be used instead of "lens assembly".

Henceforth and for simplicity, the use of the term "pop-out" before various components may be skipped, with the understanding that if defined the first time as being a "pop-out" component, that component is such throughout this description.

In various examples of cameras above and below, the window pop-up mechanism includes a window frame engageable with the optics module, wherein the window frame does not touch the optics module in the pop-out state and wherein the window frame is operable to press on the optics module to bring the camera to the collapsed state. The window frame includes a window that is not in direct contact with the lens.

In some examples, the largest air-gap d is between L_{N-1} and L_N.

In some examples, the largest air-gap d is between L_{N-2} and L_{N-1} or between L_{N-1} and L_N, and the lens assembly has a 35 mm equivalent focal length 35 eqFL between 40 mm and 150 mm. In such an example, d may be larger than TTL/5.

In some examples, $cTTL/S_D<0.55$.

In some examples, S_D is in the range of 10 mm to 15 mm.

In some examples, a camera as above or below is included in a multi-camera together with a second camera having a second total track length TTL_2 in the range of 0.9xTTL to 1.1xTTL.

In some examples, the lens assembly has a 35 mm equivalent focal length 35 eqFL larger than 24 mm.

In some examples, the lens assembly has an effective focal length EFL and ratio TTL/EFL is smaller than 1.4 and larger than 1.0.

In various examples, there are provided digital cameras comprising: an optics module comprising a lens assembly that includes N lens elements L_1-L_N starting with L_1 on an object side, wherein N≥4 and wherein the lens assembly has a back focal length BFL that is larger than any air-gap between lens elements and has an effective focal length EFL in the range of 7 mm to 18 mm; a pop-out mechanism configured to actuate the lens assembly to an operative pop-out state and to a collapsed state, wherein the lens assembly has a total track length TTL in the operative pop-out state and a collapsed total track length cTTL in the collapsed state, and wherein the pop-out mechanism is configured to control the BFL such that cTTL/EFL<0.55; and an image sensor having sensor diagonal S_D.

In some examples, a pop-out mechanism includes a window pop-out mechanism based on a pin-groove assembly, and one or more of the pins slide in vertically oriented grooves and one or more pins slide in angled grooves that have an angle of 20-80 degrees, 30-70 degrees or 40-60 degrees with respect to the vertical.

In some examples, a pop-out mechanism includes a barrel pop-out mechanism that comprises springs and a guiding and positioning mechanism that enables sufficient z-decenter and xy-decenter accuracy between lens elements in the operative pop-out state and enables repeatability in switching between operative and collapsed states, wherein the sufficient decenter accuracy is less than 0.1 mm decenter and wherein the repeatability is less than 0.05 mm decenter. In other examples, the sufficient decenter accuracy is less than 0.8 mm decenter and the repeatability is less than 0.04 mm decenter. In yet other examples, the sufficient decenter accuracy is less than 0.6 mm decenter and the repeatability is less than 0.03 mm decenter. The guiding and positioning mechanism may be based on a pin and groove assembly, on a stopper or on a kinematic coupling mechanism. In some examples, a guiding mechanism may be based on a pin-groove assembly and a positioning mechanism based on a magnetic force.

In some examples, S_D is in the range of 4.5 mm to 10 mm and the lens assembly has a 35 eqFL larger than 45 mm and smaller than 180 mm.

In some examples, S_D is in the range of 10 mm to 20 mm and the lens assembly has a 35 eqFL larger than 40 mm and smaller than 180 mm.

In some examples, ratio TTL/EFL is smaller than 1.0 and larger than 0.7.

In some examples, BFL is larger than TTL/3 and smaller than TTL/1.5.

In some examples of cameras as above or below, the lens has a lens element with a largest lens diameter d_L, wherein a penalty between a largest diameter d_{module} of the optics module and the largest lens diameter d_L is smaller than 4 mm, than 2 mm or even than 1 mm.

In various examples, there are provided multi-cameras comprising: a first camera that includes a first lens assembly with a first field of view FOV_1 and N lens elements L_1-L_N starting with L_1 on an object side wherein N≥4, a first image sensor having a sensor diagonal S_{D1}, and a pop-out mechanism that controls a largest air-gap d between two consecutive lens elements to bring the first camera to an operative pop-out state and a collapsed state, wherein the first lens assembly has a first 35 mm equivalent focal length 35 $eqFL_1$, a total track length TTL_1 in the operative state and a collapsed total track length $cTTL_1$ in the collapsed state, wherein S_{D1} is in the range 7-20 mm and wherein $cTTL_1/S_{D1}<0.6$; and a second camera having a second camera effective focal length EFL_2 of 7-18 mm and including a second lens assembly with a second field of view FOV_2 smaller than FOV_1, the second lens assembly comprising M lens elements L_1-L_M starting with L_1 on an object side wherein M≥4, and a pop-out mechanism configured to actuate the second camera to an operative state and a collapsed state, wherein the second lens assembly has a second 35 mm equivalent focal length 35 $eqFL_2$, a total track length TTL_2 in the operative state and a collapsed total track length $cTTL_2$ in the collapsed state, and wherein cTTL/EFL<0.55.

In some examples, $cTTL_1=cTTL_2\pm10\%$.

In some examples, 35 $eqFL_2\geq1.5\times35$ $eqFL_1$.

In some examples, 35 $eqFL_1$ is larger than 24 mm.

In some examples, 35 $eqFL_2$ is larger than 45 mm.

In various examples, there are provided multi-cameras comprising: a Wide camera comprising a lens barrel carrying a Wide lens assembly comprising N≥4 lens elements L_1-L_N starting with L_1 on an object side, an image sensor having a Wide sensor diagonal S_{DW}, and a first pop-out mechanism that controls an air-gap d_{N-1} between lens elements L_N and L_{N-1} to bring the camera to an operative state and a collapsed state, wherein the Wide lens assembly has a field of view FOV_W, a total track length TTL_W in the operative state and a collapsed total track length $cTTL_W$ in the collapsed state, wherein if S_{DW} is in the range 10-16 mm then $cTTL_W/S_{DW}<0.6$; and a Tele camera comprising a lens barrel carrying a Tele lens assembly comprising N is ≥4 lens elements L_1-L_N starting with L_1 on an object side, a Tele image sensor having a sensor diagonal S_{DT} and a second pop-out mechanism that controls an air-gap between lens element L_N and the Tele image sensor to bring the camera to an operative state and a collapsed state, wherein the Tele lens assembly has a field of view FOV_T smaller than FOV_W, a TTL_T in the operative state and a $cTTL_T$ in the collapsed state, wherein if S_{DT} is in the range 4.5-10 mm then $cTTL_T$<EFL_T<0.55, and wherein $cTTL_W$=$cTTL_T$±10%.

In some examples, the multi-camera is embedded in a device having a device exterior surface, and in an operative state the camera extends beyond the device exterior surface by 2 mm-7 mm and in a non-operative state the cameras extends beyond the device exterior surface by less than 2 mm.

In some examples, 7 mm<TTL_W<13 mm, 1.0<TTL_W/EFL_W<1.3 and d_{N-1} is greater than TTL/4.

In some examples, there is provided a camera comprising: a lens assembly comprising N lens elements L_1-L_N starting with L_1 on an object side wherein N≥4; a curved image sensor having a sensor diagonal S_D in the range of 7-20 mm; and a pop-out mechanism that controls an air-gap d between L_N and the image sensor to bring the camera to an operative pop-out state and a collapsed state, wherein the lens assembly has a total track length TTL in the operative pop-out state and a collapsed total track length cTTL in the collapsed state, wherein cTTL/S_D<0.6 and wherein the lens assembly has a 35 mm equivalent focal length 35 eqFL that is smaller than 18 mm.

BRIEF DESCRIPTION OF THE DRAWINGS

Non-limiting examples of embodiments disclosed herein are described below with reference to figures attached hereto that are listed following this paragraph. Identical structures, elements or parts that appear in more than one figure are generally labeled with a same numeral in all the figures in which they appear. If identical elements are shown but numbered in only one figure, it is assumed that they have the same number in all figures in which they appear. The drawings and descriptions are meant to illuminate and clarify embodiments disclosed herein and should not be considered limiting in any way. In the drawings:

FIG. **1**A illustrates schematically the definition of various entities such as TTL and EFL;

FIG. **1**B shows definitions of FOV, EFL and S for a thin lens approximation or equivalence;

FIG. **2**A shows a cross sectional view of a pop-out camera disclosed herein in an in a pop-out state and incorporated in a host device;

FIG. **2**B shows a cross sectional view of a pop-out frame of the camera of FIG. **2**A;

FIG. **2**C shows a cross sectional view of the camera of FIG. **2**A in a collapsed state;

FIG. **2**D shows a cross sectional view of the frame shown in FIG. **2**B in a collapsed state;

FIG. **3**A shows a perspective view of the camera of FIG. **2**A in the pop-out state;

FIG. **3**B shows a perspective view of the camera of FIG. **2**A in the collapsed state;

FIG. **4**A shows in cross section a lens module in the camera of FIG. **2**A;

FIG. **4**B shows the same as FIG. **4**A in a perspective view;

FIG. **4**C shows an example of an optical lens system that may be used in a pop-out camera disclosed herein;

FIG. **5**A shows in cross section a lens module in the camera of FIG. **2**A in the collapsed state;

FIG. **5**B shows the same as FIG. **5**A in a perspective view;

FIG. **6**A shows in cross section another example of a lens module in a pop-out state;

FIG. **6**B shows in cross section the pop-out lens module of FIG. **6**A in a collapsed state;

FIGS. **6**C and **6**D shows other examples of an optical lens system that may be used in a pop-out camera disclosed herein;

FIG. **7** shows a perspective view of the lens module of FIG. **6**A;

FIG. **8** shows a perspective view of the lens module of FIG. **6**B;

FIG. **9**A shows a perspective view of an actuator of the pop-out mechanism in a pop-out state;

FIG. **9**B shows a perspective view of the actuator of FIG. **9**A in a collapsed state;

FIG. **10** shows yet another example of an optical lens system that may be used in a pop-out camera disclosed herein;

FIG. **11**A shows an example of a smartphone with a dual-camera that includes a regular folded Tele camera and an upright pop-out Wide camera;

FIG. **11**B shows details of the cameras of FIG. **11**A with the Wide pop-out camera being in a pop out state;

FIG. **11**C shows the smartphone of FIG. **11**A with the Wide pop-out camera in a collapsed state;

FIG. **11**D shows details of the cameras of FIG. **11**A, with the Wide pop-out camera being in a collapsed state;

FIG. **12**A shows another example of a smartphone with a dual-camera comprising an upright Tele camera and an upright Wide camera, with both cameras in a pop-out state;

FIG. **12**B shows details of the cameras of the smartphone in FIG. **12**A in a pop-out state;

FIG. **12**C shows the smartphone of FIG. **12**A with both cameras in a collapsed state;

FIG. **12**D shows details of the cameras of the smartphone in FIG. **12**A in a collapsed state;

FIG. **13** shows yet another example of an optical lens system can be included in a pop-out camera disclosed herein;

FIG. **14**A shows in cross sectional view another example of a pop-out camera disclosed herein in a pop-out state and incorporated in a host device;

FIG. **14**B shows a perspective view of a frame in the pop-out camera of FIG. **14**A;

FIG. **14**C shows in cross section the camera of FIG. **14**A in a collapsed state;

FIG. **14**D shows a perspective view of the frame of FIG. **14**B in a collapsed state;

FIG. **15**A shows in cross-section a pop-out mechanism in the camera of FIG. **14**A;

FIG. **15**B shows the mechanism of FIG. **15**A in a collapsed state;

FIG. **16**A shows a cross sectional view of another example of a pop-out optics module in a pop-out state;

FIG. **16**B shows a perspective view of the pop-out optics module of FIG. **16**A;

FIG. **17**A shows in perspective view the pop-out optics module of FIG. **16**A in a pop-out state;

FIG. **17**B shows in perspective view the pop-out optics module of FIG. **16**A in a collapsed state;

FIG. **18**A shows a perspective view of an optics frame in the optics module of FIG. **16**A a pop-out state;

FIG. **18**B shows a perspective view of the optics frame of FIG. **18**A in a collapsed state;

FIG. **18**C shows a section of the optics frame of FIG. **18**A in more detail;

FIG. **18**D shows a section of optics frame of FIG. **18**B in more detail;

FIG. **18**E shows the optics frame of FIG. **18**A in a top view;

FIG. **18**F shows the optics frame of FIG. **18**A in an exploded view;

FIG. **19**A shows a perspective view of yet another example of an optics module in a pop-out state;

FIG. **19**B shows the optics module of FIG. **19**A in a top view;

FIG. **19**C shows the optics module of FIG. **19**A in a pop-out state in a cross-sectional view;

FIG. **19**D shows the optics module of FIG. **19**A in a collapsed state in a cross-sectional view;

FIG. **19**E shows a top cover and a magnet of the optics module of FIG. **19**A in a perspective view;

FIG. **19**F shows the top cover and magnet of the optics module of FIG. **19**E in a top view;

FIG. **20**A shows a magnet part of window position measurement mechanism in a side view;

FIG. **20**B shows the window position measurement mechanism of FIG. **20**A in a perspective view;

FIG. **20**C shows a side view of 3 magnets and a Hall sensor of the window position measurement mechanism of FIG. **20**A in a collapsed state;

FIG. **20**D shows a side view of the 3 magnets and the Hall sensor of the window position measurement mechanism of FIG. **20**A in a pop-out state;

FIG. **20**E shows an example of a design and magnetic field of the window position measurement mechanism of FIG. **20**A;

FIG. **20**F shows an example of a magnet configuration that may be included in the position measurement mechanism;

FIG. **20**G shows another example of another magnet configuration that may be included in a position measurement mechanism.

DETAILED DESCRIPTION

FIG. **2**A shows in cross sectional view (through cross section marked **2**A-**2**A in FIG. **3**A) an example numbered **200** of a pop-out camera disclosed herein incorporated in a "host" device **250** (e.g. a smartphone, tablet, etc.). In FIG. **2**A, camera **200** is shown in an operative or "pop-out" state (and thus referred to as a "camera in pop-out state"). Camera **200** has also a collapsed ("c" or "non-operative") state, shown in FIG. **2**C. In this state, the camera is not operative as a camera in pop-out state. FIG. **3**A shows camera **200** in the pop-out state and FIG. **3**B shows camera **200** in the collapsed state, both in perspective views.

Camera **200** comprises a general pop-out mechanism **210** and a pop-out optics module **240**. Optics module **240** comprises a lens barrel holder **202** carrying a pop-out lens barrel **204** with a pop-out lens assembly **206**, and in some cases ("examples") an image sensor **208**. In some examples, the image sensor may be separate from the optics module. Lens barrel **204** and window **216** are separated by an air-gap **222** of, for example, 0.15-3 mm. Air-gap **222** allows for a movement of the lens barrel by 0.1-3 mm for performing auto-focus (AF) and optical image stabilization (OIS) by moving the lens as known in the art. Optics module **240** is covered by a cover **232**. In some examples, the pop-out lens barrel (e.g. a lens barrel **602**) may be divided into two or more sections, e.g. in a fixed lens barrel section and a collapsible barrel section.

General pop-out mechanism **210** comprises a "window" pop-out mechanism (external to the optics module) and a "barrel" pop-out mechanism with some parts external to and some parts internal to the optics module. The window

pop-out mechanism raises and lowers the window. The barrel pop-out mechanism enables the pop-out and collapsed lens barrel states.

The window pop-out mechanism includes parts shown in detail for example in FIG. **9**A-B, FIG. **14**B, FIG. **14**D, FIGS. **15**A-B, FIGS. **20**A-F. Specifically, the window pop-out mechanism comprises an actuator like **212** or **212'**, a pop-out frame **220** (see e.g. FIG. **2**B) that includes a window frame **214** carrying a window **216** that covers an aperture **218** of the camera, and an external module seal **224**. External module seal **224** prevents particles and fluids from entering the camera and host device **250**. In some embodiments (e.g. in a frame **220'** described with reference to FIGS. **14**A-D), a pop-out frame may include additional parts such as a cam follower (e.g. **1402** in FIG. **14**A), a side limiter (e.g. **1406** in FIG. **14**A) and a window position measurement mechanism (e.g. **1420** in FIG. **14**B).

The barrel pop-out mechanism includes parts shown in detail for example in FIGS. **4**A, **5**A, **6**A-B, **14**A, **14**C, **16**A, **17**A-B, **18**A-F and **19**A-F. Specifically, the barrel pop-out mechanism may include one or more springs **230**, pop-out lens barrel **204** with pop-out lens assembly **206**, one or more springs **230** and a guiding and positioning mechanism (see e.g. FIGS. **19**A-**19**B and description below). The one or more springs push optics module **240** towards frame **220**, i.e. when frame **220** moves upwards for switching from a collapsed state to a pop-out state, no further actuation mechanism within the optics module is required.

The guiding and positioning mechanism positions the lens groups and optical components in fixed distance and orientation. In an example, the guiding and positioning mechanism comprises a pin **242** and a groove **244** (see FIG. **2**C, FIG. **4**A and FIG. **5**A). In some examples, the guiding and positioning mechanism may include a stopper **618** (see FIG. **6**A-B), a kinematic coupling mechanism (see FIG. **18**A-D) or a magnet-yoke assembly (see FIGS. **19**A-F). In some examples, the guiding and positioning mechanism works by means of an interplay between an optics module and another component of a camera such as camera **200** (see e.g. FIGS. **6**A-B and FIGS. **19**A-F). Pin **242** and groove **244** provide a first example of a pin-groove assembly. Groove **244** may comprise a v-shaped groove or another groove, with groove **244** having legs at an angle of e.g. 30-150 degrees. Other pin-groove assemblies are described below. In some examples, the guiding and positioning mechanism is included in an optics module in its entirety (see e.g. FIG. **4**A and FIG. **5**A and FIG. **18**A-D).

The pin-groove assembly with pin **242** and groove **244** provides mechanical stability and repeatability in the X-Z plane and in the Y plane of the coordinate system shown. Stopper **618** provides mechanical stability and repeatability in Y plane. In some examples, other pins such as pins **1206** (see FIG. **12**B and FIG. **12**D) may be used for providing mechanical stability and repeatability in the X-Z plane.

The lens, the image sensor and (optionally) an optical window or "filter" (e.g. IR filter) **234** form a pop-out optical lens system **260** (see e.g. FIG. **4**C). The image sensor may have a sensor diagonal S_D in the range 3.5-30 mm. For a lens having an EFL of 5 mm to 25 mm, this typically represents a 35 eqFL in the range 10-300 mm. Sensor diagonal S_D connects to a sensor width W and a height H via $S_D=\sqrt{(W^2+H^2)}$. In other examples EFL may be 8 mm to 28 mm.

To switch between pop-out and collapsed states, pop-out mechanism **210** causes the following movements in frame **220** (where all movements are defined relative to the host device and the coordinate systems shown): a horizontal (i.e.

in the X-Z plane) movement of the cam follower and a vertical (i.e. in the Y direction) movement of the window frame. The movement in frame **220** causes a vertical (Y direction) movement of the lens barrel (for a single group or "1G" lens) or of a collapsible section of the lens barrel (in a two group or "2G" lens) in optics module **240**. The image sensor and the side limiter do not move. Importantly, the barrel pop-out mechanism does not include an actuator.

In the pop-out state shown in FIG. **2**B, camera **200** forms a significant pop-out bump **226** with respect to an exterior surface **228** of host device **250**. Here, "significant" may be for example 1.5 mm-8 mm. In the pop-out state, camera **200** increases the height of host device **250** to a "height in a pop-out state".

The pop-out lens may be a Tele lens, for example as in FIG. **4**C or FIG. **10** or FIG. **6**D, or a Wide lens as in FIG. **6**C or FIG. **13**. Depending on the type of lens, a pop-out camera operates as a pop-out Tele camera or as a pop-out Wide camera. A pop-out Tele camera may have a FOV_T of 20-50 deg. A pop-out Wide camera may have a FOV_W of 50-120 deg. The TTL of the lens, measured from the first surface of the first lens element in the lens to the image sensor may be for example 6 mm-18 mm.

FIG. **2**D shows a cross sectional view of frame **220** in the collapsed state. Actuator **212** brings the camera to a collapsed state by performing work against the spring. In the collapsed state, the spring is in a compressed state, see also FIG. **4**B. To switch camera **200** to the collapsed state, actuator **212** moves window frame **214** to apply pressure on lens barrel **204**. This translates into a movement of lens barrel **204** towards the image sensor. In the collapsed state, the TTL is a collapsed TTL (cTTL) and may be for example 5-12 mm. cTTL is always measured between a first surface of lens element **L1** on the image side (marked S**2**) and an imaging surface of the image sensor along the optical axis marked S**16**. The difference between cTTL and TTL stems from a modified BFL with respect to the pop-out state. Camera **200** is designed such that there is a large BFL in the operative state. This large BFL can be collapsed to bring the camera to a collapsed state, achieving a slim camera design. In the collapsed state, the camera forms a collapsed bump (c-bump) **236** with respect to device exterior surface **228**. The c-bump may have for example a size (height) of 0-3 mm. In the collapsed state, the height of host device **250** is a "height in the collapsed state" that is much smaller than the height in the pop-out state but still larger than the host device height by the c-bump **236**.

Camera **200** may be designed to support, in some examples, accuracy tolerances for decenter of e.g. ±20 μm in the X-Z plane and of e.g. ±10 μm in the Y direction, as well as for a tilt of ±0.5°. The planes and directions are as in the coordinate systems shown in the figures. Repeatability tolerances for decenter may be e.g. ±10 μm in the X-Z plane and of e.g. ±5 μm in the Y direction, as well as for a tilt of ±0.25°. In other examples, accuracy tolerances for decenter may be e.g. ±10 μm in the X-Z plane and of e.g. ±5 μm in the Y direction, as well as e.g. ±0.15°. Repeatability tolerances for decenter may be e.g. ±5 μm in the X-Z plane and of e.g. ±2.5 μm in the Y direction, as well as for a tilt of ±0.08°. In yet other examples, accuracy tolerances for decenter may be e.g. ±5 μm in the X-Z plane and of e.g. ±2.5 μm in the Y direction, as well as e.g. ±0.1°. Repeatability tolerances for decenter may be e.g. ±1.5 μm in the X-Z plane and of e.g. ±0.8 μm in the Y direction, as well as for a tilt of ±0.05°.

Similar accuracy tolerances and repeatability tolerances hold for optics frame **1650** (see e.g. FIG. **16**A) and optics module **600**" (see e.g. FIG. **19**A).

"Accuracy tolerances" refer here to a maximum variation of the distances between optical elements and between mechanical elements. "Repeatability tolerances" refer here to a maximum variation of the distances between optical elements and between mechanical elements in different pop-out cycles, i.e. the capability of the mechanical and optical elements to return to their prior positions after one or many pop-out (or collapse) events.

Tolerances in the Y direction may be less important, as variations in Y can be compensated by optical feedback and moving the lens for auto-focus.

FIG. **4**A shows in cross section optics module **240** in the pop-out state. FIG. **4**B shows optics module **240** in the same state in perspective. The diameter of the smallest circle that entirely surrounds the optics module defines a "largest diameter" "d_{module}" of the optics module. That is "d_{module}" marks the largest diagonal of an optics module (here and in e.g. FIGS. **7**, **17**B, **18**A, **18**B, **18**E and **19**A) except when stated otherwise (e.g. as re. FIG. **16**A).

FIG. **4**C shows details of a first exemplary lens system **400** that can be used in camera **200** in a pop-out state. Lens system **400** comprises a lens **420** that includes, in order from an object side to an image side, a first lens element **L1** with object-side surface S**2** and image-side surface S**3**; a second lens element L**2** with object-side surface S**4**, with an image side surface marked S**5**; a third lens element **L3** with object-side surface S**6** with image-side surface S**7**; a fourth lens element **L4** object-side surface marked S**8** and an image-side surface marked S**9**; a fifth lens element **L5** with object-side surface marked S**10** and an image-side surface marked S**11**; and a sixth lens element **L6** with object-side surface marked S**12** and an image-side surface marked S**13**. S**1** marks a stop. Lens system **400** further comprises optical window **234** disposed between surface S**13** and image sensor **208**. Distances between lens elements and other elements are given in tables below along an optical axis of the lens and lens system.

In lens system **400**, TTL=11.55 mm, BFL=5.96 mm, EFL=13 mm, F number=2.20 and the FOV=29.7 deg. A ratio of TTL/EFL=0.89. The optical properties of lens **420** do not change when switching between a pop-out state and a collapsed state (i. e. gaps between lens elements are constant).

In the collapsed state (see FIG. **5**A), cTTL may be 5.64-8.09 mm. The difference between cTTL and TTL stems from a modified BFL which is now a collapsed BFL, "c-BFL" (see FIG. **5**A). c-BFL may be 0.051-2.5 mm. All distances between lens elements **L1**-**L6** and lens surfaces S**2**-S**13** remain unchanged.

Detailed optical data of lens system **400** is given in Table 1, and the aspheric surface data is given in Table 2 and Table 3, wherein the units of the radius of curvature (R), lens element thickness and/or distances between elements along the optical axis and diameter are expressed in mm. "Index" is the refraction index. The equation of the aspheric surface profiles is expressed by:

$$z(r) = \frac{cr^2}{1+\sqrt{1-(1+k)c^2r^2}} + D_{con}(u)$$

$$D_{con}(u) = u^4 \sum_{n=0}^{N} A_n Q_n^{con}(u^2)$$

-continued

$$u = \frac{r}{r_{norm}},\ x = u^2$$

$$Q_0^{con}(x) = 1 \quad Q_1^{con} = -(5 - 6x) \quad Q_2^{con} = 15 - 14x(3 - 2x)$$

$$Q_3^{con} = -\{35 - 12x[14 - x(21 - 10x)]\}$$

$$Q_4^{con} = 70 - 3x\{168 - 5x[84 - 11x(8 - 3x)]\}$$

$$Q_5^{con} = -[126 - x(1260 - 11x\{420 - x[720 - 13x(45 - 14x)]\})]$$

where {z, r} are the standard cylindrical polar coordinates, c=1/R is the paraxial curvature of the surface, k is the conic parameter, and r_{norm} is generally one half of the surface's clear aperture. An are the polynomial coefficients shown in lens data Table 2 and Table 3 (as well as in Table 5 and Table 6, and in Table 10 and Table 11). The Z-axis is defined to be positive towards the image. Also note that in Table 1 (as well as in Table 4 and Table 9), the distances between various elements (and/or surfaces) refer to the element thickness and are measured on the optical axis Z, wherein the stop is at z=0. Each number is measured from the previous surface. Thus, the first distance −1.197 mm is measured from the stop to surface **S2**. The reference wavelength is 555.0 nm. Units are in mm (except for refraction index "Index" and Abbe #). The largest lens diameter d_L of a lens such as lens **240** is given by the largest diameter present among all the lens elements of a lens such as lens **240**.

TABLE 3

Surface #	A4	A5	A6
2	−4.91E−05	6.04E−06	−7.27E−07
3	8.15E−05	−5.60E−05	6.51E−06
4	1.04E−04	−1.60E−04	1.22E−05
5	−4.30E−04	9.99E−05	−1.20E−05
6	6.30E−05	−4.83E−06	1.65E−07
7	2.23E−05	2.33E−05	2.32E−07
8	−2.32E−04	−1.73E−05	−4.47E−05
9	−4.98E−05	−1.12E−04	−7.34E−05
10	1.23E−04	1.53E−04	7.13E−05
11	−6.11E−04	1.56E−04	1.39E−05
12	−5.99E−04	1.37E−04	7.47E−06
13	5.73E−05	−1.33E−05	1.30E−06

FIG. **5**A shows a pop-out optics module **240** in a collapsed state in a cross sectional view. FIG. **5**B shows a perspective view of the same.

FIG. **6**A shows a cross sectional view (through cross section marked **6**A-**6**A in FIG. **7**) of another example numbered **600** of a pop-out optics module in a pop-out state. Optics module **600** may be integrated into a pop-out mechanism such as **210** (not shown here). Optics module **600** includes a lens barrel **602** with a collapsible lens barrel section (first barrel section) **604** carrying a first lens group **606**, and a fixed lens barrel section (second barrel section) **608** carrying a second lens group **610**. The two lens groups form a lens **620** that includes altogether N lens elements L**1**-LN, arranged with a first lens element L**1** on an object side and a last lens element LN on an image side. Optics module **600** is covered by cover **232**. Lens **620**, and optional optical window **234** and an image sensor **208** form a lens system **630**.

In general, N≥4. In other examples, the lens barrel may comprise more than two barrel sections with more lens groups each, e.g. 3, 4, 5 lens barrel sections with each barrel section carrying a lens group. The lens barrel sections may be divided into fixed barrel sections and movable barrel sections. Air-gaps may be formed between lens groups according to their relative movement. In examples with more than two barrel sections, some or all barrel sections may be movable and have respective air-gaps formed between the lens groups. The air-gaps between lens groups may collapse in a non-operative camera state. The sum of

TABLE 1

Lens system 400 EFL = 13.00 mm, F number = 2.20, FOV = 29.7 deg.									
Surface #	Comment	Type	Curvature Radius	Thickness	Aperture Radius (D/2)	Material	Index	Abbe #	Focal Length
1	A.S	Plano	Infinity	−1.197	2.940				
2	Lens 1	QT1	3.453	0.927	2.940	Plastic	1.54	56.18	14.80
3			5.463	0.008	2.895				
4	Lens 2	QT1	4.054	1.545	2.811	Plastic	1.54	56.18	8.78
5			22.988	0.025	2.620				
6	Lens 3	QT1	6.508	0.218	2.346	Plastic	1.67	19.4	−9.67
7			3.215	1.134	2.003				
8	Lens 4	QT1	−9.074	0.206	1.774	Plastic	1.64	23.53	−17.62
9			−45.557	0.902	1.648				
10	Lens 5	QT1	66.987	0.417	1.665	Plastic	1.67	19.44	9.93
11			−7.435	0.009	1.788				
12	Lens 6	QT1	−19.713	0.197	1.845	Plastic	1.54	56.18	−10.85
13			8.487	5.450	1.984				
14	Filter	Plano	Infinity	0.210	—	Glass	1.52	64.2	
15			Infinity	0.300	—				
16	Image	Plano	Infinity	—	—				

TABLE 2

Aspheric Coefficients					
Surface #	R_{norm}	A0	A1	A2	A3
2	2.385	−6.99E−02	−1.08E−02	−1.52E−03	−4.14E−04
3	2.385	−6.86E−02	−1.05E−02	2.44E−03	−3.81E−04
4	2.385	9.39E−03	−1.24E−02	2.84E−03	8.52E−04
5	2.385	1.26E−03	−2.19E−02	1.12E−03	8.76E−04
6	1.692	−5.76E−02	2.59E−02	−4.13E−03	−2.03E−05
7	1.692	−2.94E−02	3.95E−02	−5.32E−03	−6.02E−04
8	1.648	3.50E−01	8.82E−03	−2.47E−03	6.30E−04
9	1.648	4.12E−01	2.61E−02	2.43E−03	2.06E−03
10	1.587	−2.16E−01	−3.53E−02	−8.04E−03	4.41E−04
11	1.587	−1.36E−01	−2.45E−02	−2.51E−03	1.25E−03
12	1.609	−2.58E−01	2.37E−02	−1.19E−03	2.12E−04
13	1.609	−2.99E−01	2.60E−02	−4.30E−03	−2.04E−04

such air-gaps may be 1-8.5 mm. The largest air-gaps present between two consecutive lens elements may be used to define lens groups. For example, the largest air-gap present between two consecutive lens elements may be used to divide a lens into two lens groups, the largest air-gap and the second largest air-gap present between two consecutive lens elements may be used to define three lens groups, etc. This statement is true for all lens and camera examples below. In the pop-out state, air-gap d_{N-1} may be 1-3.5 mm. A spring **614** pushes the first lens barrel section **604** towards a window frame like frame **214**. In the operative state, stopper **618** and another stopper **618'** may act as a stopper mechanism that keeps the lens groups in fixed distance and orientation. In some examples, an camera in pop-out state disclosed herein may be designed to support tolerances for decenter of e.g. ±20 μm in the X-Z plane and of e.g. ±10 μm in the Y direction, as well as for a tilt of ±0.2° of the lens barrel with respect to image sensor **208**. In other examples tolerances for decenter may be e.g. ±3-10 μm in the X-Z plane and of e.g. ±3-10 μm in the Y direction, as well as e.g. ±0.05°-0.15° for a tilt of lens barrel with respect to the image sensor Y. In yet other examples, tolerances for decenter may be smaller than 1 μm in the X-Z plane, e.g. 0.8 μm. In yet other examples, tolerances for decenter in a Y plane may be smaller than 1 μm, e.g. 0.8 μm, to support the properties of a lens system like system **630**, **650** or **1000**, especially for air-gaps between lens elements such as d_{N-1} (see FIG. **6**C) or d_{1006} (see FIG. **10**). In some examples, pins such as pins **1208** (see FIG. **12**B and FIG. **12**D) may be used for providing mechanical stability and repeatability in X-Z plane.

The TTL of the lens, measured from the first (object side) surface of L**1** to the image sensor may be 5-18 mm. The image sensor diagonal may be 6 mm<sensor diagonal<30 mm. The 35 eqFL may be 15 mm<equivalent focal length<200 mm. The TTL/EFL ratio may vary in the range 0.7<TTL/EFL<1.5.

FIG. **6**B shows a cross sectional view (through cross section marked **6**B-**6**B in FIG. **8**) of optics module **600** in a collapsed state. To switch optics module **600** to the collapsed state, actuator **212** decreases the air-gap between the first surface of LN and the second surface of LN−1 by moving the window frame (not shown here) to apply pressure to the lens barrel that translates into a movement of the collapsible lens barrel section towards the image sensor. In the collapsed state, cTTL may be 5-12 mm, and collapsed air-gap $c\text{-}d_{N-1}$ may be 0.05-0.85 mm. The difference between cTTL and TTL stems from a modified distance between the first lens group **606** in first collapsible lens barrel section **604** and second lens group **610** in second fixed lens barrel section **608**. The distance between first lens group **606** and the image sensor changed with respect to the pop-out state, but the distance between second lens group **610** and the image sensor did not change. The optical properties of lens **620** change when switching between a pop-out state and a collapsed state.

FIG. **6**C shows an example of another lens system **650** that may be used in optics module **600** or another pop-out optics module **600'** below. Lens system **650** is shown in a pop-out state. The design data is given in Tables 4-6. Lens system **650** includes a lens **620'** with seven lens elements L**1**-L**7** arranged as shown, optical window **234** and image sensor **208**. Lens elements L**1**-L**6** form the first lens group **606**, and lens element L**7** forms the second lens group **610**. The TTL is 8.49 mm and the BFL is 1.01 mm. Focal length is EFL=6.75 mm, F number=1.80 and the FOV=80.6 deg. Air-gap d_{N-1} is 2.1 mm.

TABLE 4

Lens system 650
EFL = 6.75 mm, F number = 1.80, FOV = 80.6 deg.

Surface #	Comment	Type	Curvature Radius	Thickness	Aperture Radius (D/2)	Material	Index	Abbe #	Focal Length
1	A.S	Plano	Infinity	−0.727	1.880				
2	Lens 1	QT1	2.844	0.861	1.880	Plastic	1.54	55.9	7.55
3			8.156	0.128	1.797				
4	Lens 2	QT1	6.089	0.250	1.769	Plastic	1.67	19.4	−19.67
5			4.106	0.323	1.677				
6	Lens 3	QT1	7.530	0.384	1.678	Plastic	1.54	55.9	26.14
7			15.633	0.489	1.685				
8	Lens 4	QT1	19.241	0.257	1.726	Plastic	1.66	20.4	−34.80
9			10.465	0.397	1.974				
10	Lens 5	QT1	−9.931	0.601	2.060	Plastic	1.57	37.4	−6.53
11			6.067	0.187	2.315				
12	Lens 6	QT1	4.294	0.738	2.725	Plastic	1.54	55.9	3.66
13			−3.522	2.097	2.984				
14	Lens 7	QT1	−5.605	0.770	4.901	Plastic	1.54	55.9	−5.10
15			5.824	0.188	5.579				
16	Filter	Plano	Infinity	0.2100	—	Glass	1.52	64.2	
17			Infinity	0.610	—				
18	Image	Plano	Infinity	—	—				

In the collapsed state (see FIG. **6**B or FIG. **14**C), cTTL may be 6.44-7.24 mm. The difference between cTTL and TTL stems from a modified air-gap between L**6** and L**7**, which is a collapsed air-gap $c\text{-}d_{N-1}$ and which may be 0.05-0.85 mm. The BFL did not change with respect to the pop-out state.

The optical properties of lens **620'** change when switching between a pop-out state and the collapsed state. The optical properties presented here refer to the lens elements in a "maximal" pop-out state, i.e. when the lens has the largest TTL.

TABLE 5

Aspheric Coefficients						
Surface #	R_{norm}	A0	A1	A2	A3	A4
2	2.03E+00	8.98E−02	1.91E−02	4.39E−03	8.23E−04	1.41E−04
3	1.88E+00	1.36E−02	1.31E−02	1.21E−03	2.44E−04	8.58E−05
4	1.87E+00	−6.37E−02	3.51E−02	1.73E−03	6.81E−04	1.38E−04
5	1.85E+00	−1.16E−02	5.94E−02	1.65E−02	6.10E−03	1.62E−03
6	1.85E+00	−1.04E−01	1.94E−02	1.51E−02	4.62E−03	9.45E−04
7	1.78E+00	−1.51E−01	−4.96E−03	3.05E−03	1.05E−03	2.14E−04
8	1.78E+00	−5.18E−01	−1.59E−02	5.30E−03	2.48E−03	8.04E−04
9	2.10E+00	−5.53E−01	7.15E−02	3.81E−02	1.81E−02	6.25E−03
10	2.30E+00	−3.60E−01	2.33E−01	1.57E−01	1.29E−01	5.32E−02
11	2.42E+00	−1.68E+00	1.98E−01	−4.68E−03	5.58E−02	2.62E−02
12	2.59E+00	−1.52E+00	8.24E−02	−8.59E−03	1.18E−02	2.67E−04
13	2.86E+00	4.37E−01	2.30E−03	4.67E−02	−1.08E−02	−3.33E−03
14	5.06E+00	1.44E+00	5.66E−01	−2.72E−01	1.01E−01	−2.51E−02
15	5.55E+00	−5.28E+00	5.59E−01	−2.71E−01	4.18E−02	−4.23E−02

TABLE 6

Aspheric Coefficients					
Surface #	A5	A6	A7	A8	A9
2	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
3	−2.45E−05	0.00E+00	0.00E+00	0.00E+00	0.00E+00
4	−7.03E−05	0.00E+00	0.00E+00	0.00E+00	0.00E+00
5	2.33E−04	0.00E+00	0.00E+00	0.00E+00	0.00E+00
6	5.32E−05	0.00E+00	0.00E+00	0.00E+00	0.00E+00
7	3.84E−05	0.00E+00	0.00E+00	0.00E+00	0.00E+00
8	3.66E−04	0.00E+00	0.00E+00	0.00E+00	0.00E+00
9	1.49E−03	−2.63E−04	0.00E+00	0.00E+00	0.00E+00
10	1.64E−02	1.44E−03	0.00E+00	0.00E+00	0.00E+00
11	1.59E−02	3.54E−03	1.08E−03	0.00E+00	0.00E+00
12	3.18E−03	−5.00E−04	−3.15E−04	0.00E+00	0.00E+00
13	2.65E−03	2.67E−05	−4.87E−04	1.84E−04	0.00E+00
14	−2.21E−04	−1.68E−05	6.75E−04	−6.40E−04	2.77E−04
15	5.53E−03	−7.30E−03	−2.11E−04	−1.38E−03	2.11E−04

FIG. **6**D shows an example of yet another lens system **660** that may be used in optics module **600** or **600'**. Lens system **660'** is shown in a pop-out state. The design data is given in Tables 7-9. Lens system **660** includes a lens **620"** with six lens elements L**1**-L**6** arranged as shown, optical window **234** and image sensor **208**. Lens elements L**1**-L**3** form the first lens group **606**, and lens elements L**4**-L**6** form the second lens group **610**. The TTL is 13.5 mm and the BFL is 5.49 mm. Focal length is EFL=15.15 mm, F number=2.0 and the FOV=32.56 deg. Air-gap d_{607} is 1.78 mm. A ratio of TTL/EFL=0.89.

In the collapsed state (see FIG. **6**B), cTTL may be 5-11 mm. The difference between cTTL and TTL stems from a modified air-gap between L**3** and L**4**, which is a collapsed air-gap c-d_{607} and which may be 0.05-1.0 mm and a modified BFL which is a c-BFL and may be 0.1-1.5 mm. The optical properties of lens **620"** change when switching between a pop-out state and the collapsed state. For lens system **660**, a ratio TTL/EFL is 0.89, i.e. EFL>TTL. The ratio cTTL/EFL may be 0.35-0.75.

TABLE 7

Lens system 660 EFL = 15.15 mm, F number = 2.0, FOV = 32.56 deg.									
Surface #	Comment	Type	Curvature Radius	Thickness	Aperture Radius (D/2)	Material	Index	Abbe #	Focal Length
1	A.S	Plano	Infinity	−1.823	3.731				
2	Lens 1	ASP	4.314	1.837	3.731	Plastic	1.54	55.91	9.50
3			21.571	0.048	3.560				
4	Lens 2	ASP	4.978	0.265	3.419	Plastic	1.67	19.44	−17.41
5			3.422	0.113	3.139				
6	Lens 3	ASP	5.764	1.473	3.113	Plastic	1.67	19.44	20.20
7			11.201	1.780	2.909				
8	Lens 4	ASP	−6.075	0.260	2.143	Plastic	1.67	19.44	−14.33
9			−17.446	1.230	2.008				
10	Lens 5	ASP	−18.298	0.688	2.264	Plastic	1.54	55.91	184.98
11			−16.202	0.040	2.468				
12	Lens 6	ASP	10.235	0.273	2.679	Plastic	1.54	55.91	−93.97
13			8.454	4.783	2.848				
14	Filter	Plano	Infinity	0.210	—	Glass	1.52	64.17	
15			Infinity	0.500	—				
16	Image	Plano	Infinity	—	—				

TABLE 8

Aspheric Coefficients					
Surface #	Conic	A4	A6	A8	A10
2	0	−4.57E−04	−5.55E−05	2.46E−05	−4.65E−06
3	0	8.55E−04	7.37E−04	−1.07E−04	9.78E−06
4	0	−1.51E−02	3.43E−03	−6.33E−04	8.54E−05
5	0	−2.21E−02	5.71E−03	−1.50E−03	2.85E−04
6	0	−3.61E−03	3.56E−03	−1.08E−03	2.29E−04
7	0	−1.74E−04	2.47E−04	5.66E−05	−3.21E−05
8	0	1.75E−02	2.27E−03	−2.24E−03	7.99E−04
9	0	1.79E−02	5.45E−03	−3.71E−03	1.37E−03
10	0	−4.37E−03	−1.59E−02	1.33E−02	−6.54E−03
11	0	−7.77E−02	4.02E−02	−1.21E−02	1.65E−03
12	0	−1.39E−01	7.50E−02	−2.44E−02	4.78E−03
13	0	−5.32E−02	1.90E−02	−4.73E−03	6.11E−04

TABLE 9

Aspheric Coefficients			
Surface #	A12	A14	A16
2	4.92E−07	−2.88E−08	5.71E−10
3	−6.44E−07	1.90E−08	−1.21E−10
4	−6.96E−06	3.18E−07	−6.61E−09
5	−3.28E−05	2.13E−06	−6.25E−08
6	−2.83E−05	1.87E−06	−5.35E−08
7	5.30E−06	−4.54E−07	1.54E−08
8	−1.70E−04	1.94E−05	−9.28E−07
9	−2.64E−04	2.29E−05	−1.78E−07
10	1.83E−03	−2.76E−04	1.73E−05
11	−8.43E−06	−2.54E−05	2.18E−06
12	−5.86E−04	4.30E−05	−1.42E−06
13	−2.86E−05	−1.51E−06	1.53E−07

FIG. **7** shows a perspective view of optics module **600** in a pop-out state. FIG. **8** shows a perspective view of optics module **600** in a collapsed state.

FIG. **9**A shows a perspective view of actuator **212** in a pop-out state. FIG. **9**B shows a perspective view of actuator **212** in a collapsed state. Cross sections **2**B-**2**B and **2**D-**2**D refer to respectively FIGS. **2**B and **2**D. Actuator **212** comprises a pop-out actuator **902** with moving parts for actuation. A pop-out actuator—window frame coupling **904** with a switch **906** translates the pop-out actuation to a movement of the window frame. Switch **906** couples actuator **902** with window frame **214**. As indicated above, the window frame movement is used to switch the camera to the collapsed state. In FIG. **9**A, switch **906** is "down" to provide the pop-out state. In FIG. **9**B, switch **906** is "up" to provide the collapsed state.

FIG. **10** shows another lens system numbered **1000** that can be included in a pop-out Tele camera in a maximal pop-out state. Lens system **1000** includes a lens **1020** with five lens elements as shown, optical window **234** and image sensor **208**. The Tele pop-out camera with lens system **1000** may be incorporated in a host device (e.g. a smartphone, tablet, etc., not shown here). Similar to the shown in FIG. **6**A and FIG. **6**B, in lens system **1000** switching between the pop-out and the collapsed states is obtained by modifying an air-gap d**1006** between a first lens group **1016** and a second lens group **1018**.

In lens system **1000**, a first lens group **1016** includes lens elements **1002**, **1004** and **1006** and a second lens group **1018** includes lens elements **1008** and **1010**. In the pop-out state, air-gap d**1006** between surface **1008***a* of lens element **1008** and surface **1006***b* of the immediately preceding lens element **1006** is 2.020 mm (see Table 10). The TTL of the lens system is 5.904 mm. The division into a first lens group and a second lens group is done according to the largest air-gap between two consecutive lens elements.

Lens system **1000** may provide a FOV of 25-50 degrees, and EFL=6.9 mm, a F number=2.80 and a TTL=5.904 mm. The ratio TTL/EFL is 0.86, i.e. EFL>TTL. The ratio cTTL/EFL may be 0.58-0.69. For air-gap d**1006**=TTL/2.95, so d**1006**>TTL/3. In other examples, for a largest air-gap that divides the lens elements into first and a second lens groups the air-gap may fulfill air-gap>TTL/5 and EFL>TTL.

The optical properties of lens system **1000** change when switching to the collapsed state (not shown). In the collapsed state, cTTL may be 3.97-10 mm and collapsed air-gap c-d**1006** may be 0.05-0.85 mm. The difference between cTTL and TTL stems from a modified distance between first lens group **1016** and second lens group **1018**. The distance between first lens group **1016** and image sensor **208** changed with respect to the pop-out state, but distance between second lens group **1016** and the image sensor **1014** did not change.

In lens system **1000**, all lens element surfaces are aspheric. Detailed optical data is given in Table 10, and the aspheric surface data is given in Table 11, wherein the units of the radius of curvature (R), lens element thickness and/or distances between elements along the optical axis and diameter are expressed in mm. "Nd" is the refraction index. The equation of the aspheric surface profiles is expressed by:

$$z = \frac{cr^2}{1+\sqrt{1-(1+k)c^2r^2}} + \alpha_1 r^2 + \alpha_2 r^4 + \alpha_3 r^6 + \alpha_4 r^8 + \alpha_5 r^{10} + \alpha_6 r^{12} + \alpha_7 r^{14}$$

where r is distance from (and perpendicular to) the optical axis, k is the conic coefficient, c=1/R where R is the radius of curvature, and α are coefficients given in Table 2. In the equation above as applied to examples of a lens assembly disclosed herein, coefficients α_1 and α_7 are zero. Note that the maximum value of r "max r"=Diameter/2. Also note that Table 1 the distances between various elements (and/or surfaces) are marked "Lmn" (where m refers to the lens element number, n=1 refers to the element thickness and n=2 refers to the air-gap to the next element) and are measured on the optical axis z, wherein the stop is at z=0. Each number is measured from the previous surface. Thus, the first distance −0.466 mm is measured from the stop to surface **1002***a*, the distance L**11** from surface **1002***a* to surface **1002***b* (i.e. the thickness of first lens element **1002**) is 0.894 mm, the gap L**12** between surfaces **1002***b* and **1004***a* is 0.020 mm, the distance L**21** between surfaces **1004***a* and **1004***b* (i.e. thickness d**2** of second lens element **1004**) is 0.246 mm, etc. Also, L**21**=d_2 and L**51**=d_5.

TABLE 10

Lens system 1000 EFL = 6.9 mm, F number = 2.80, FOV = 44 degrees					
#	Comment	Radius R [mm]	Distances [mm]	Nd/Vd	Diameter [mm]
1	Stop	Infinite	−0.466		2.4
2	L11	1.5800	0.894	1.5345/57.095	2.5
3	L12	−11.2003	0.020		2.4
4	L21	33.8670	0.246	1.63549/23.91	2.2
5	L22	3.2281	0.449		1.9
6	L31	−12.2843	0.290	1.5345/57.095	1.9

TABLE 10-continued

Lens system 1000 EFL = 6.9 mm, F number = 2.80, FOV = 44 degrees					
#	Comment	Radius R [mm]	Distances [mm]	Nd/Vd	Diameter [mm]
7	L32	7.7138	2.020		1.8
8	L41	−2.3755	0.597	1.63549/23.91	3.3
9	L42	−1.8801	0.068		3.6
10	L51	−1.8100	0.293	1.5345/57.095	3.9
11	L52	−5.2768	0.617		4.3
12		Infinite	0.410		3.0

TABLE 11

#	Conic coefficient k	α_2	α_3	α_4	α_5	α_6
2	−0.4668	7.9218E−03	2.3146E−02	−3.3436E−02	2.3650E−02	−9.2437E−03
3	−9.8525	2.0102E−02	2.0647E−04	7.4394E−03	−1.7529E−02	4.5206E−03
4	10.7569	−1.9248E−03	8.6003E−02	1.1676E−02	−4.0607E−02	1.3545E−02
5	1.4395	5.1029E−03	2.4578E−01	−1.7734E−01	2.9848E−01	−1.3320E−01
6	0.0000	2.1629E−01	4.0134E−02	1.3615E−02	2.5914E−03	−1.2292E−02
7	−9.8953	2.3297E−01	8.2917E−02	−1.2725E−01	1.5691E−01	−5.9624E−02
8	0.9938	−1.3522E−02	−7.0395E−03	1.4569E−02	−1.5336E−02	4.3707E−03
9	−6.8097	−1.0654E−01	1.2933E−02	2.9548E−04	−1.8317E−03	5.0111E−04
10	−7.3161	−1.8636E−01	8.3105E−02	−1.8632E−02	2.4012E−03	−1.2816E−04
11	0.0000	−1.1927E−01	7.0245E−02	−2.0735E−02	2.6418E−03	−1.1576E−04

Advantageously, the Abbe number of the first, third and fifth lens element is 57.095. Advantageously, the first air-gap between lens elements **1002** and **1004** (the gap between surfaces **1002***b* and **1004***a*) has a thickness (0.020 mm) which is less than a tenth of thickness d_2 (0.246 mm). Advantageously, the Abbe number of the second and fourth lens elements is 23.91. Advantageously, the third air-gap between lens elements **1006** and **1008** has a thickness (2.020 mm) greater than TTL/5 (5.904/5 mm). Advantageously, the fourth air-gap between lens elements **108** and **110** has a thickness (0.068 mm) which is smaller than $d_5/2$ (0.293/2 mm).

The focal length (in mm) of each lens element in lens system **1000** is as follows: f1=2.645, f2=−5.578, f3=−8.784, f4=9.550 and f5=−5.290. The condition 1.2×|f3|>|f2|<1.5×f1 is clearly satisfied, as 1.2×8.787>5.578>1.5×2.645. f1 also fulfills the condition f1<TTL/2, as 2.645<2.952.

FIG. **11**A shows an example of a host device **1100** such as a smartphone with a dual-camera comprising a regular (non pop-up) folded Tele camera **1102** and a Wide pop-out camera **1104**. The Wide camera **1104** is in an operative pop-out state and extends the device's exterior surface **228**. Bump **226** is visible. A large image sensor such as **208** (not visible here) and a pop-out frame such as frame **220** (not fully visible here) required for switching between a collapsed and a pop-out camera state define a minimum area of the device's exterior surface **228** that is covered by the pop-out camera (in X-Z). The minimum pop-out camera area may be larger than that of folded Tele cameras or that of regular (i.e. non pop-out) upright Wide cameras that are typically included in a device.

FIG. **11**B shows details of folded Tele camera **1102** and the upright Wide camera **1104** in a pop out state. The folded Tele camera comprises a prism **1108** and a folded Tele lens and sensor module **1112**. In FIG. **11**A and FIG. **11**B only prism **1108** is visible.

FIG. **11**C shows host device **1100** with Wide camera **1104** in a collapsed state, illustrating the small height of the c-bump.

FIG. **11**D shows details of the folded Tele camera and the upright Wide camera in a collapsed state.

FIG. **12**A shows another example of a host device **1200** such as a smartphone with a dual-camera comprising a Tele pop-out camera **1202** as disclosed herein and a Wide pop-out camera **1204** in an operative pop-out state. Pop-out bump **226** is visible. A pop-out mechanism cover **1206** covers both the Tele and the Wide camera. A frame like **220** (not shown) switches the Tele and the Wide camera between a pop-out state and a collapsed state together and simultaneously. Pins **1208** may provide mechanical stability and repeatability in the X-Z plane. In some examples, 2 pins may be included. In other examples, 3 or more pins may be used.

FIG. **12**B shows details of upright Tele camera **1202** and upright Wide camera **1204**, with both cameras in the pop-out state.

FIG. **12**C shows host device **1200** with the cameras in a collapsed state. A c-bump **236** is shown. FIG. **12**E shows details of upright Tele camera **1202** and upright Wide camera **1204**, with both cameras in the collapsed state.

FIG. **13** shows yet another example of a lens system numbered **1300** comprising a lens **1320** including seven lens elements L**1**-L**7**, optionally optical window **234**, and an image sensor **208**. Here, image sensor **208** is a curved image sensor, meaning that its light collecting surface is curved with a radius of curvature R=−19.026 mm wherein the "−" sign refers to a curvature with center at the object side of the image sensor. Use of a curved image sensor may be beneficial as undesired effects such as field curvature and shading toward the sensor edges may be less than for a planar image sensor. Lens system **1300** may be used in a camera such camera **200** in a pop-out state. The design data is given in Tables 12-14.

In lens system **1300**, TTL=8.28 mm, BFL=3.24 mm, EFL=6.95 mm, F number=1.85 and the FOV=80.52 deg.

In the collapsed state (see FIG. **2**C), cTTL may be 6.54-10 mm. The difference between cTTL and TTL stems from a modified BFL which is now a collapsed "c-BFL" (see FIG. **5**A). c-BFL may be 1.494-2.5 mm. The optical properties of lens **1320** do not change when switching between a pop-out state and a collapsed state (i. e. all distances between the lens elements L**1**-L**7** and the lens surfaces S**2**-S**15** did not change).

TABLE 12

Lens system 1300 EFL = 6.95 mm, F number = 1.85, FOV = 80.52 deg.										
Surface #	Comment	Type	Curvature Radius	Thickness	Aperture Radius (D/2)	Material	Index	Abbe #	Focal Length
	A.S	Plano	Infinity	−0.681	1.879				
	Lens 1	QT1	2.791	0.577	1.879	Plastic	1.54	55.9	11.01
			4.824	0.426	1.824				
	Lens 2	QT1	5.964	0.244	1.824	Plastic	1.67	19.4	−12.73
			3.463	0.005	1.795				
	Lens 3	QT1	3.871	0.959	1.814	Plastic	1.54	55.9	7.85
			35.947	0.452	1.783				
	Lens 4	QT1	−134.398	0.458	1.812	Plastic	1.66	20.37	63.02
			−32.072	0.342	2.004				
	Lens 5	QT1	−4.466	0.252	2.096	Plastic	1.57	37.4	−10.23
			−19.576	0.286	2.280				
	Lens 6	QT1	7.072	0.364	2.336	Plastic	1.54	55.9	3.95
			−3.047	0.203	2.714				
	Lens 7	QT1	−7.929	0.478	3.717	Plastic	1.54	55.9	−5.59
			5.074	1.744	4.108				
	Filter	Plano	Infinity	0.2100	—	Glass	1.52	64.2	
			Infinity	1.284	—				
	Image	Plano	−19.026	—	—				

TABLE 13

Aspheric Coefficients					
Surface #	R_{norm}	A0	A1	A2	A3
2	2.170	6.97E−02	6.81E−02	4.19E−02	1.88E−02
3	2.170	1.92E−01	1.54E−01	7.45E−02	2.62E−02
4	1.891	−1.23E−01	2.32E−02	−4.13E−04	−3.43E−04
5	1.891	−1.59E−01	2.59E−02	−6.88E−03	−1.68E−03
6	1.891	−3.55E−03	3.14E−02	3.10E−03	1.53E−03
7	1.891	−4.64E−02	2.24E−02	1.31E−02	5.48E−03
8	2.225	−5.98E−01	1.18E−01	8.64E−02	1.98E−02
9	2.225	−4.13E−01	9.40E−02	1.01E−01	5.25E−02
10	2.670	−3.70E−01	−1.57E−01	−7.49E−02	3.70E−02
11	2.670	−6.73E−01	2.51E−01	−1.90E−01	−1.12E−02
12	3.671	−3.40E+00	9.02E−01	−4.45E−01	5.39E−02
13	3.671	3.45E+00	5.97E−02	−2.00E−01	−2.88E−02
14	5.340	2.76E+00	7.29E−02	5.26E−04	−1.83E−01
15	5.340	−6.08E+00	7.69E−02	−7.67E−01	−1.27E−01

TABLE 14

Aspheric Coefficients					
Surface #	A4	A5	A6	A7	A8
2	6.08E−03	1.29E−03	1.49E−04	0.00E+00	0.00E+00
3	6.42E−03	1.10E−03	1.85E−04	0.00E+00	0.00E+00
4	−6.39E−04	−2.45E−04	−7.91E−05	0.00E+00	0.00E+00
5	−1.32E−03	−9.94E−06	1.50E−04	0.00E+00	0.00E+00
6	−4.90E−05	3.53E−04	2.69E−04	0.00E+00	0.00E+00
7	2.07E−03	6.57E−04	1.36E−04	0.00E+00	0.00E+00
8	3.42E−04	−1.45E−03	−8.99E−04	0.00E+00	0.00E+00
9	1.74E−02	5.33E−03	4.89E−04	0.00E+00	0.00E+00
10	−3.53E−02	−1.64E−02	−1.34E−02	0.00E+00	0.00E+00
11	−4.40E−02	−4.87E−03	−1.09E−02	0.00E+00	0.00E+00
12	−1.00E−02	−3.74E−02	−5.14E−02	9.87E−03	1.02E−02
13	−2.41E−02	5.41E−02	4.32E−02	1.11E−02	7.56E−03
14	−2.94E−01	−1.73E−01	−8.86E−02	−1.66E−02	−2.06E−03
15	−1.78E−01	−7.01E−02	−5.97E−02	−2.09E−02	−8.43E−03

In other examples, optical window **234** may be curved. A radius of curvature R_W of the optical window may be of same sign as the radius of curvature R of curved image sensor **208** (i.e. with a center at the object side of the optical window) and may be curved in a similar way, so R_W may e.g. be R_W=−15 to −25 mm. In another example may be R_W=R, with R being radius of curvature of the curved image sensor. This may allow for a smaller cTTL. cTTL may be 5.64-7.54 mm and c-BFL may be 0.594-2.5 mm.

FIG. **14**A shows in cross sectional view another example numbered **1400** of a pop-out camera disclosed herein in a pop-out state and incorporated in a "host" device **250** (e.g. a smartphone, tablet, etc.). Camera **1400** comprises a pop-out frame **220'** and an optics module **600'** that includes a lens **620**. As shown in FIG. **14**B, frame **220'** comprises a window frame **214'**, a cam follower **1402** and a side limiter **1406**. Cam follower **1402** is coupled via springs **1408** to a pop-out actuator **1408**. Optics module **600'** includes a lens barrel **602** with a collapsible lens barrel section (first barrel section) **604** carrying a first lens group **606**, and a fixed lens barrel section (second barrel section) **608** carrying a second lens group **610**. The two lens groups form a lens **620** that includes altogether N lens elements L**1**-LN, arranged with a first lens element L**1** on an object side and a last lens element LN on an image side. Lens **620**, an optional optical window **234** and image sensor **208** form a lens system **630**.

Camera **1400** comprises an external module seal **224** and an internal module seal **1404**. External seal **224** prevents particles and fluids from entering device **250**. Seal **224** may support a IP68 class ranking of device **250**. Internal seal **1404** prevents particles from entering optics module **600'**.

"External" and "internal" refer to the fact that seal **224** prevents contamination of the camera from outside the host device, while seal **1404** prevents contamination of the camera from inside the host device.

Optics module **600'** and window frame **214** form an air-gap **222'** between the lens barrel and window **216**, which may be for example 0.1 mm-3 mm. Air-gap **222'** allows for a movement of the lens barrel by 0.1-3 mm for performing auto-focus (AF) and optical image stabilization (OIS) by moving lens **620** or parts of lens **620** or optics module **600'** or sensor **208** as known in the art.

Camera **1400** forms a significant pop-out bump **226** with respect to an exterior surface **228** of device **250**. Here, "significant" may be for example 1.5 mm-12 mm. In the pop-out state, camera **1400** increases the height of host device **250** to a height in a pop-out state.

Lens **620** may have N≥4 lens elements, and, as mentioned, comprises a barrel with two lens barrel sections. In other examples, the lens barrel may comprise more than two barrel sections with more lens groups, e.g. 3, 4, 5 lens barrel sections with each barrel section carrying a lens group. The lens barrel sections may be divided into fixed barrel sections and movable barrel sections. In the example shown, first lens group **606** includes lenses L**1**-LN−1 and second lens group **610** includes lens LN (see FIG. **14**A). Air-gaps may be formed between lens groups according to their relative movement. In examples with more than two barrel sections, some or all barrel sections may be movable and have respective air-gaps formed between the lens groups. The air-gaps between lens groups may collapse in a non-operative camera state. The sum of such air-gaps may be 1-12 mm. In the pop-out state, air-gap d_{N-1} may be 1-5.5 mm. Three springs **614** (not all visible here) push first lens barrel section **604** towards a mechanical stop. The mechanical stop may be provided by a kinematic coupling mechanism as shown in FIGS. **18**A-B and FIGS. **19**A-B. In other examples and as shown in FIG. **20**C, the mechanical stop may be provided by a top cover **1606'**. In some examples, the camera in pop-out state may be designed to support tolerances for decenter of e.g. ±20 μm in the X-Z plane and of e.g. ±10 μm in the Y direction, as well as a tilt of ±0.2° of the lens barrel with respect to image sensor **208**. In other examples, tolerances for decenter may be e.g. ±2-10 μm in the X-Z plane and of e.g. ±2-10 μm in the Y direction, as well as e.g. ±0.05°-0.15° for a tilt of lens barrel with respect to the image sensor Y.

The TTL of the lens may be 5-22 mm. The image sensor diagonal may be 6 mm<sensor diagonal<30 mm. The 35 eqFL may be 15 mm<equivalent focal length<200 mm. The TTL/EFL ratio may vary in the range 0.7<TTL/EFL<1.5.

A window position measurement mechanism **1420** shown in FIG. **14**B comprises one or more magnets and one or more Hall sensors shown in FIGS. **20**C-E. The magnets are fixedly coupled to a cam follower **1402**, and the Hall sensor(s) is (are) fixedly coupled to a side limiter **1406**. Mechanism **1420** senses the position of the cam follower relative to side limiter **1406** and host device **250**. The camera is mechanically coupled to the host device and the side limiter is mechanically coupled to the camera.

FIG. **14**B shows a perspective view of frame **220'** in a pop-out state. A pop-out camera such as **1400** is formed when optics module **600'** is inserted into frame **220'**. Window frame **214'**, cam follower **1402** and side limiter **1406** move with respect to each other. Window frame **214'** and cam follower **1402** also move with respect to host device **250**, but side limiter **1406** does not move with respect to host device **250**. Camera **1400** is switched from a pop-out state to a collapsed state by moving window frame **214'** in a positive X direction with respect of host device **250** and side limiter **1406**. Window frame **214'** is moved by actuator **212'** via cam follower **1402**. The movement of cam follower **1402** is substantially parallel to the X axis, and this movement is translated in a movement of window frame **214'** substantially parallel to the Y axis. This translation of movements in X direction and in Y direction is described in FIG. **15**A-B. As for the movement along Y, window frame **214'** applies pressure to the lens barrel that translates into a movement of the collapsible lens barrel section towards the image sensor. Cam follower **1402** is coupled via springs **1408** to a pop-out actuator **1412**. Actuator **1412** moves cam follower **1402** e.g. via a screw stepper motor or another actuation method. The movement is mediated by the springs **1408**. Springs **1408** may act as shock absorber for camera **1400**. E.g. when host device **250** is dropped and hits another object, a large force may act on window frame **214'**. By means of springs **1408**, this large force may be translated into a collapse of the pop-out camera, thereby mediating a large portion of the large force. Internal module seal **1404** may act as an additional shock absorber.

FIG. **14**C shows a cross sectional view of camera **1400** in a collapsed ("c") or non-operative state. FIG. **14**D shows a perspective view of frame **220'** in a collapsed state. To switch optics module **600'** to the collapsed state, actuator **212'** decreases air-gap d_{N-1} by moving the window frame **214'** to apply pressure to the lens barrel that translates into a movement of the collapsible lens barrel section towards the image sensor. In the collapsed state, cTTL may be 5-12 mm. and collapsed air-gap c-d_{N-1} may be 0.05-1.5 mm.

FIG. **15**A shows the frame **220'** of example **1400** in a cross sectional view via X-Y plane in a pop-out state. A switching pin **1502** and a switching pin **1504** are rigidly coupled to cam follower **1402**. A side limiter pin **1512** is fixedly coupled to side limiter **1406** and slides within a vertically oriented limiter groove **1514**. Switching pin **1502** and **1504** slide within switching grooves **1506** and **1508**. Switching pins **1502** and **1504** have a diamond shape which is superposed by a curvature having a large radius of curvature for minimizing contact stress acting between the pins and window frame **214'**. Side limiter pin **1512** has a rectangular shape superposed by a curvature having a large radius of curvature for minimizing contact stress.

When cam follower **1402** is moved in a negative X direction, the inclination of switching grooves **1506** and **1508** leads to a downward movement (in a negative Y direction) of window frame **214'**. This downward movement is used to switch the camera to the collapsed state. The downward movement is limited and guided by side limiter pin **1512**. The inclination of switching grooves **1506** and **1508** may e.g. be between 20-80 degrees with respect to a vertical Y axis.

FIG. **15**B shows the frame **220'** of FIG. **15**A in a collapsed state. To switch the camera from the collapsed state to a pop-out state, cam follower **1402** is moved in a positive X direction and the inclination of switching grooves **1506** and **1508** leads to an upward movement (in a positive Y direction) of window frame **214'**.

FIG. **16**A shows a cross sectional view and FIG. **16**B show a perspective view of optics module **600'** in a pop-out state. Module **600'** comprises an optics frame **1650**, first collapsible lens barrel section **604** (lens elements not shown here), second fixed lens barrel section **608**, three springs **614** (not all visible here), a side cover **1604**, a top cover **1606** and three stoppers **1608** (not all visible here). Each spring sits on one of three spring holders **1612** (not all visible here). Optics frame **1650** holds all components of optics module **600'** except the lens elements that are included in the first and the second lens barrel sections. Stoppers **1608** are rigidly coupled to top cover **1606** and ensure that the collapsible lens barrel section (**604**) is not in direct contact with window frame **214**.

A "penalty" p for a diameter of an optics module is defined as the difference between the diameter of the optics module and the largest diameter of a lens included in the optics module. For optics module **600'**, d_{module} is slightly larger than the largest diameter of lens **620**, represented by the diameter of L_N. Therefore, for optics module **600'**, penalty p is $p=p_1+p_2$ and may be 0.5 mm-8 mm.

FIG. **17**A shows optics module **600'** in cross-section and FIG. **17**B shows the module in perspective in a collapsed state. In the collapsed state, springs **614** are compressed.

FIGS. **18**A-**18**F shows optics frame **1650** in various positions and with various details of its components. FIG. **18**A shows optics frame **1650** in a pop-out state and FIG. **18**B shows optics frame **1650** in a collapsed state, both in a perspective view. Collapsible lens barrel section **604** is coupled to optics frame **1650** via a "Maxwell kinematic coupling" mechanism. The Maxwell kinematic coupling mechanism comprises three v-groove/pin pairs **1810** that act as a guiding and positioning mechanism that ensures that collapsible lens barrel section **604** is kept in a fixed position relative to the other optical elements such as image sensor **208** with high accuracy. Each v-groove/pin pair **1810** is identical and includes a hemispherical pin **1812** and a v-groove **1814**. More details of a v-groove/pin pair **1810** are given in FIG. **18**C (for the pop-out state) and FIG. **18**D (for the collapsed state). In other examples, the pins may be round or diamond-shaped or canoe-shaped. The v-grooves shown in FIG. **18**A-**18**D have an angle of about 90 degrees. In other examples, the angle of the v-grooves may vary between 30 to 150 degrees.

Pairs **1810** are distributed at equal distance from each other. By means of the three v-groove/pin pairs **1810**, optics frame **1650** supports narrow tolerances in terms of accuracy as well as repeatability for decenter in X-Z and Y as well as for tilt. Here and in the description of FIGS. **19**A and **19**B, "tolerances" refer to tolerances between collapsible lens barrel section **604** the fixed lens barrel section **608**.

Optics frame **1650** as well as optics module **600"** below may be designed to support accuracy tolerances for decenter and reliability tolerances like those of camera **200**.

FIG. **18**E shows optics frame **1650** in a top view. FIG. **18**F shows optics frame **1650** in an exploded view showing the single parts that **1650** may be assembled from. Three spring holders **1612** keep the three respective springs **614** in a fixed position. One may assemble optics frame **1650** from the bottom to the top. One may start an assembly process with inserting L_N into fixed lens barrel section **608**, then insert springs **614** into spring holders **1612**, then put top cover **1606**, then put side cover **1604** and then insert the collapsible lens barrel **604** on top. In some examples such as shown in FIGS. **2**A-D and FIGS. **4**A-B, a lens such as lens **420** may be included in an optics frame such as **1650**. Lens **420** includes a single group of lens elements only and may be included entirely in a collapsible lens barrel such as **604**. In some examples, collapsible lens barrel **604** and top cover **1606** may be one single unit.

FIG. **19**A and FIG. **19**B show (in perspective and cross section respectively) another optics module numbered **600"**. Optics module numbered **600"** comprises a guiding and positioning mechanism for keeping collapsible lens barrel section **604** in a fixed position with high accuracy. The guiding and positioning mechanism is based on a yoke-magnet pair. A yoke **2002** is fixedly coupled to top cover **1606'**, and a permanent magnet **2004** is fixedly coupled to the side cover **1604'**. Through the use of yoke **2002** and magnet **2004**, top cover **1606** and the side cover **1604** are attracted to each other, keeping a constant distance and orientation to each other. Optics module **1650'** thus supports narrow tolerances in terms of accuracy as well as repeatability for decenter in X-Z and Y and for tilt.

FIG. **19**C shows optics module **600"** in a pop-out state in a cross-sectional view. A side cover **1604'** acts also as a second and fixed lens barrel section carrying a second group of lens elements, i.e. no additional component acting as second lens barrel section is required. FIG. **19**D shows optics module **600"** in a collapsed state in a cross-sectional view.

FIG. **19**E shows a perspective view and FIG. **19**F shows a top view of top cover **1606'** and magnet **2004**.

FIG. **20**A shows a side view and FIG. **20**B shows a perspective view of a magnet part of window position measurement mechanism **1420** in a collapsed state. Two side magnets **2102***a* and **2102***b* are located on both sides of an inner (auxiliary) magnet **2104**. All magnets are fixedly coupled to cam follower **1402**. Magnets **2102***a*, **2102***b* and **2104** create a magnetic field that is sensed by Hall sensor **2106**. Hall sensor **2106** is fixedly coupled to side limiter **1406** (not shown here). The magnetic field sensed by Hall sensor **2106** depends on the relative position of cam follower **1402** and side limiter **1406**. That is, mechanism **1420** allows sensing of the relative position of cam follower **1402** and side limiter **1406** continuously along a stroke that may be in a range 1-10 mm.

FIG. **20**C shows a side view of magnets **2102***a*, **2102***b*, **2104** and Hall sensor **2106**, with camera **1400** shown in a collapsed state. FIG. **20**D shows a side view of magnets **2102***a*, **2102***b*, **2104** and Hall sensor **2106**, with camera **1400** shown in a pop-out state. The stroke extends between the extreme positions shown here, i.e. between the collapsed state and the pop-out state. In some examples, mechanism **1420** may measure the relative position of **1402** and **1406** with the same accuracy along the entire stroke. In other examples and beneficially, mechanism **1420** may measure the relative position of **1402** and **1406** with a higher accuracy close to the extreme positions shown here, and with a lower accuracy in other positions.

FIG. **20**E shows an example of (a) a design and (b) the magnetic field of mechanism **1420**, with magnetization of magnets **2102***a*, **2102***b* and **2104** shown.

FIG. **20**F shows an example of a magnet configuration **2110** that may be included in a position measurement mechanism such as **1420**. A configuration of magnets **2102***a*, **2102***b* and **2104** is shown in (a), and magnetic flux density versus a position X as created by the magnet configuration of (a) is shown in (b). A large and substantially identical slope $\Delta B/\Delta X$ may be achieved along a linear range. The linear range of **2110** may extend between 1-10 mm.

FIG. **20**G shows another example of a magnet configuration **2120** that may be included in a position measurement mechanism such as **1420**. A configuration of magnets **2102***a*, **2102***b* and **2104** is shown in (a), and magnetic flux density versus a position X as created by the magnet configuration of (a) is shown in (b). The linear range is divided into three sub-ranges A1, B and A2. In the sub-ranges A1 and A2, slope $\Delta B/\Delta X$ is larger than the slope in sub-range B. For example, a slope in the sub-ranges A1 and A2, $\Delta B/\Delta X(A)$, may be 5 times, 10 times or 25 times larger than a slope in the sub-range B, $\Delta B/\Delta X(B)$. For example, $\Delta B/\Delta X(A)\sim 500$ mT/mm, and $\Delta B/\Delta X(B)\sim 50$ mT/mm, so that a ratio of $[\Delta B/\Delta X(A)]/[\Delta B/\Delta X(B)]=10$. The division in of the linear range in sub ranges with different slopes may be beneficial for a position measurement mechanism such as **1420**, as a higher accuracy may be required in the extreme regions close to the positions of the pop-out state and the collapsed state.

In summary, disclosed herein are digital cameras with a pop-out mechanisms that allow for large EFLs and large image sensor sizes and low camera heights in a collapsed mode.

While this disclosure has been described in terms of certain examples and generally associated methods, alterations and permutations of the examples and methods will be apparent to those skilled in the art. The disclosure is to be understood as not limited by the specific examples described herein, but only by the scope of the appended claims.

It is appreciated that certain features of the presently disclosed subject matter, which are, for clarity, described in the context of separate examples, may also be provided in combination in a single example. Conversely, various features of the presently disclosed subject matter, which are, for brevity, described in the context of a single example, may also be provided separately or in any suitable sub-combination.

Furthermore, for the sake of clarity the term "substantially" is used herein to imply the possibility of variations in values within an acceptable range. According to one example, the term "substantially" used herein should be interpreted to imply possible variation of up to 10% over or under any specified value. According to another example, the term "substantially" used herein should be interpreted to imply possible variation of up to 5% over or under any specified value. According to a further example, the term "substantially" used herein should be interpreted to imply possible variation of up to 2.5% over or under any specified value.

Unless otherwise stated, the use of the expression "and/or" between the last two members of a list of options for selection indicates that a selection of one or more of the listed options is appropriate and may be made.

It should be understood that where the claims or specification refer to "a" or "an" element, such reference is not to be construed as there being only one of that element.

All patents and patent applications mentioned in this specification are herein incorporated in their entirety by reference into the specification, to the same extent as if each individual patent or patent application was specifically and individually indicated to be incorporated herein by reference. In addition, citation or identification of any reference in this application shall not be construed as an admission that such reference is available as prior art to the present disclosure.

What is claimed is:

1. A camera integrated in a mobile device, comprising:

a lens assembly that includes N lens elements L**1**-LN starting with L_1 on an object side, wherein N≥6;

an image sensor having a sensor diagonal SD in the range of 11-22 mm; and

a pop-out mechanism configured to control an air-gap between a lens element and the image sensor to bring the camera to an operative pop-out state and to a collapsed state,

wherein the pop-out mechanism is configured to raise and lower a moving window element relative to a stationary surface of the mobile device and comprises an external module seal configured to prevent liquids and particles from penetrating the mobile device,

wherein the external module seal is coupled to the moving window element and to the stationary surface so as to adapt to a motion of the moving window element,

wherein the lens assembly has a total track length TTL in the operative pop-out state and a collapsed total track length cTTL in the collapsed state, and

wherein cTTL/SD<0.65.

2. The camera of claim **1**, wherein the external module seal has a collapsible structure, an inner edge and an outer edge, and wherein the external module seal collapses into a peripheral bulge formed between its inner and outer edges in response to the pop-out mechanism lowering the moving window element.

3. The camera of claim **2**, wherein the peripheral bulge of the external module seal unfolds in response to the pop-out mechanism raising the moving window element.

4. The camera of claim **3**, wherein in the pop-out state, the external module seal extends to form a crown with a tubular shape.

5. The camera of claim **3**, wherein in the collapsed state, the peripheral bulge extends inwards into the mobile device.

6. The camera of claim **1**, wherein the external module seal facilitates completely preventing ingress of dust into the mobile device and enabling the mobile device to be continuously immersed in water under predefined pressure conditions.

7. The camera of claim **6**, wherein the external module seal supports an ingress protection ranking IP68 of the mobile device.

8. The camera of claim **1**, wherein the pop-out mechanism includes a stepper motor.

9. The camera of claim **1**, wherein cTTL/SD<0.625.

10. The camera of claim **1**, wherein N≥6 includes N=6 or 7.

11. The camera of claim **1**, wherein the lens has a f number f/# in the range 1.7-2.5.

12. The camera of claim **1**, wherein the lens has a f number f/# in the range 1.5-1.7.

13. The camera of claim **1**, wherein the camera includes an optics module comprising the lens, wherein the pop-up mechanism includes a window frame engageable with the optics module, wherein the window frame does not touch the optics module in the operative pop-out state, and wherein the window frame is operable to press on the optics module to bring the camera to the collapsed state.

14. The camera of claim **13**, wherein the moving window element is the window frame.

15. The camera of claim **13**, wherein the optics module has an optics module diameter dO, wherein the lens assembly has a maximum lens diameter dL, and wherein dO<dL+8 mm.

16. The camera of claim **15**, wherein dO<dL+4 mm.

17. The camera of claim **15**, wherein dO<dL+2 mm.

18. The camera of claim **15**, wherein dO<dL+1 mm.

19. The camera of claim **1**, wherein the pop-out mechanism comprises a guiding and positioning mechanism based on a kinematic coupling mechanism.

20. The camera of claim **1**, wherein the mobile device is a smartphone.

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